

Qalypso 2026 Summer School

Semidefinite Programming as a Tool for Quantum Information

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Paris, France

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How to use SDPs

Is your problem an SDP?

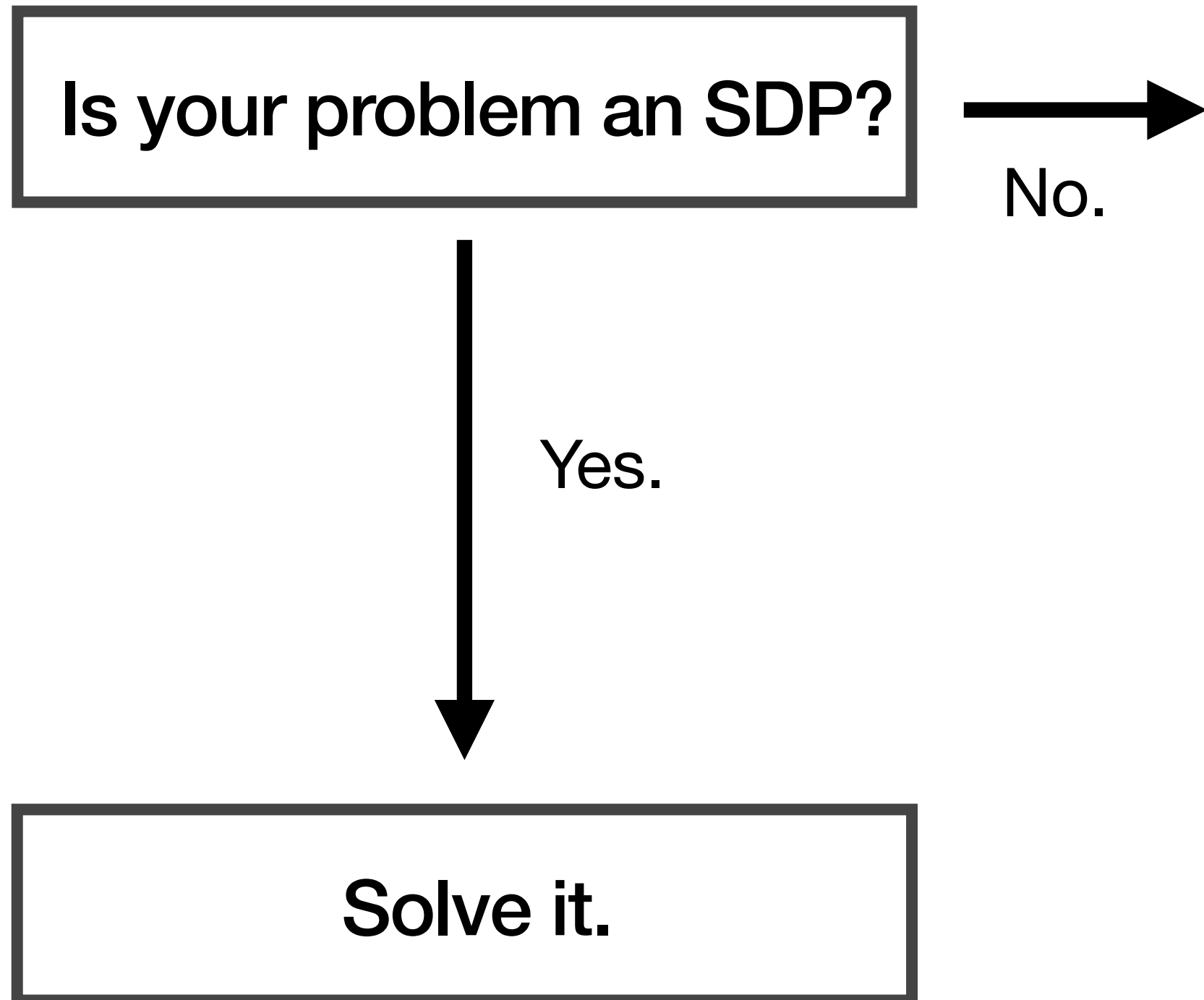
How to use SDPs

Is your problem an SDP?

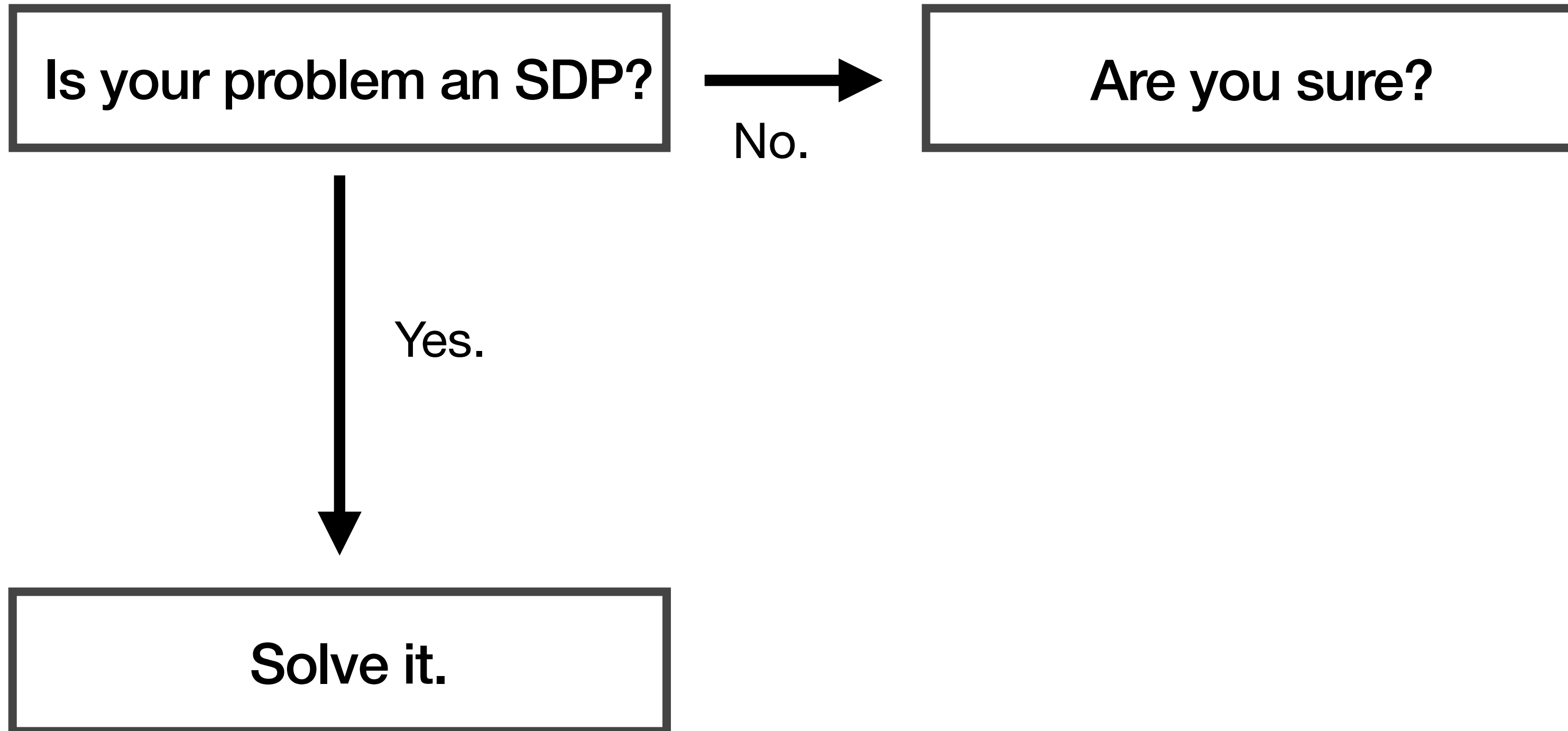
Yes.

Solve it.

How to use SDPs



How to use SDPs



How to use SDPs

PART I: Marco Túlio

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Yes.

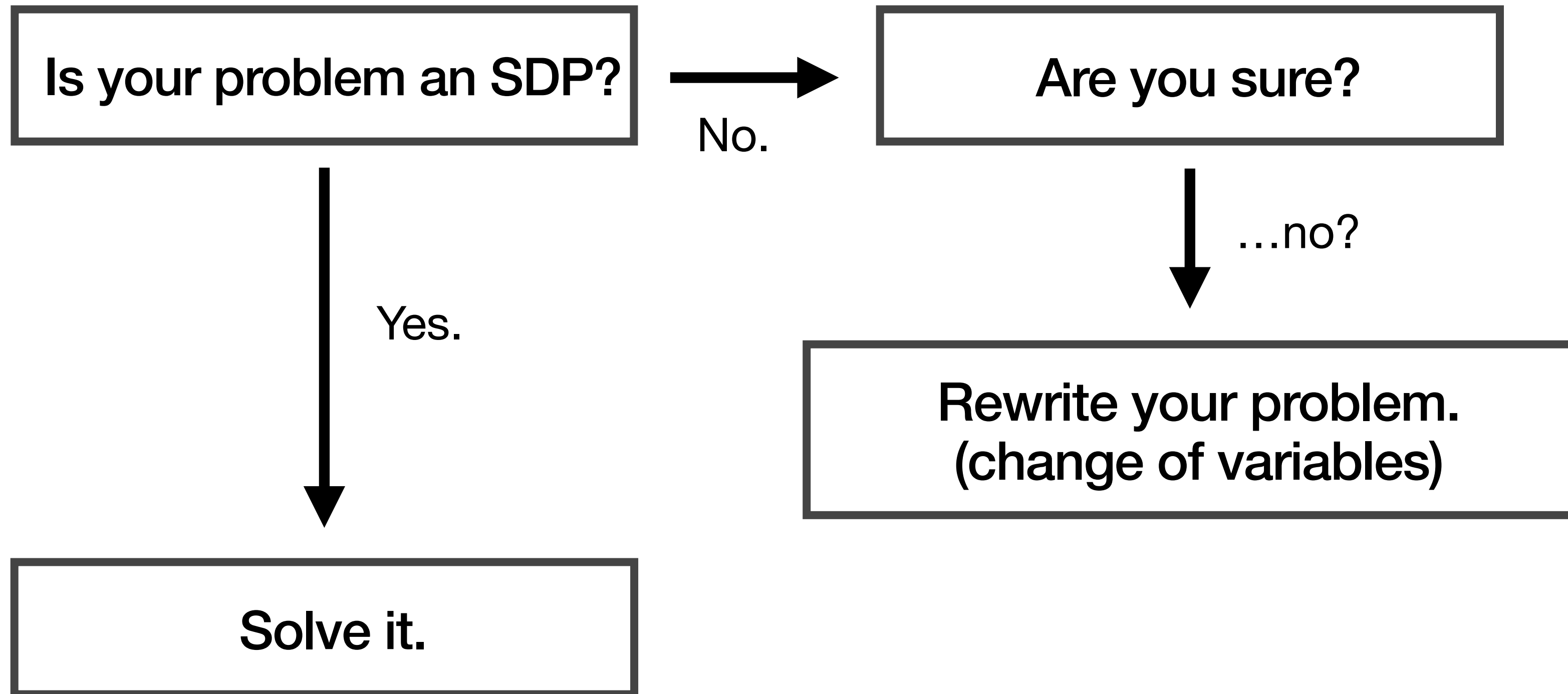
Solve it.

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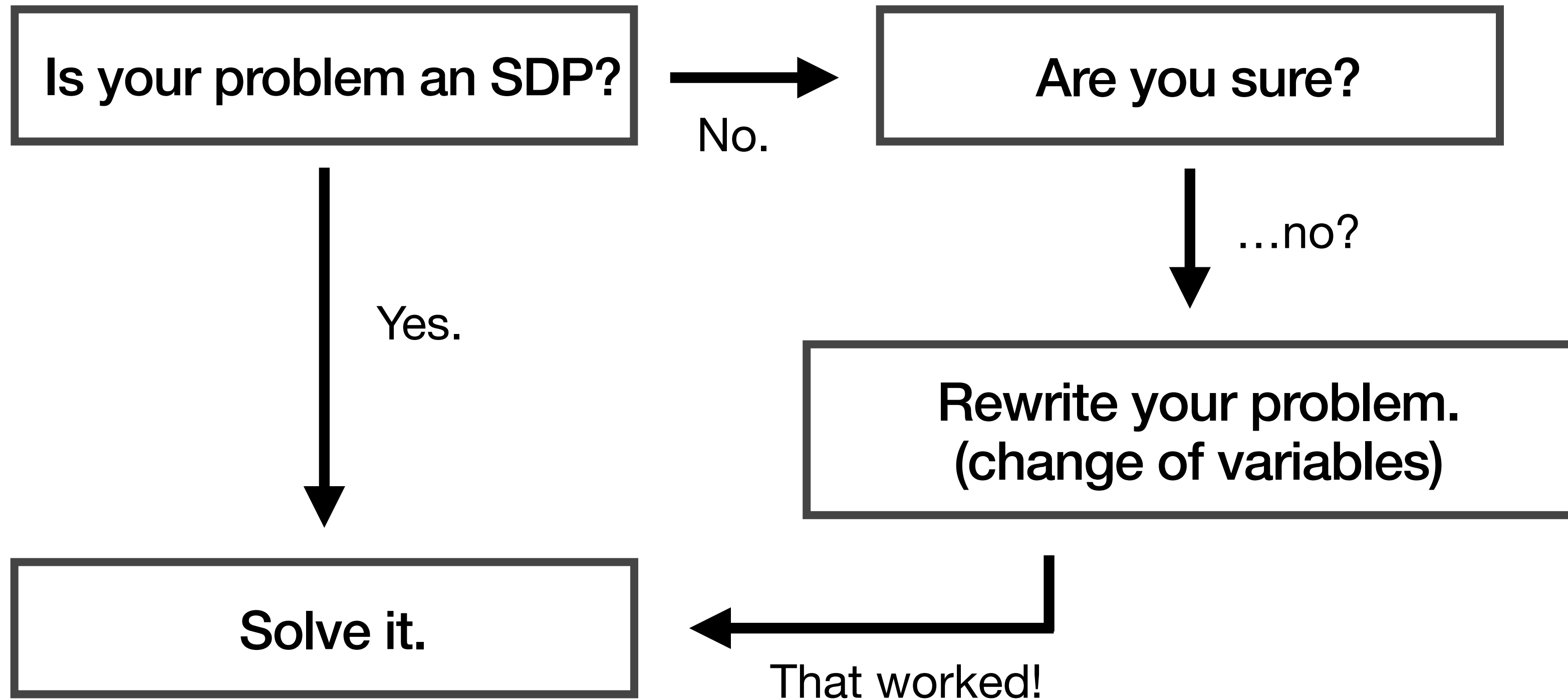
Are you sure?

PART II: Jessica

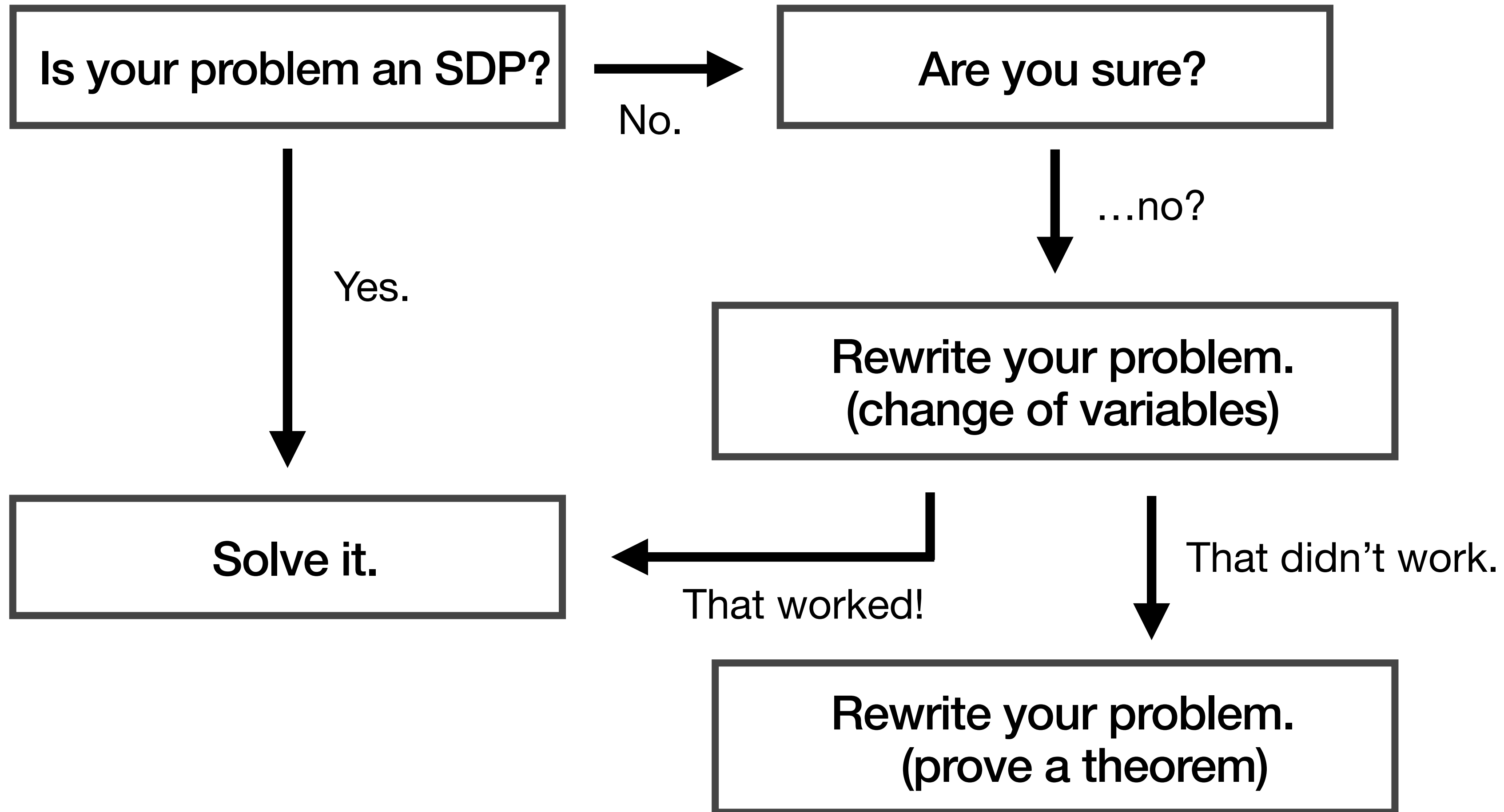
How to use SDPs



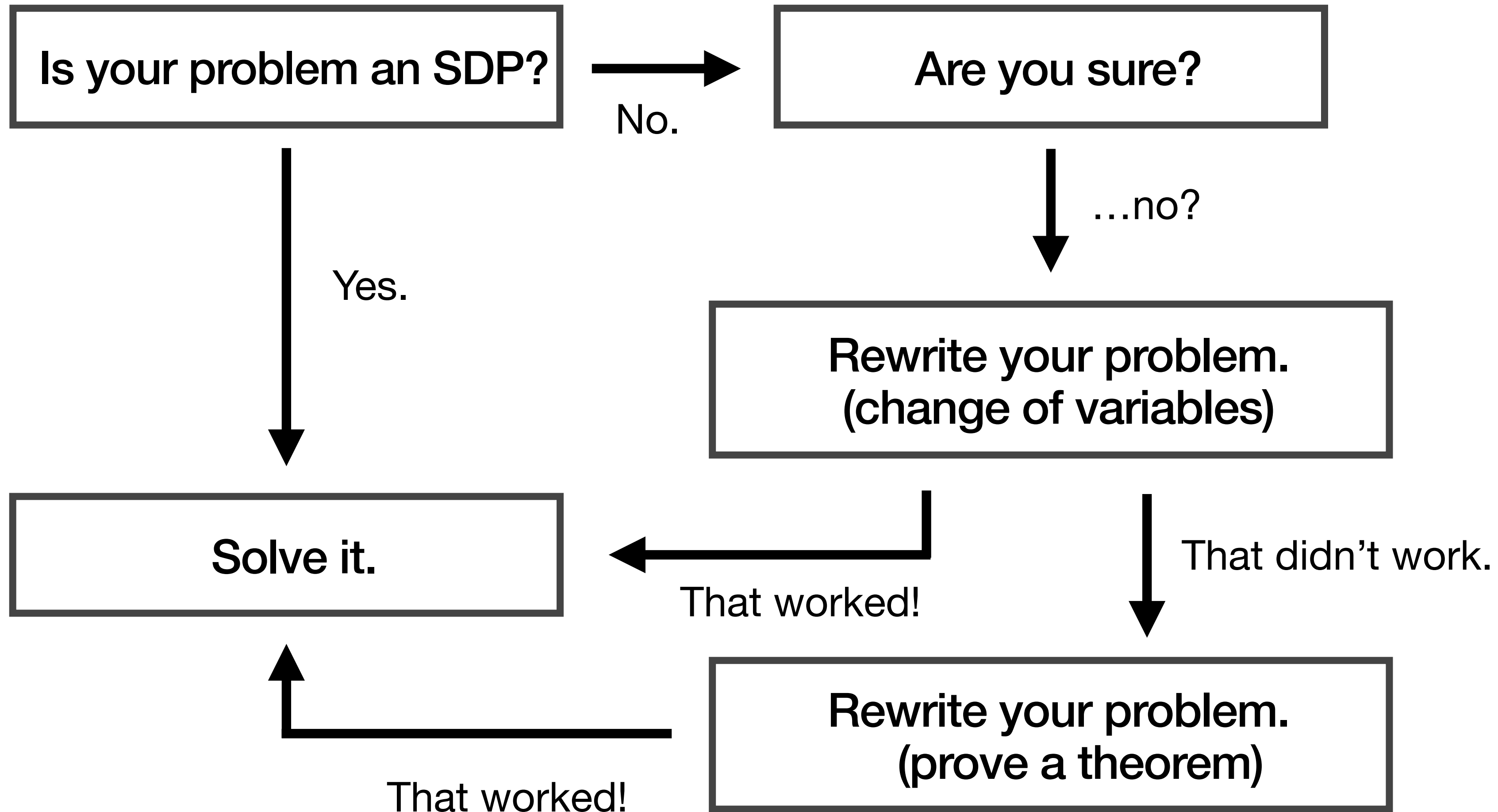
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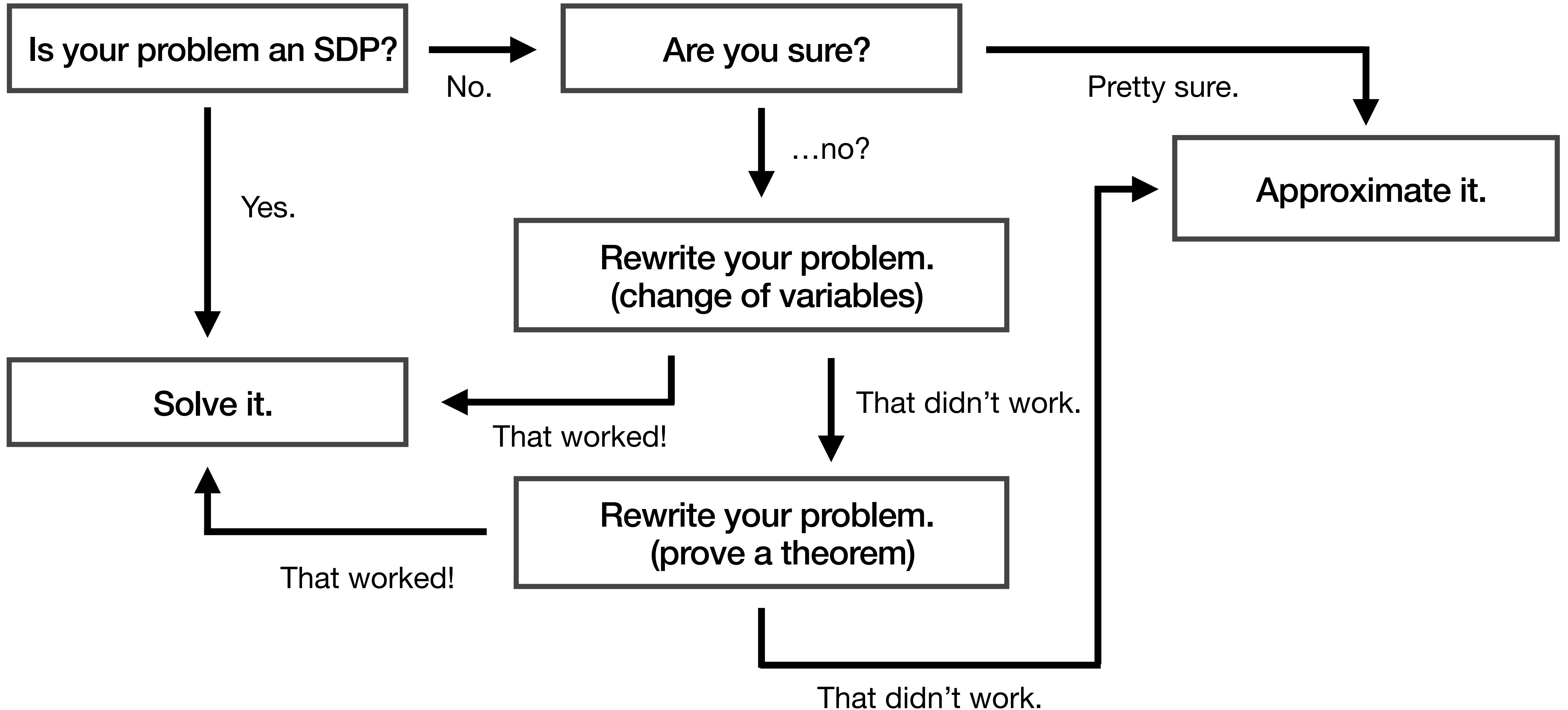
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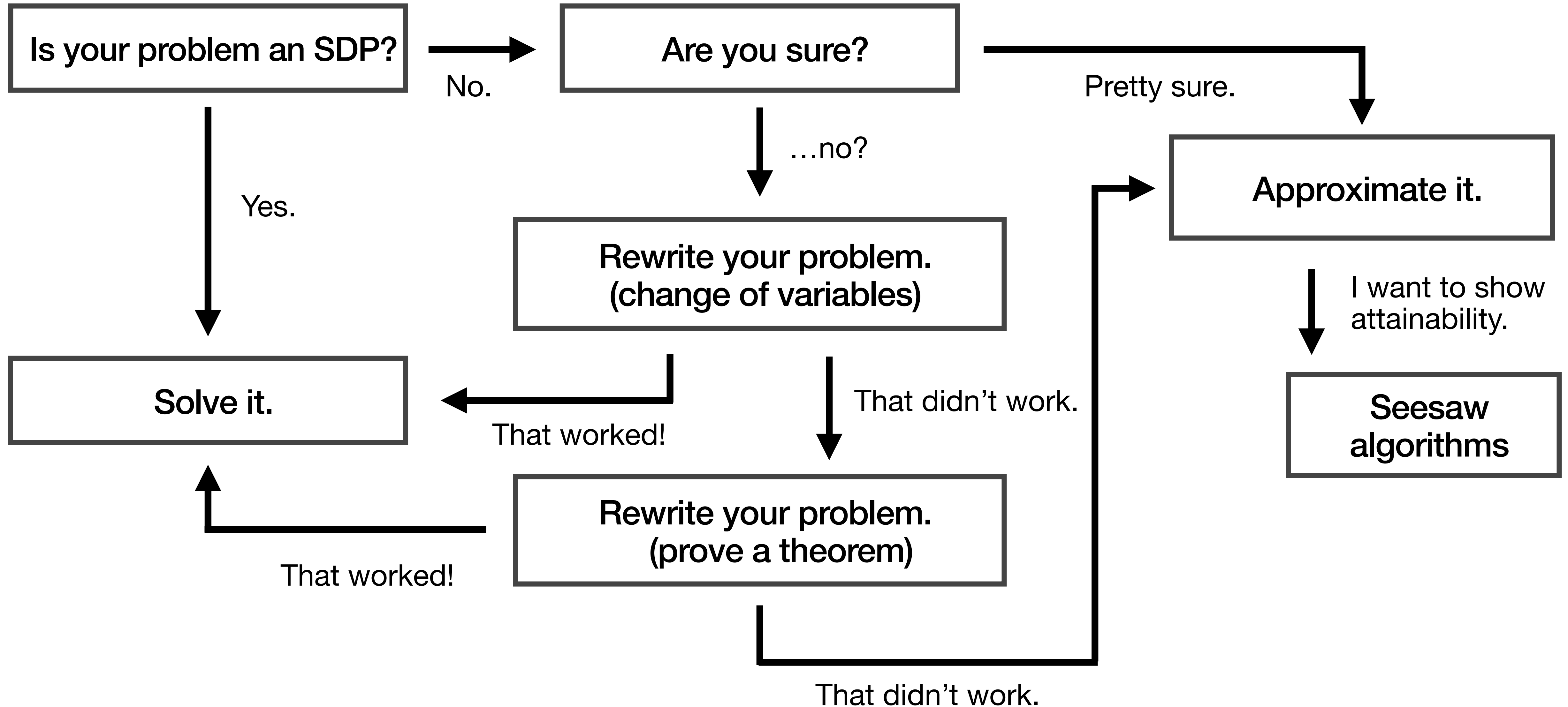
How to use SDPs



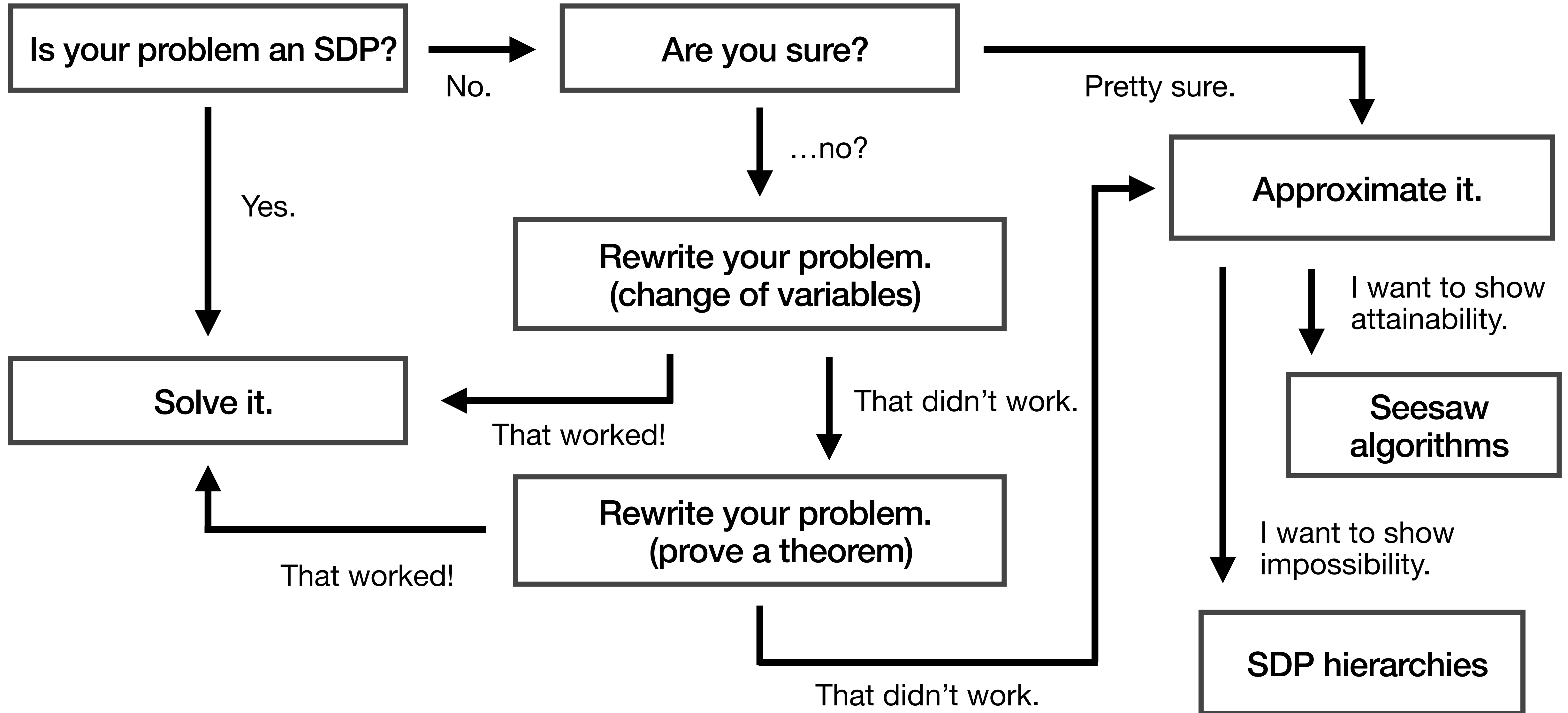
How to use SDPs



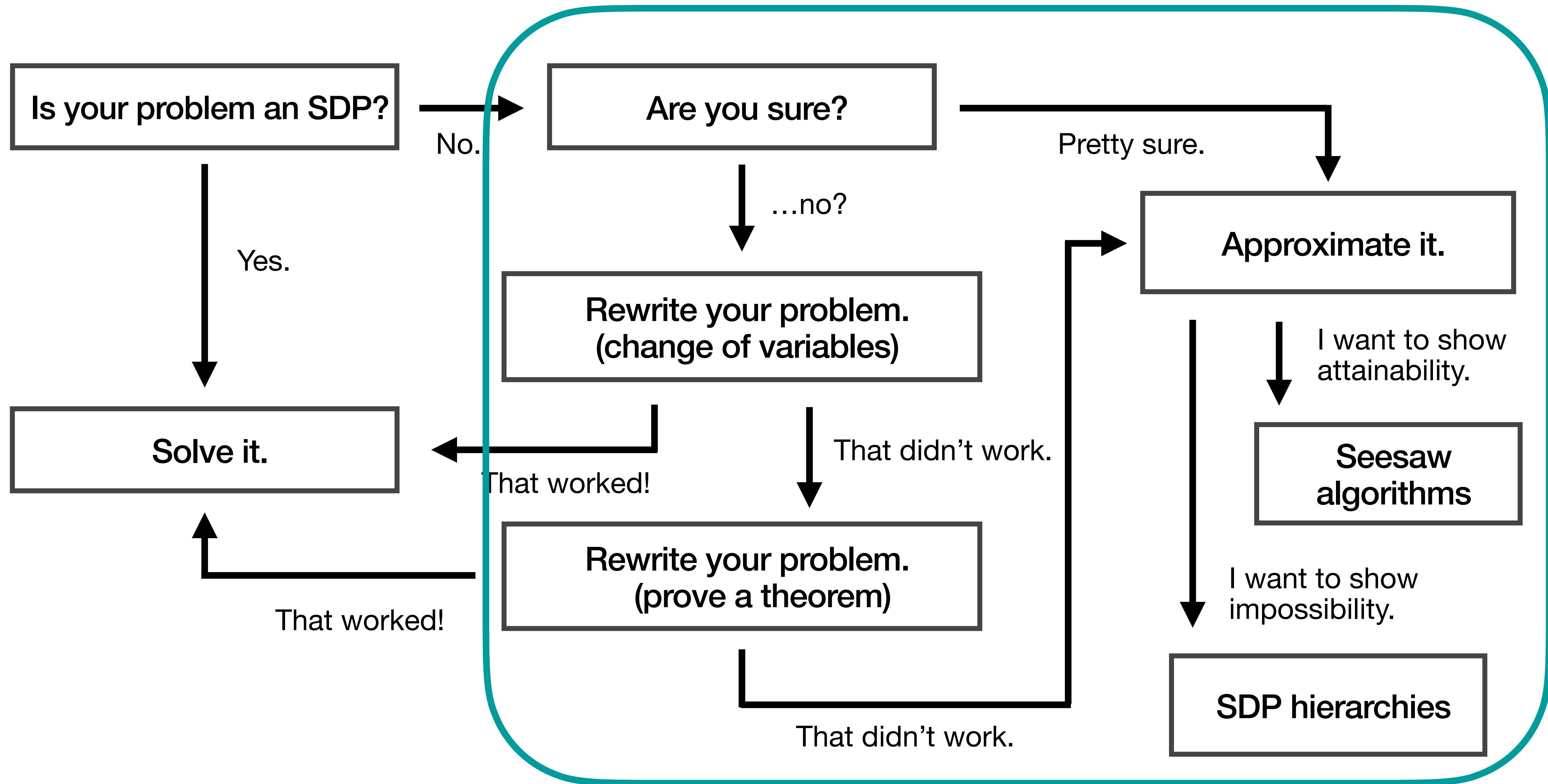
How to use SDPs



How to use SDPs



How to use SDPs



Course outline

- **I. Rewriting**

- Variable change (easy version): **Generalized noise robustness of incompatibility**
- External proof (advanced version): **Channel discrimination** and the diamond distance

- **II. Approximating**

- **Attainability**

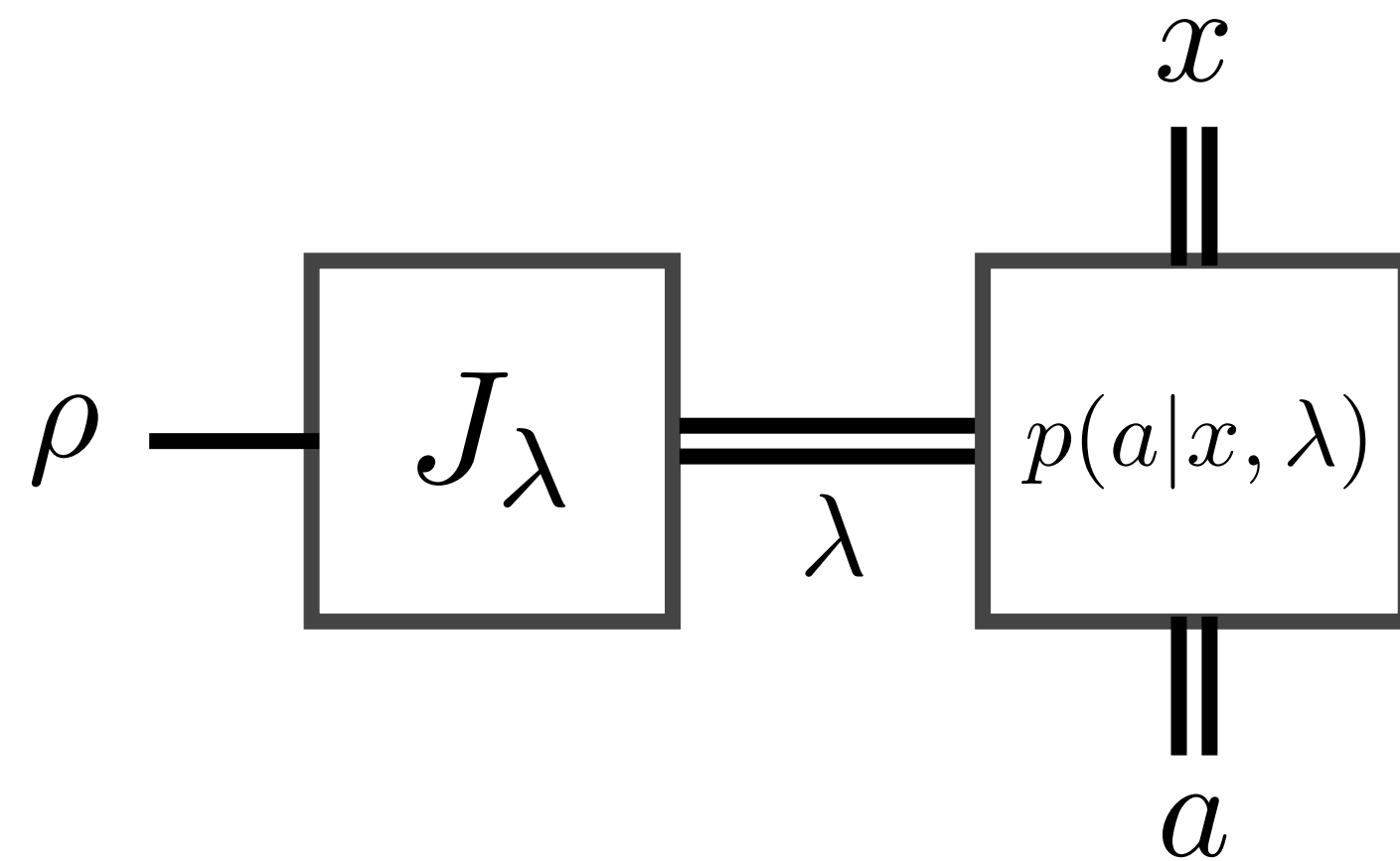
- Seesaw algorithms (easy version): **Channel discrimination** without memory
- Seesaw algorithms (advanced version): **Most incompatible measurements**

- **Impossibility**

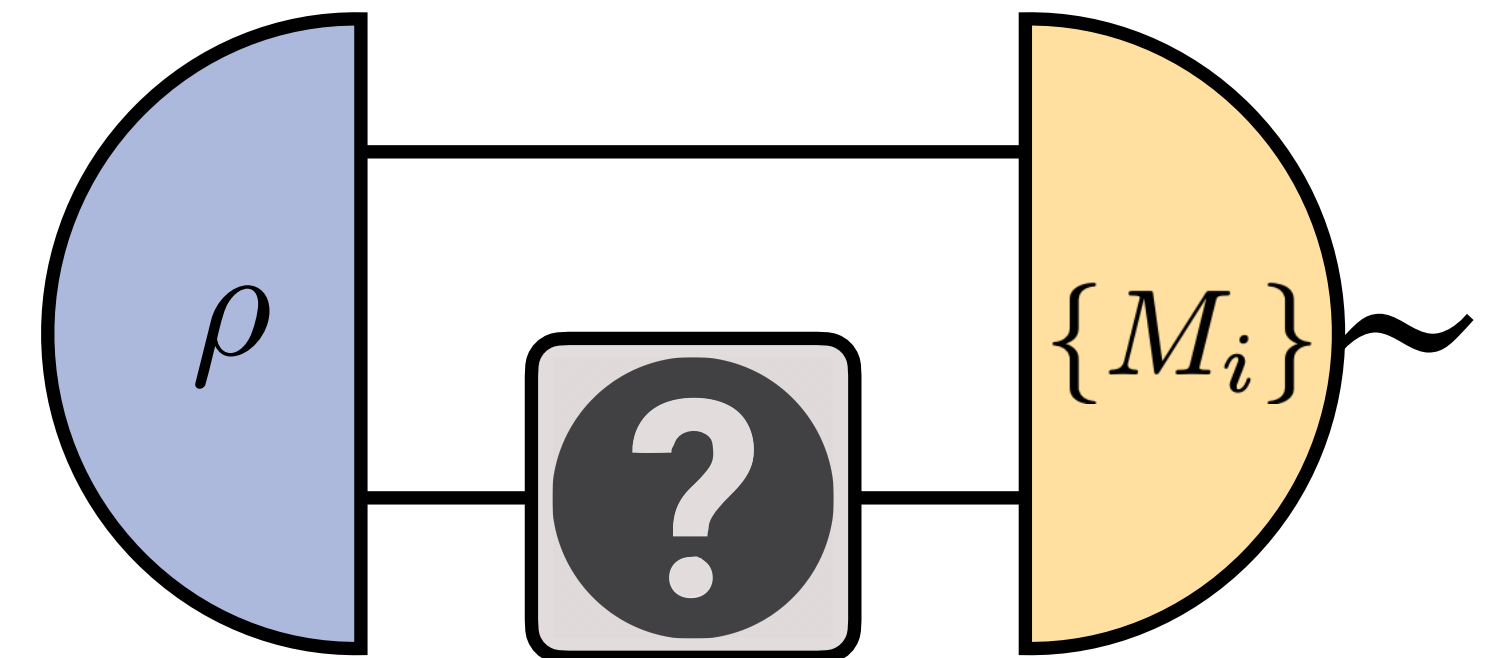
- Outer approx. (easy version): **Channel discrimination** without memory
- SDP hierarchy (advanced version): **Entanglement** and the k-symmetric extension hierarchy

Case studies

Measurement incompatibility



Channel discrimination



Entanglement

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

I. Rewriting

I. Rewriting your problem as an SDP

Generalized noise robustness of measurement incompatibility

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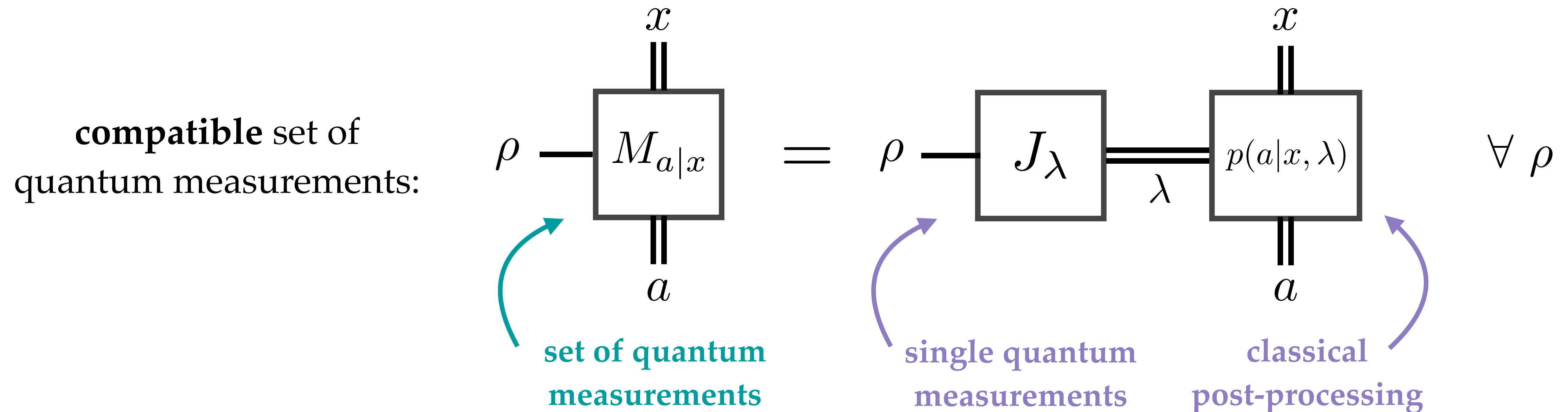
Generalized noise robustness of measurement incompatibility

Definition. A set of quantum measurements is **compatible** when it can be expressed as a single quantum measurement and classical post-processing.

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set of quantum measurements: $\{M_{a|x}\}, \quad M_{a|x} \geq 0 \quad \forall a, x, \quad \sum_a M_{a|x} = \mathbb{1} \quad \forall x$

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compatible set of quantum measurements: $\exists \{J_\lambda\}, \quad J_\lambda \geq 0 \quad \forall \lambda, \quad \sum_\lambda J_\lambda = \mathbb{1}$
 $\exists \{p(a|x, \lambda)\}, \quad p(a|x, \lambda) \geq 0 \quad \forall a, x, \lambda, \quad \sum_a p(a|x, \lambda) = 1 \quad \forall x, \lambda$

$$M_{a|x} = \sum_\lambda p(a|x, \lambda) J_\lambda \quad \forall a, x$$

I. Rewriting your problem as an SDP

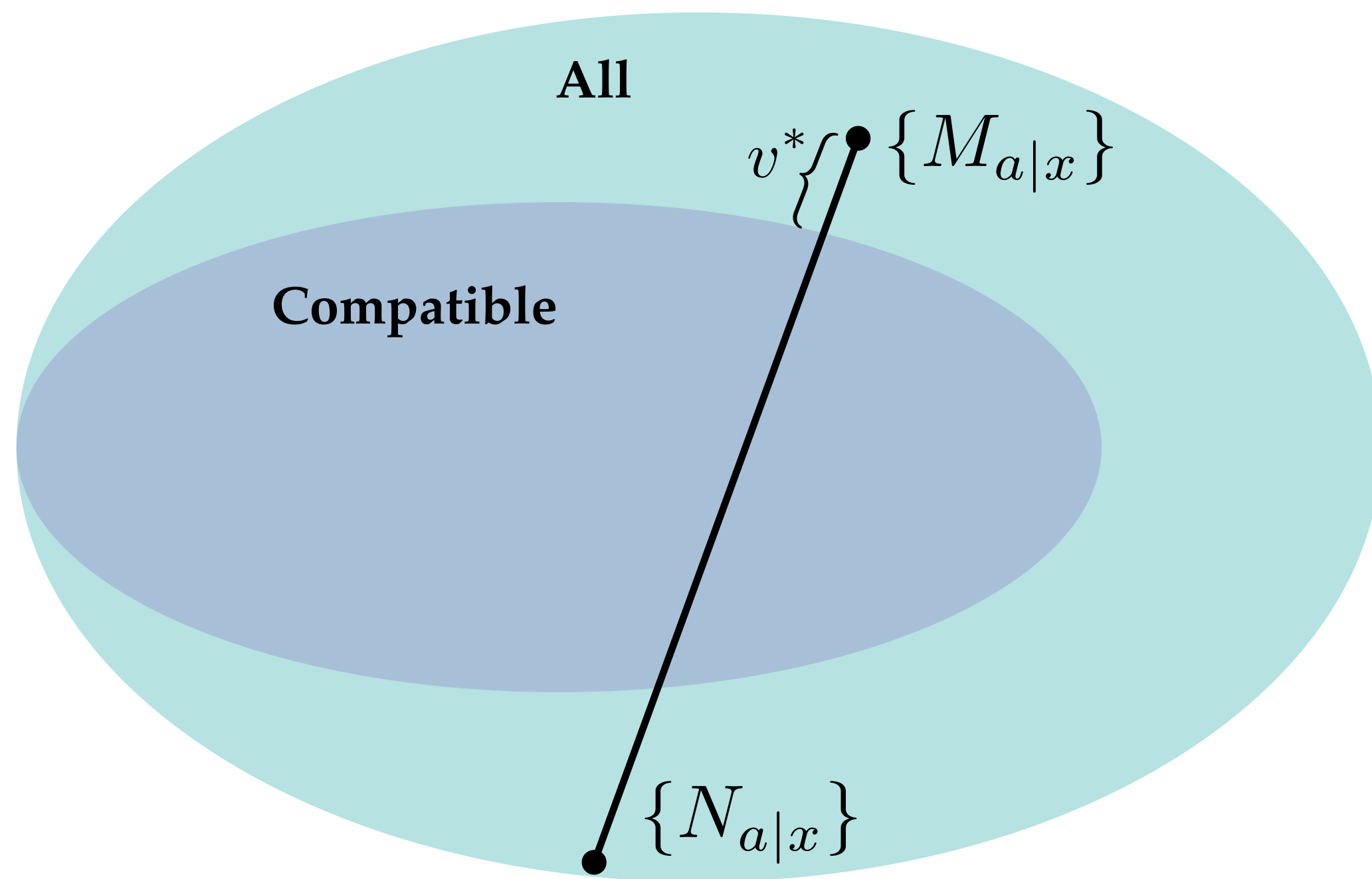
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Question. How much “worst-case” noise can I mix into a set of measurements before it becomes compatible?

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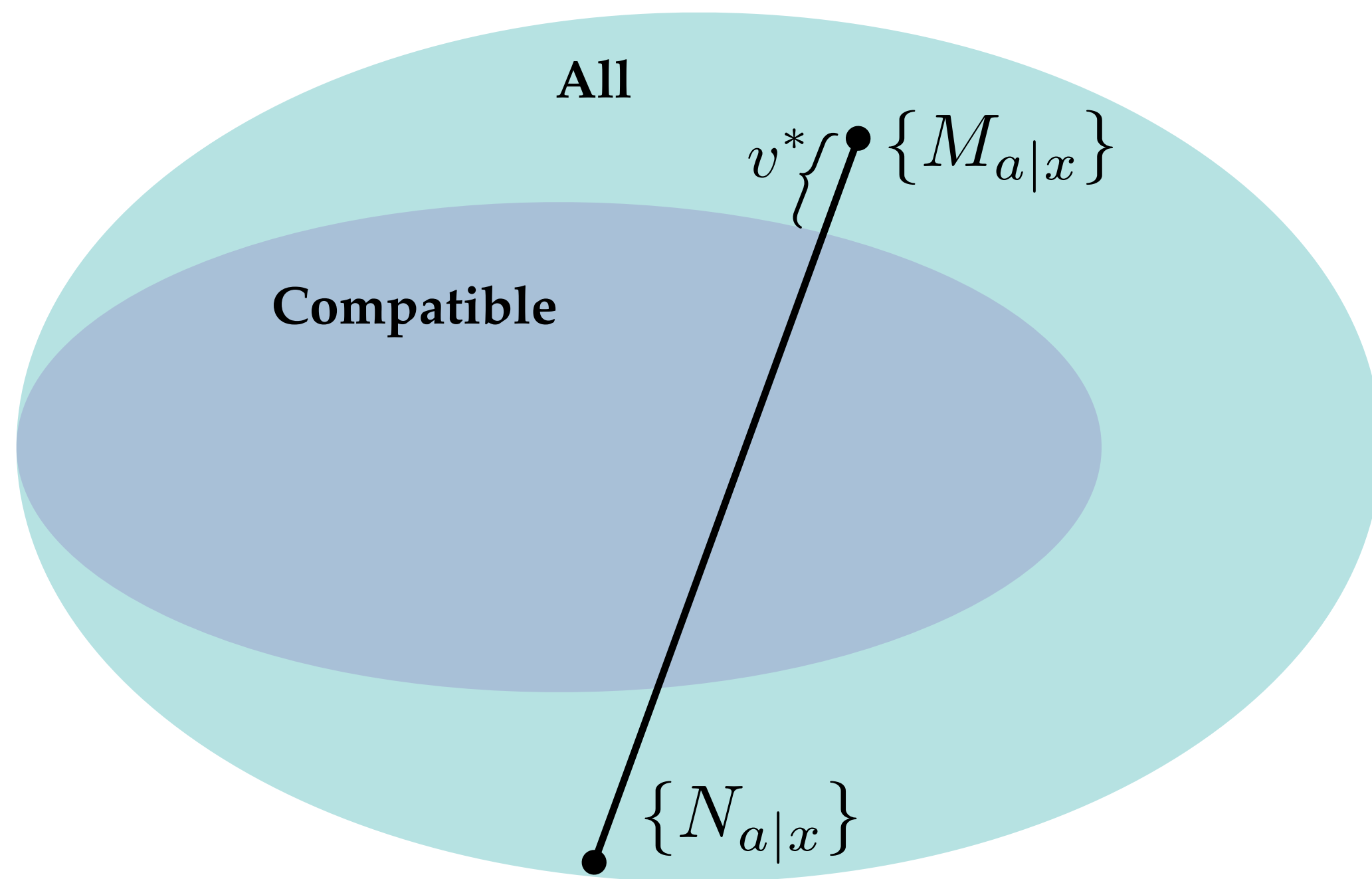
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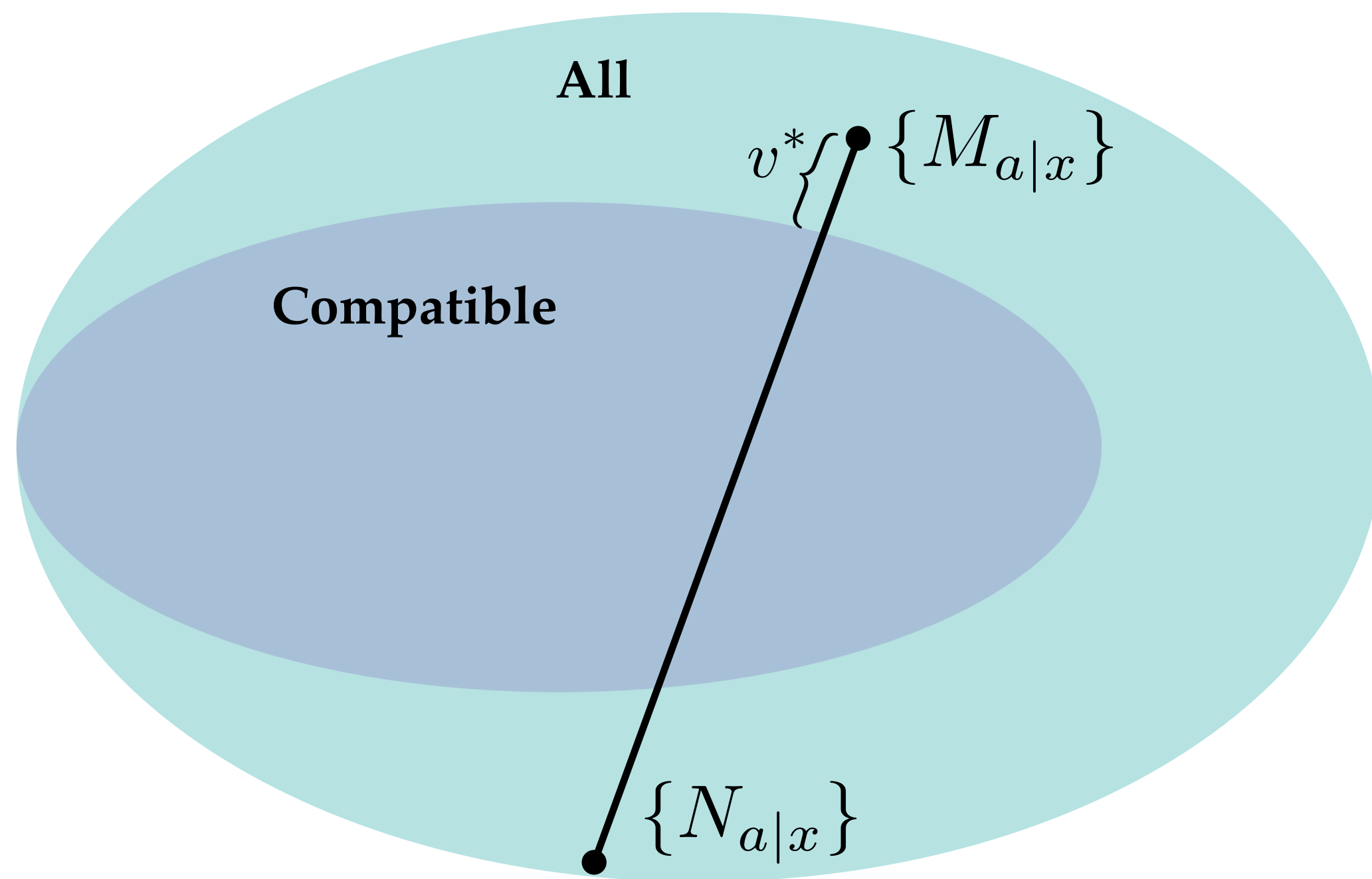
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Change of variable

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$$J_{\lambda} \mapsto G_{\mu} = \sum_{\lambda} \pi(\mu|\lambda) J_{\lambda}$$

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I. Rewriting your problem as an SDP

Generalized noise robustness of measurement incompatibility

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$$v \mapsto \frac{1}{t+1}$$

$$0 \leq v \leq 1 \iff t \geq 0$$

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$$1) \quad N_{a|x} \geq 0 \iff (t+1) \sum_{\mu} D(a|x, \mu) G_{\mu} - M_{a|x} \geq 0$$

$$2) \quad \sum_a N_{a|x} = \mathbb{1} \iff \sum_a \frac{(t+1) \sum_{\mu} D(a|x, \mu) G_{\mu} - M_{a|x}}{t} = \frac{(t+1) \mathbb{1} - \mathbb{1}}{t} = \mathbb{1} \quad \checkmark$$

I. Rewriting your problem as an SDP

Generalized noise robustness of measurement incompatibility

given $\{M_{a|x}\}$

$$t^* = \min t$$

$$\text{s.t. } (t + 1) \sum_{\mu} D(a|x, \mu) G_{\mu} - M_{a|x} \geq 0$$

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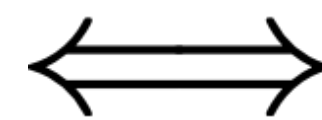
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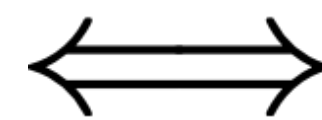
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SDP!

Rewritten

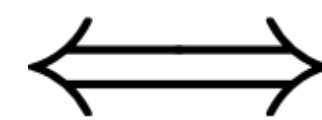
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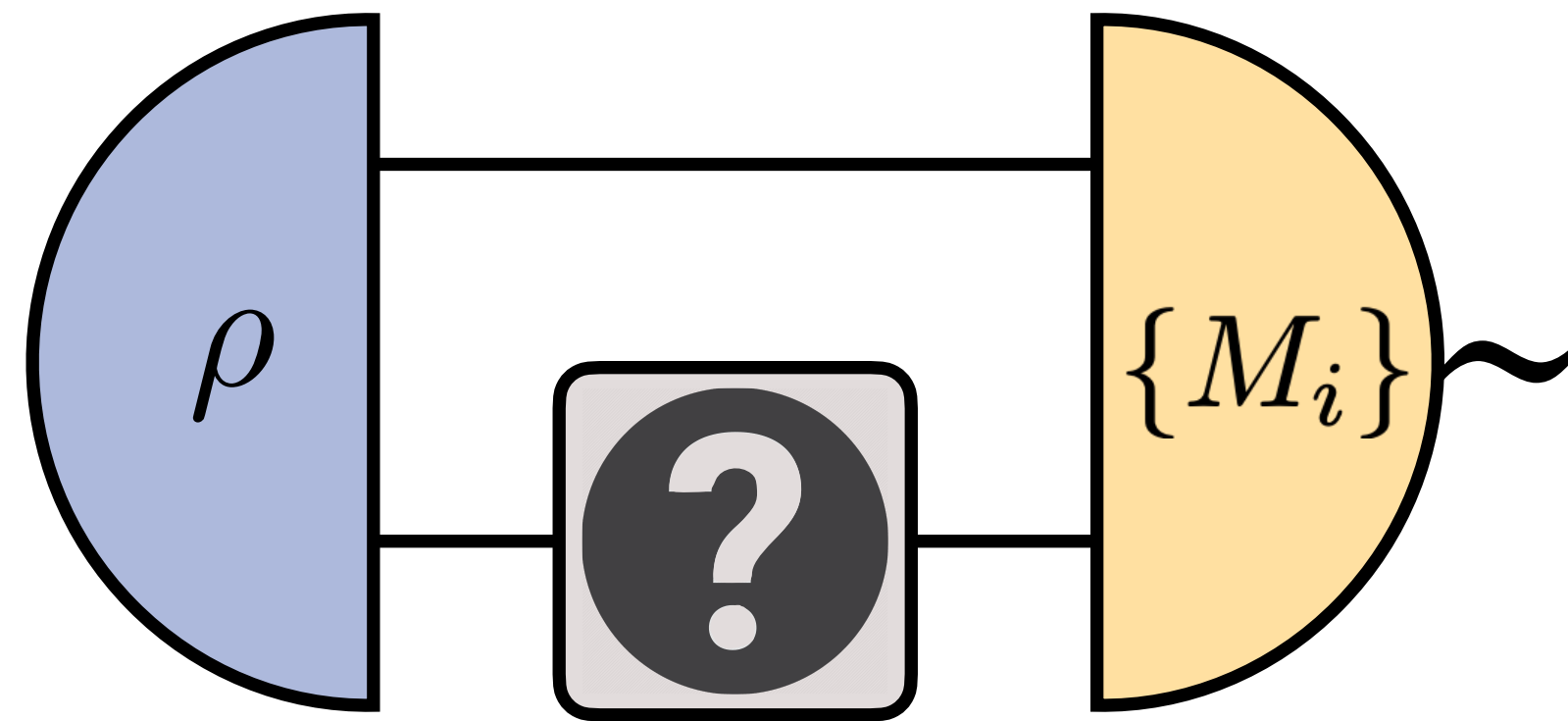
I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance

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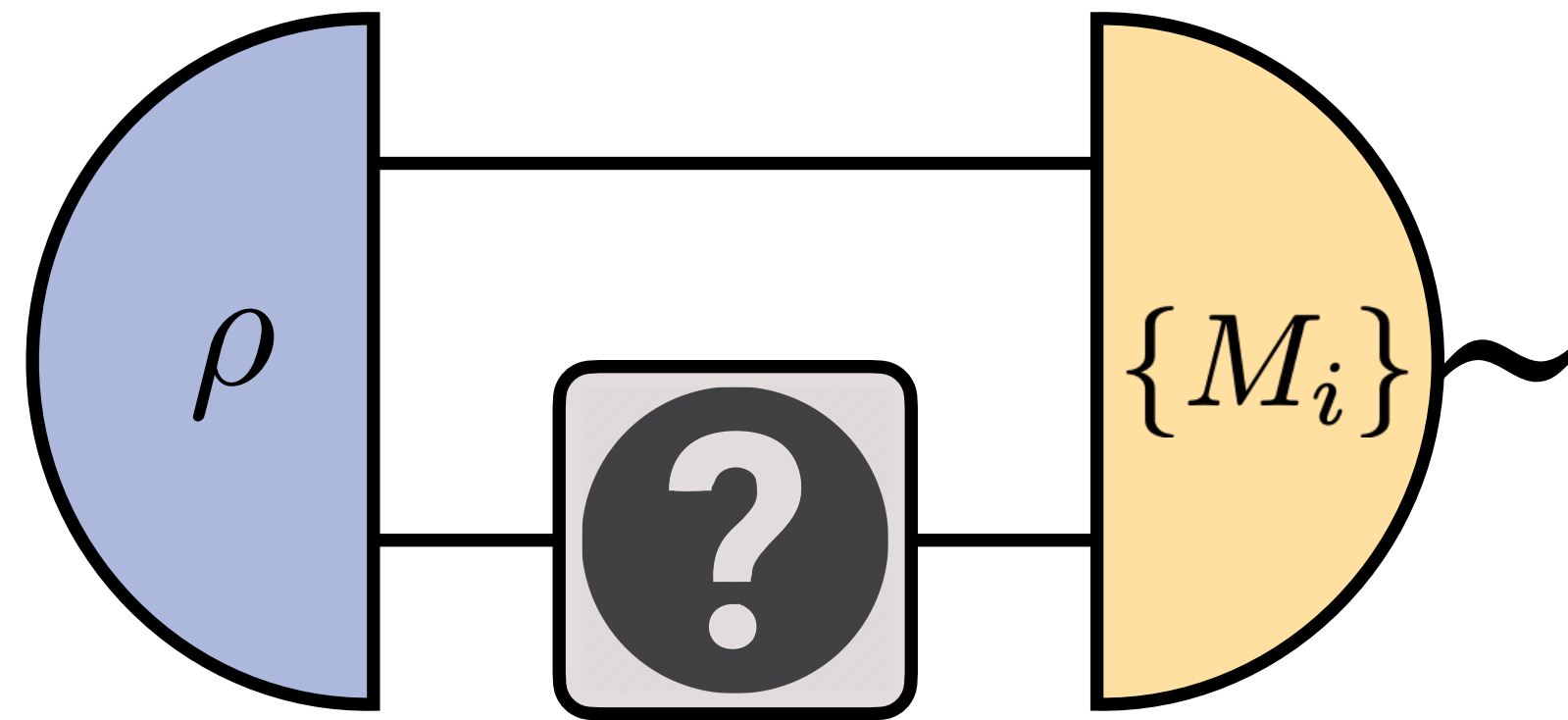
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Channel discrimination and the diamond distance

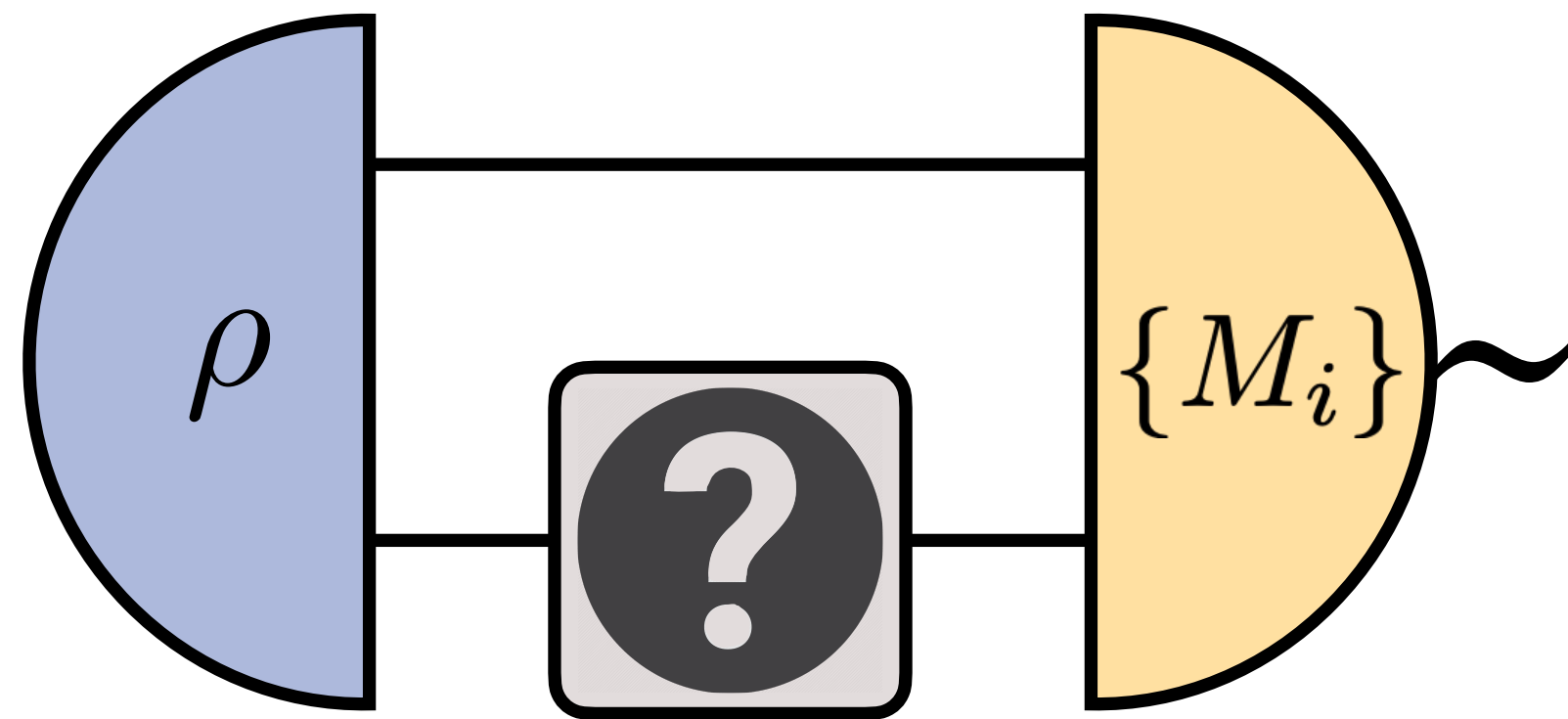
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$$P = \max_{\rho, \{M_i\}} \sum_i p_i \operatorname{tr} \left[\tilde{C}_i \otimes \tilde{\mathbb{1}}(\rho) M_i \right]$$

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Channel discrimination and the diamond distance



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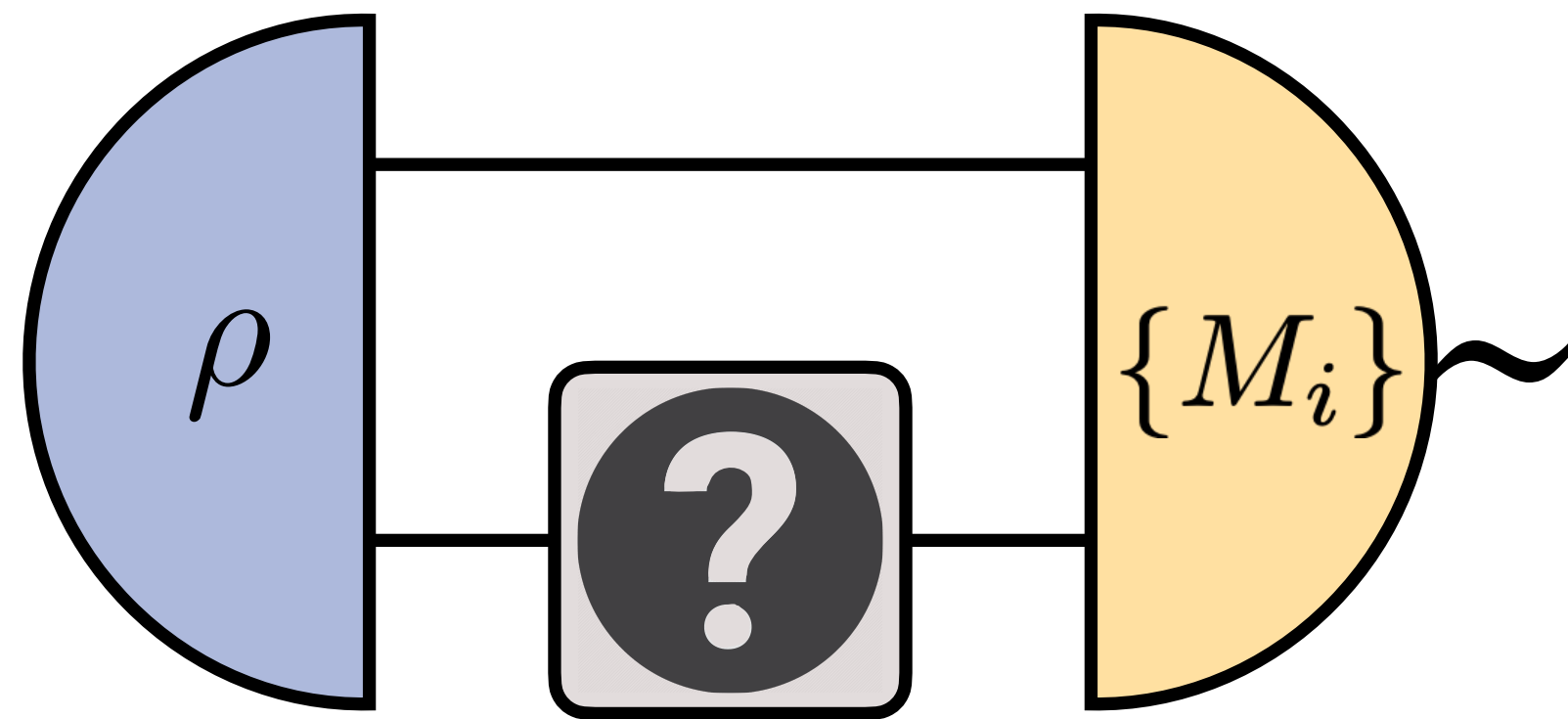
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Channel discrimination and the diamond distance

Choi operator: Every linear map can be represented by a linear operator, acting on the joint input and output space of the map, defined as

$$\tilde{C} : \mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O) \longrightarrow \text{linear map}$$

$$C := \sum_{ij} |i\rangle\langle j| \otimes \tilde{C}(|i\rangle\langle j|) \in \mathcal{L}(\mathcal{H}_I \otimes \mathcal{H}_O) \longrightarrow \text{linear operator}$$

Output operator: Consequently, the output of a linear map can be expressed as function of the Choi operator of the map and its input according to

$$\tilde{C}(\rho) = \rho' = \text{tr}_I[C (\rho^T \otimes \mathbb{1}_O)]$$

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Choi operator: Every linear map can be represented by a linear operator, acting on the joint input and output space of the map, defined as

$$\tilde{C} : \mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O) \longrightarrow \text{linear map}$$

$$C := \sum_{ij} |i\rangle\langle j| \otimes \tilde{C}(|i\rangle\langle j|) \in \mathcal{L}(\mathcal{H}_I \otimes \mathcal{H}_O) \longrightarrow \text{linear operator}$$

Output operator: Consequently, the output of a linear map can be expressed as function of the Choi operator of the map and its input according to

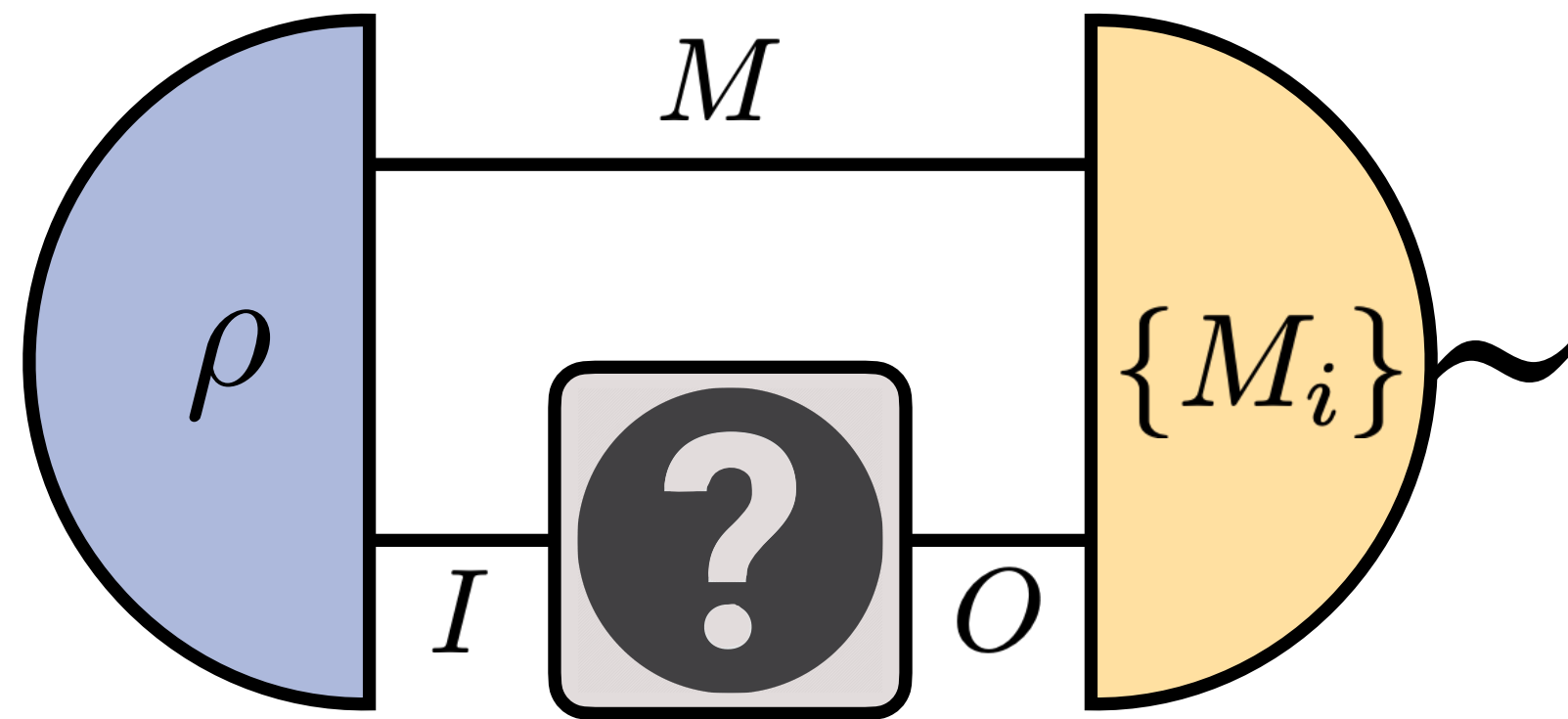
$$\tilde{C}(\rho) = \rho' = \text{tr}_I[C(\rho^T \otimes \mathbb{1}_O)]$$

Quantum channels:

$$\tilde{C} \in \text{CPTP} \iff C \geq 0, \quad \text{tr}_O C = \mathbb{1}_I$$

I. Rewriting your problem as an SDP

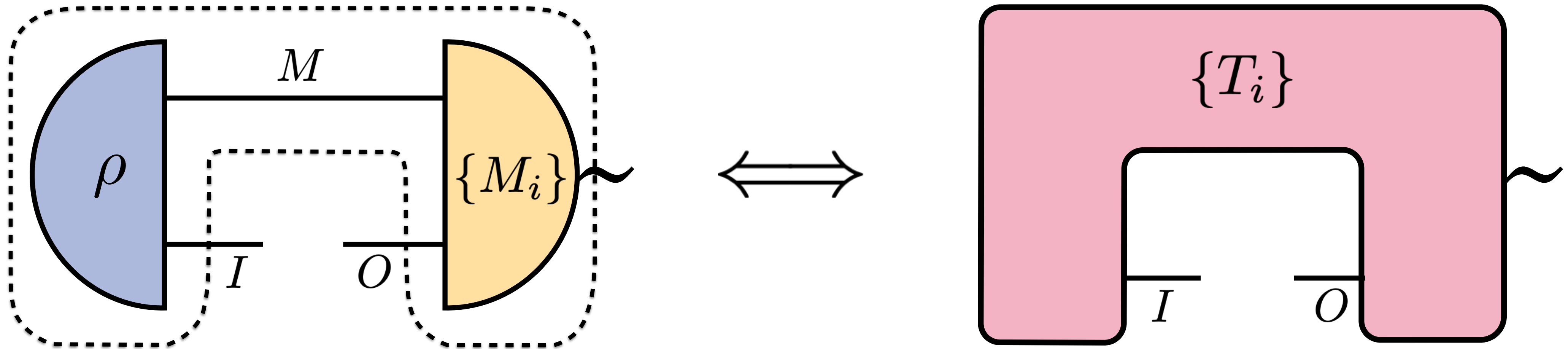
Channel discrimination and the diamond distance



$$\begin{aligned} & \text{given } \{p_i, C_i\} \\ P = & \max_{\rho, \{M_i\}} \sum_i p_i \operatorname{tr}[(C_i \otimes \mathbb{1}_M)(\rho^{T_I} \otimes \mathbb{1}_O)(\mathbb{1}_I \otimes M_i)] \\ \text{s.t. } & \rho \geq 0, \quad \operatorname{tr}(\rho) = 1 \\ & M_i \geq 0, \quad \sum_i M_i = \mathbb{1} \end{aligned}$$

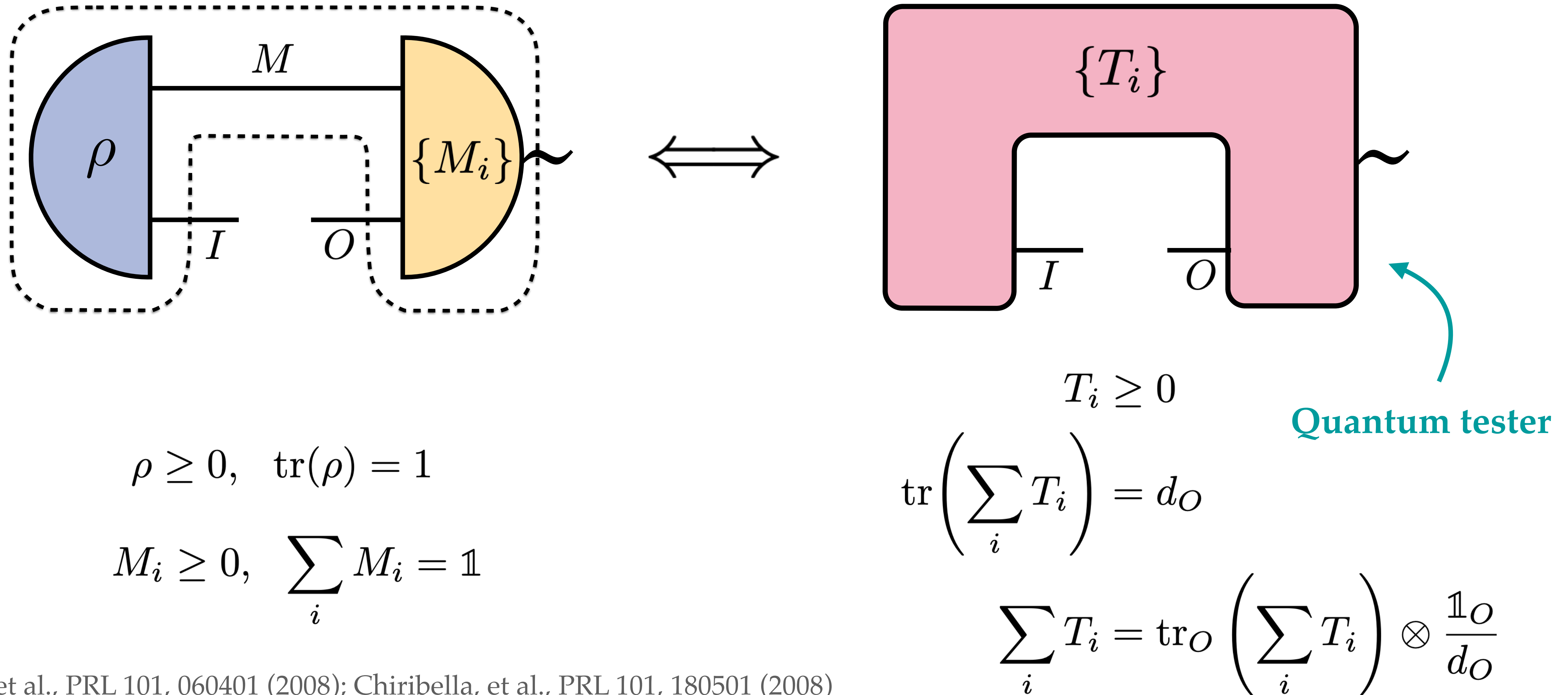
I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance



I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance

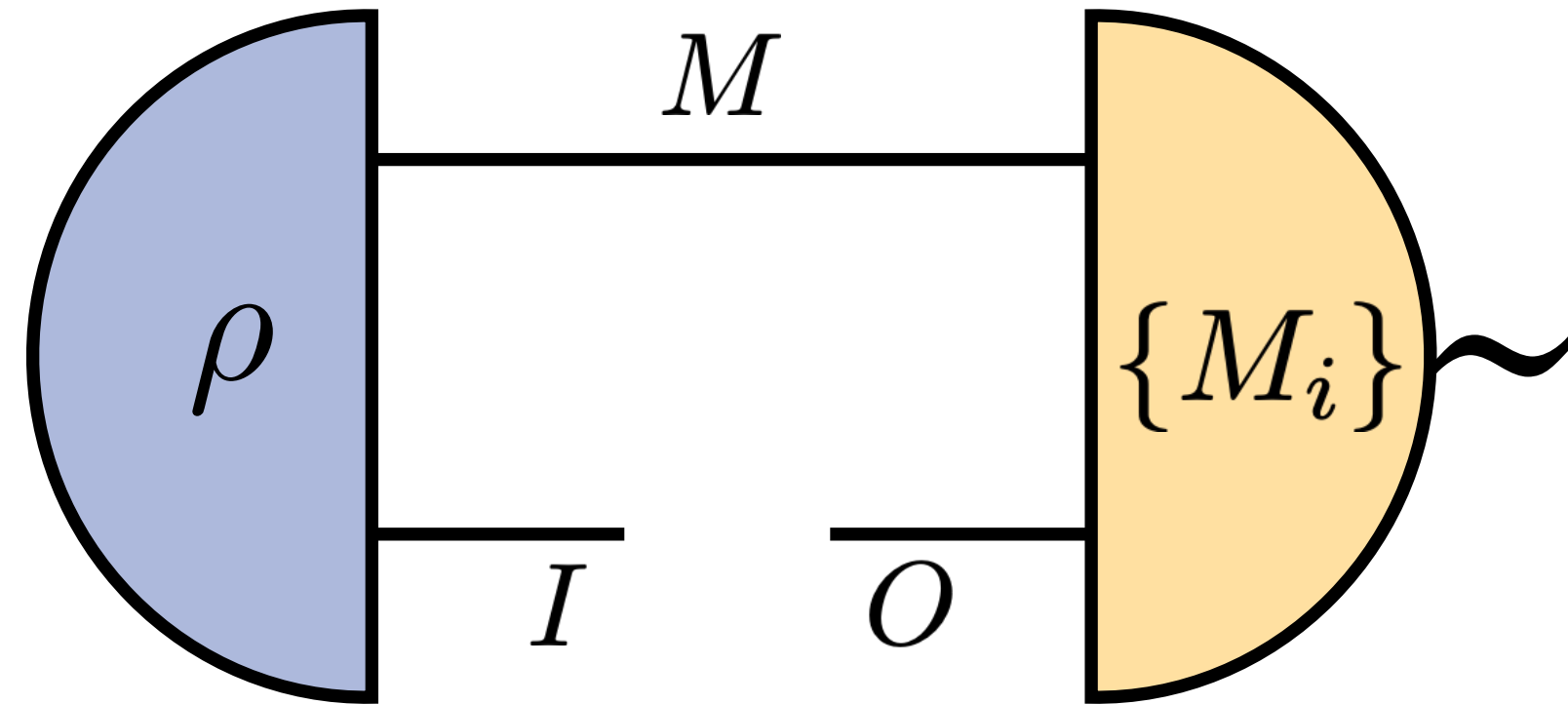


Chiribella et al., PRL 101, 060401 (2008); Chiribella, et al., PRL 101, 180501 (2008)

Ziman, PRA 77, 062112 (2008)

I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance

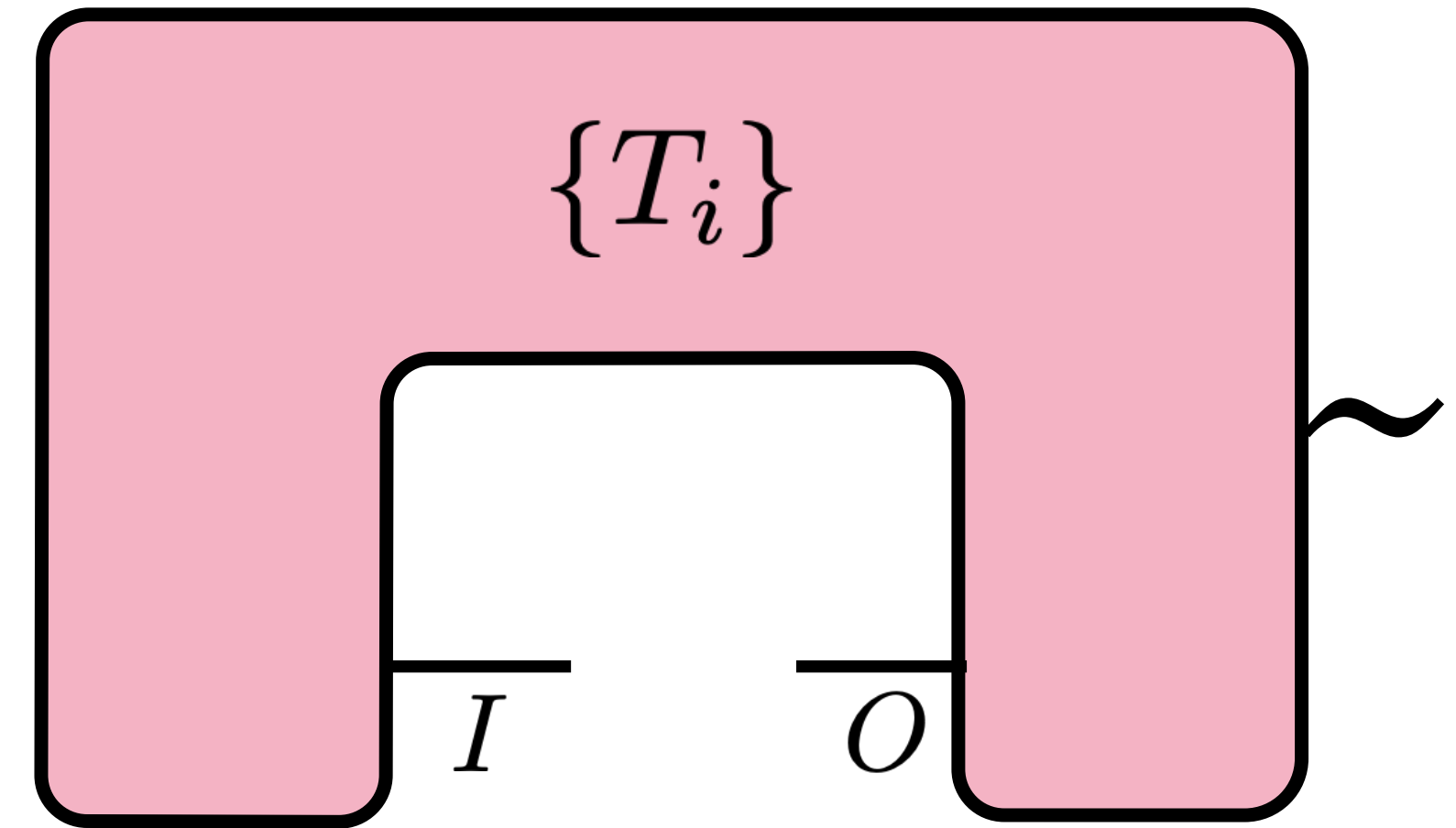


$\rho, \{M_i\}$ as a function of $\{T_i\}$

$$\rho = \left(\mathbb{1}_I \otimes \sqrt{\sigma}^T \right) \sum_{ij} |ii\rangle\langle jj| \left(\mathbb{1}_I \otimes \sqrt{\sigma}^T \right)$$

$$M_i^T = (\sqrt{\sigma}^{-1} \otimes \mathbb{1}_O) T_i (\sqrt{\sigma}^{-1} \otimes \mathbb{1}_O)$$

where $\sigma := \frac{\text{tr}_O \sum_i T_i}{d_O}$



$\{T_i\}$ as a function of $\rho, \{M_i\}$

$$T_i = \text{tr}_M [(\rho^{T_M} \otimes \mathbb{1}_O)(\mathbb{1}_I \otimes M_i^T)]$$

Chiribella et al., PRL 101, 060401 (2008); Chiribella, et al., PRL 101, 180501 (2008)

Ziman, PRA 77, 062112 (2008)

I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance

Original

$$\begin{aligned} &\text{given } \{p_i, \tilde{C}_i\} \\ P = &\max_{\rho, \{M_i\}} \sum_i p_i \operatorname{tr} \left[\tilde{C}_i \otimes \tilde{\mathbb{1}}(\rho) M_i \right] \\ \text{s.t. } &\rho \geq 0, \quad \operatorname{tr}(\rho) = 1 \\ &M_i \geq 0, \quad \sum_i M_i = \mathbb{1} \end{aligned}$$

$\Longleftrightarrow$

Rewritten

$$\begin{aligned} &\text{given } \{p_i, C_i\} \\ P = &\max_{\{T_i\}} \sum_i p_i \operatorname{tr} (C_i^T T_i) \\ \text{s.t. } &T_i \geq 0 \\ &\operatorname{tr} \left(\sum_i T_i \right) = d_O \\ &\sum_i T_i = \operatorname{tr}_O \left(\sum_i T_i \right) \otimes \frac{\mathbb{1}_O}{d_O} \end{aligned}$$

SDP!

I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance

Diamond distance: The diamond distance between two quantum channels is related to the probability of successfully discriminating this pair of channels with a uniform prior.

$$\left\| \tilde{C}_1 - \tilde{C}_2 \right\|_{\diamond} := \max_{\rho} \left\| \tilde{C}_1 \otimes \tilde{\mathbb{1}}(\rho) - \tilde{C}_2 \otimes \tilde{\mathbb{1}}(\rho) \right\|_1$$

I. Rewriting your problem as an SDP

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$$P = \frac{1}{2} + \frac{1}{4} \left\| \tilde{C}_1 - \tilde{C}_2 \right\|_{\diamond}$$

I. Rewriting your problem as an SDP

Channel discrimination and the diamond distance

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I. Rewriting

MAIN MESSAGE:

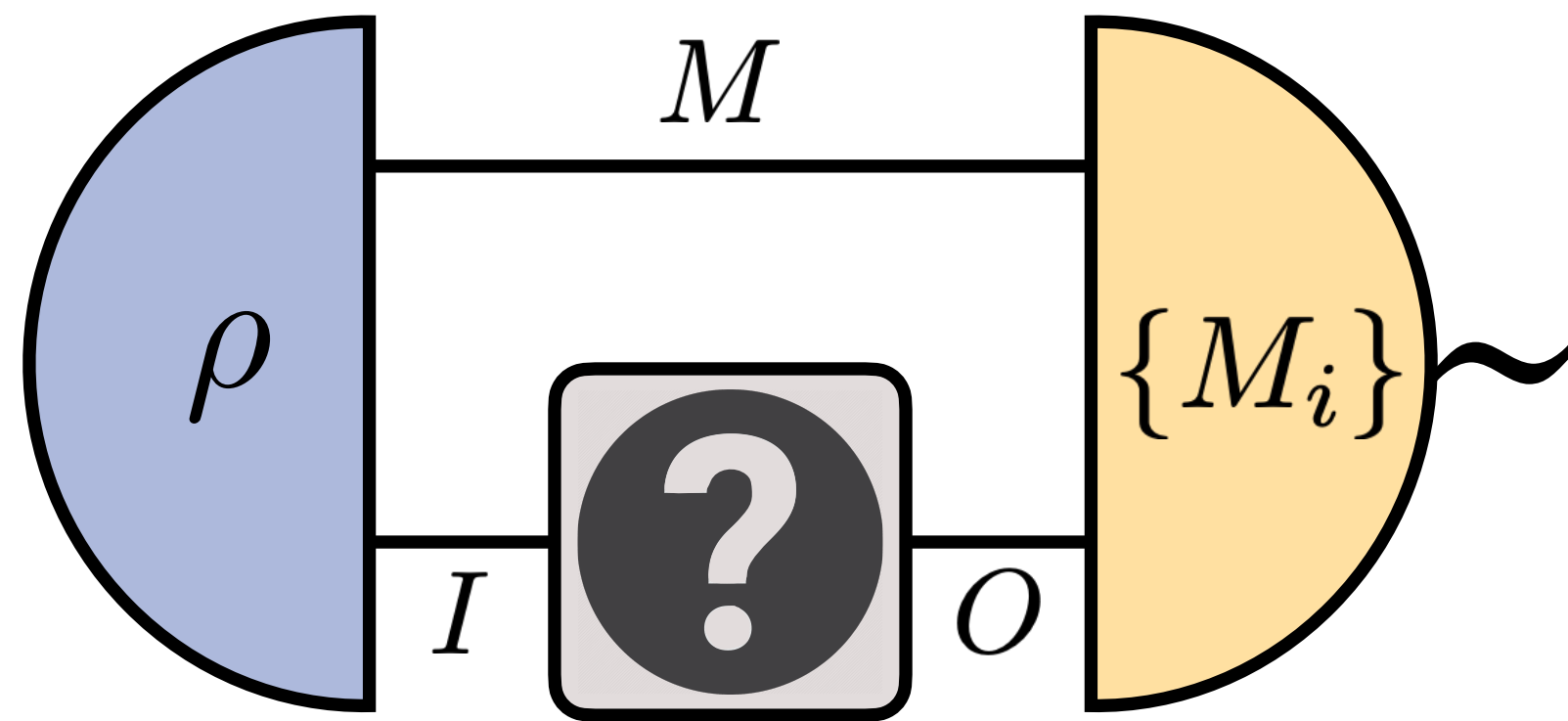
Not every SDP is created in an SDP form.

Properties of the specific problem and changes of variables
are useful tools to recognize that your problem is an SDP.

II. Approximating: **Attainability**

II. Approximation: Attainability

Channel discrimination without memory



II. Approximation: Attainability

Channel discrimination without memory



II. Approximation: Attainability

Channel discrimination without memory



memoryless

given $\{p_i, \tilde{C}_i\}$

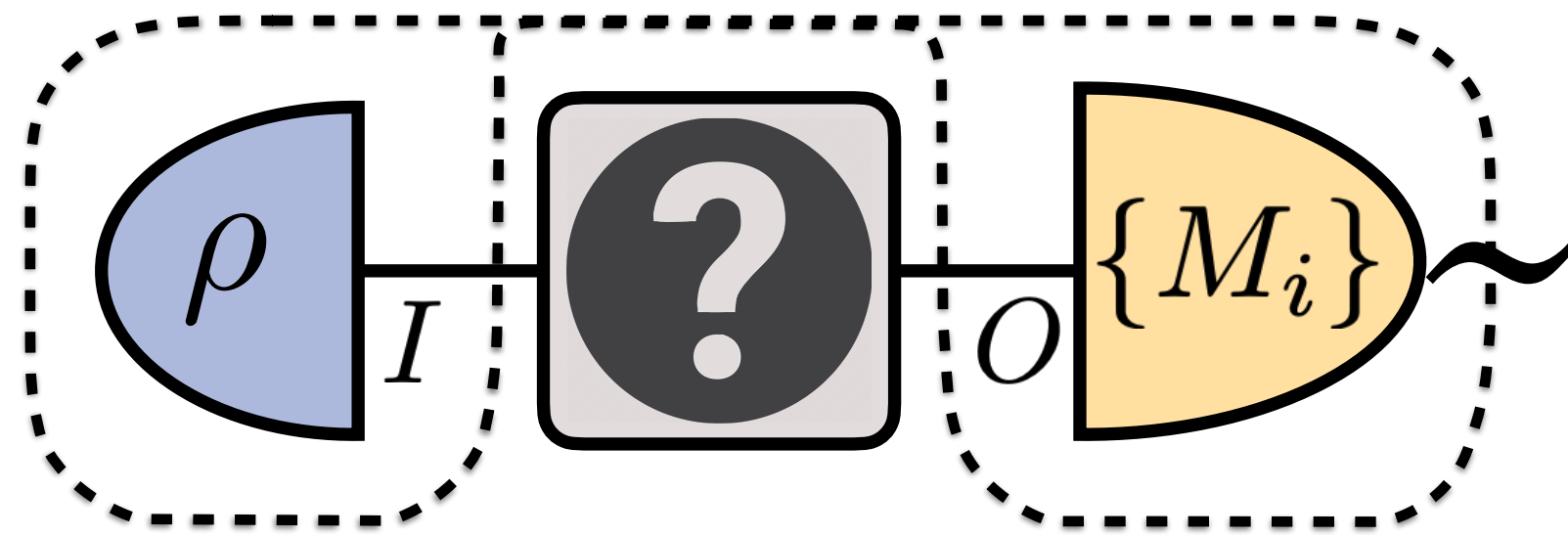
$$P_{\text{ML}} = \max_{\rho, \{M_i\}} \sum_i p_i \operatorname{tr} [\tilde{C}_i(\rho) M_i]$$

$$\text{s.t. } \rho \geq 0, \quad \operatorname{tr}(\rho) = 1$$

$$M_i \geq 0, \quad \sum_i M_i = \mathbb{1}$$

II. Approximation: Attainability

Channel discrimination without memory



$$T_i = \rho \otimes M_i^T$$

given $\{p_i, C_i\}$

$$P_{\text{ML}} = \max_{\{T_i\}} \sum_i p_i \text{tr}(C_i^T T_i)$$

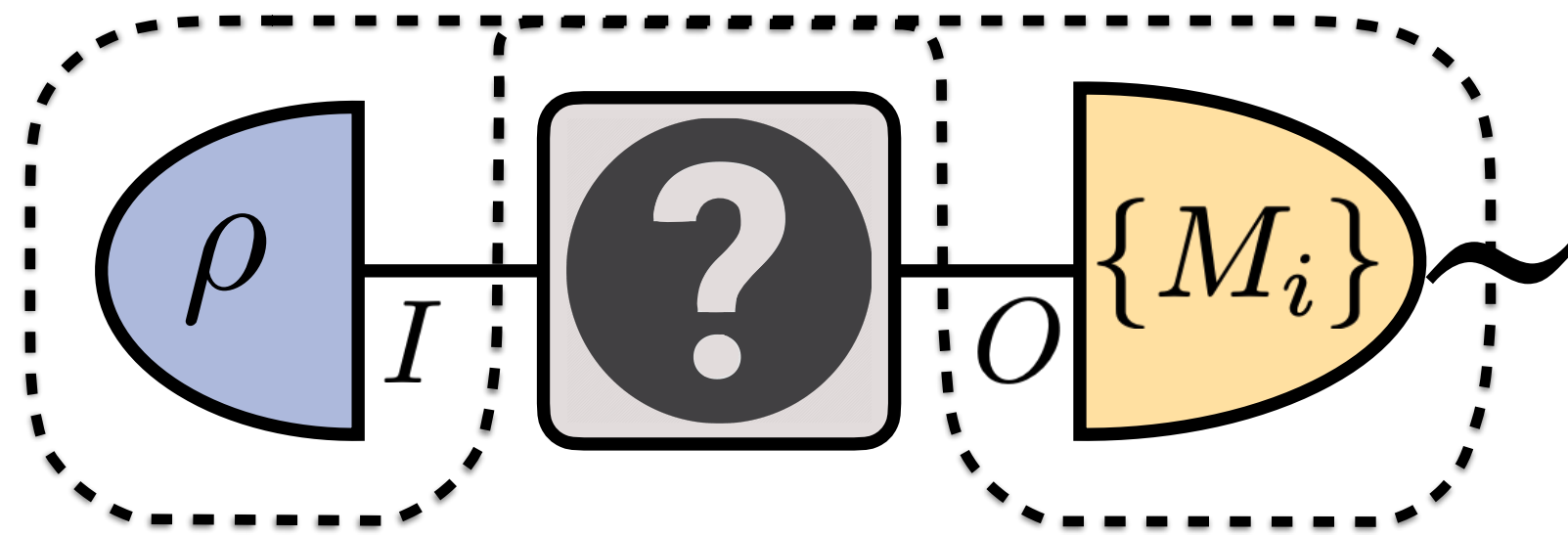
$$\text{s.t. } T_i = \rho \otimes M_i^T$$

$$\rho \geq 0, \quad \text{tr}(\rho) = 1$$

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II. Approximation: Attainability

Channel discrimination without memory



$$T_i = \rho \otimes M_i^T$$

**no known
SDP characterization**

given $\{p_i, C_i\}$

$$P_{\text{ML}} = \max_{\rho} \max_{\{T_i\}} \sum_i p_i \text{tr}[C_i^T (\rho \otimes M_i^T)]$$

$$\text{s.t. } \rho \geq 0, \quad \text{tr}(\rho) = 1$$

$$M_i \geq 0, \quad \sum_i M_i = \mathbb{1}$$

II. Approximation: Attainability

Channel discrimination without memory

Seesaw algorithm

II. Approximation: Attainability

Channel discrimination without memory

Seesaw algorithm

given $\{p_i, C_i\}, \rho$

$$\begin{aligned} P_1 = \max_{\{M_i\}} \quad & \sum_i p_i \operatorname{tr}(C_i^T T_i) \\ \text{s.t.} \quad & T_i = \rho \otimes M_i^T \\ & M_i \geq 0, \quad \sum_i M_i = \mathbb{1} \end{aligned}$$

SDP 1

II. Approximation: Attainability

Channel discrimination without memory

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Channel discrimination without memory

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SDP 1



given $\{p_i, C_i\}, \{M_i\}$

$$\begin{aligned} P_2 = \max_{\rho} \quad & \sum_i p_i \operatorname{tr}(C_i^T T_i) \\ \text{s.t.} \quad & T_i = \rho \otimes M_i^T \\ & \rho \geq 0, \quad \operatorname{tr}(\rho) = 1 \end{aligned}$$

SDP 2

II. Approximation: Attainability

Channel discrimination without memory

Seesaw algorithm

given $\{p_i, C_i\}, \rho$

$$P_1 = \max_{\{M_i\}} \sum_i p_i \operatorname{tr}(C_i^T T_i)$$

s.t. $T_i = \rho \otimes M_i^T$

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SDP 1



given $\{p_i, C_i\}, \{M_i\}$

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SDP 2

II. Approximation: Attainability

Channel discrimination without memory

Seesaw algorithm

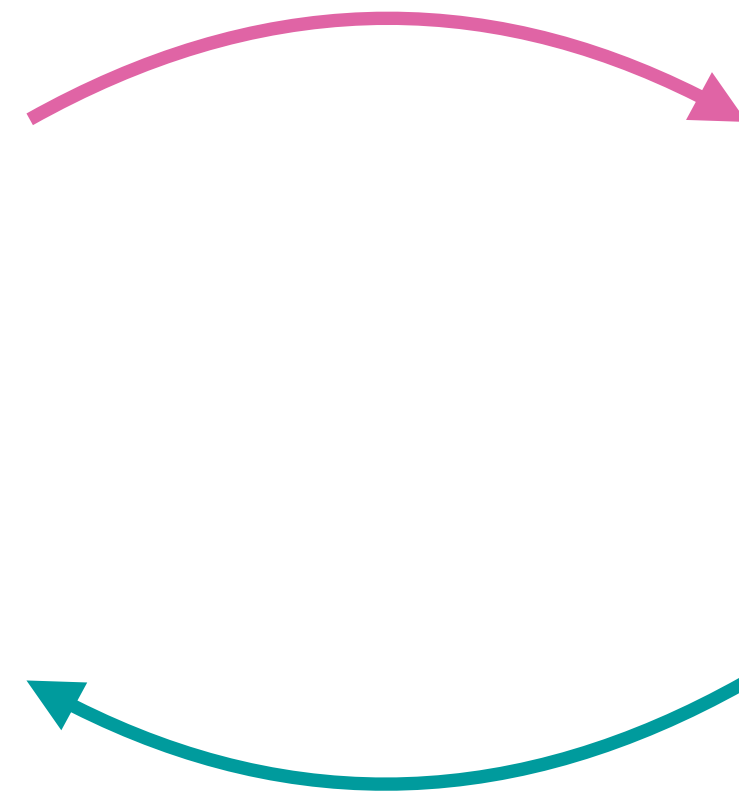
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SDP 1



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SDP 2

II. Approximation: Attainability

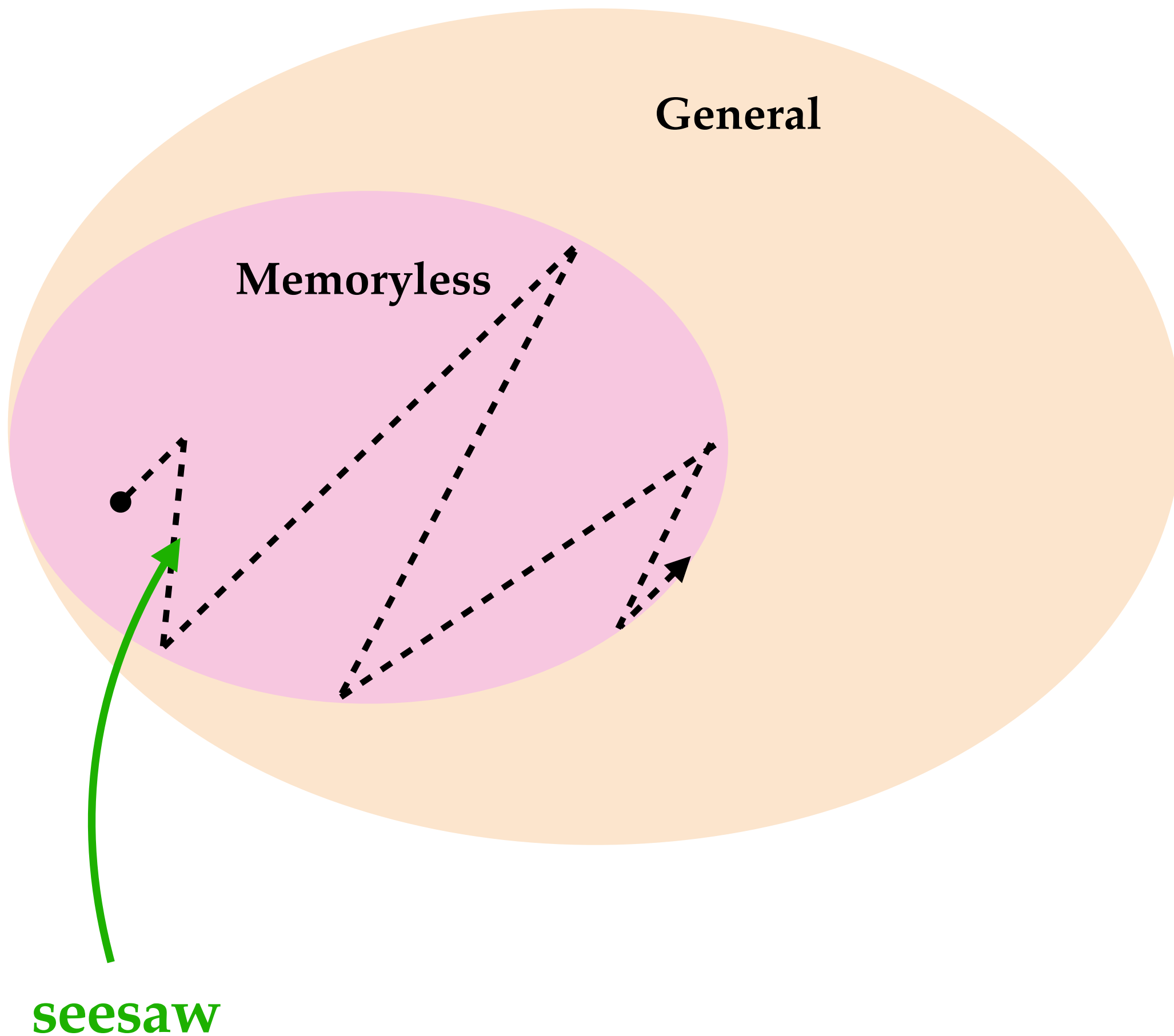
Channel discrimination without memory

Algorithm 1 Seesaw SDP algorithm

- 1: Initialize ρ
 - 2: $\epsilon \leftarrow 10^{-5}$
 - 3: $\Delta \leftarrow \infty$
 - 4: **while** $\Delta > \epsilon$ **do**
 - 5: $(P_1, \{M_i\}) \leftarrow \text{SDP 1}(\{p_i, C_i\}, \rho)$
 - 6: $(P_2, \rho) \leftarrow \text{SDP 2}(\{p_i, C_i\}, \{M_i\})$
 - 7: $\Delta \leftarrow |P_1 - P_2|$
 - 8: **end while**
 - 9: $P_{\text{seesaw}} \leftarrow P_2$
 - 10: **return** $P, \rho, \{M_i\}$
-

II. Approximation: Attainability

Channel discrimination without memory

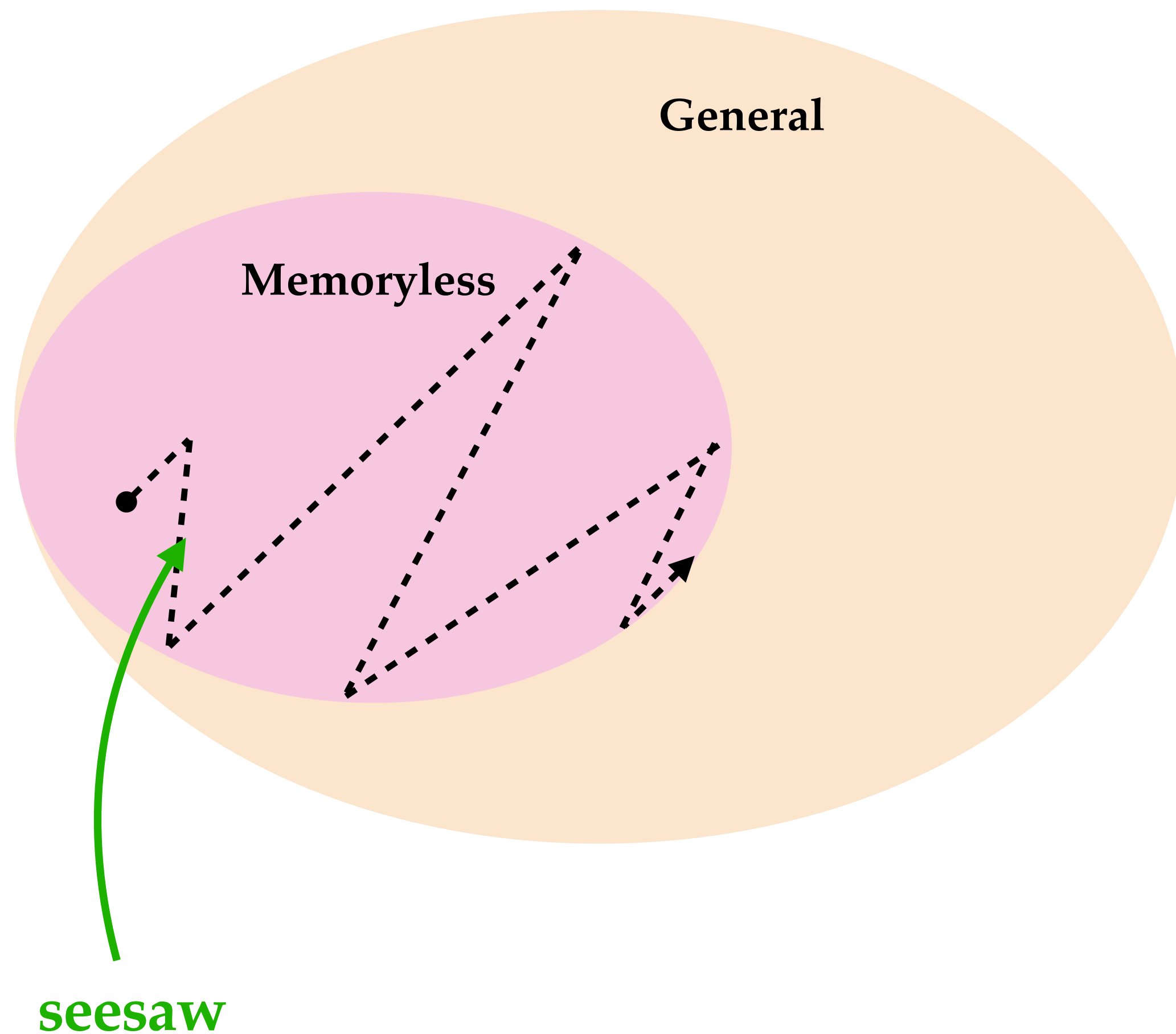


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II. Approximation: Attainability

Channel discrimination without memory



Algorithm 1 Seesaw SDP algorithm

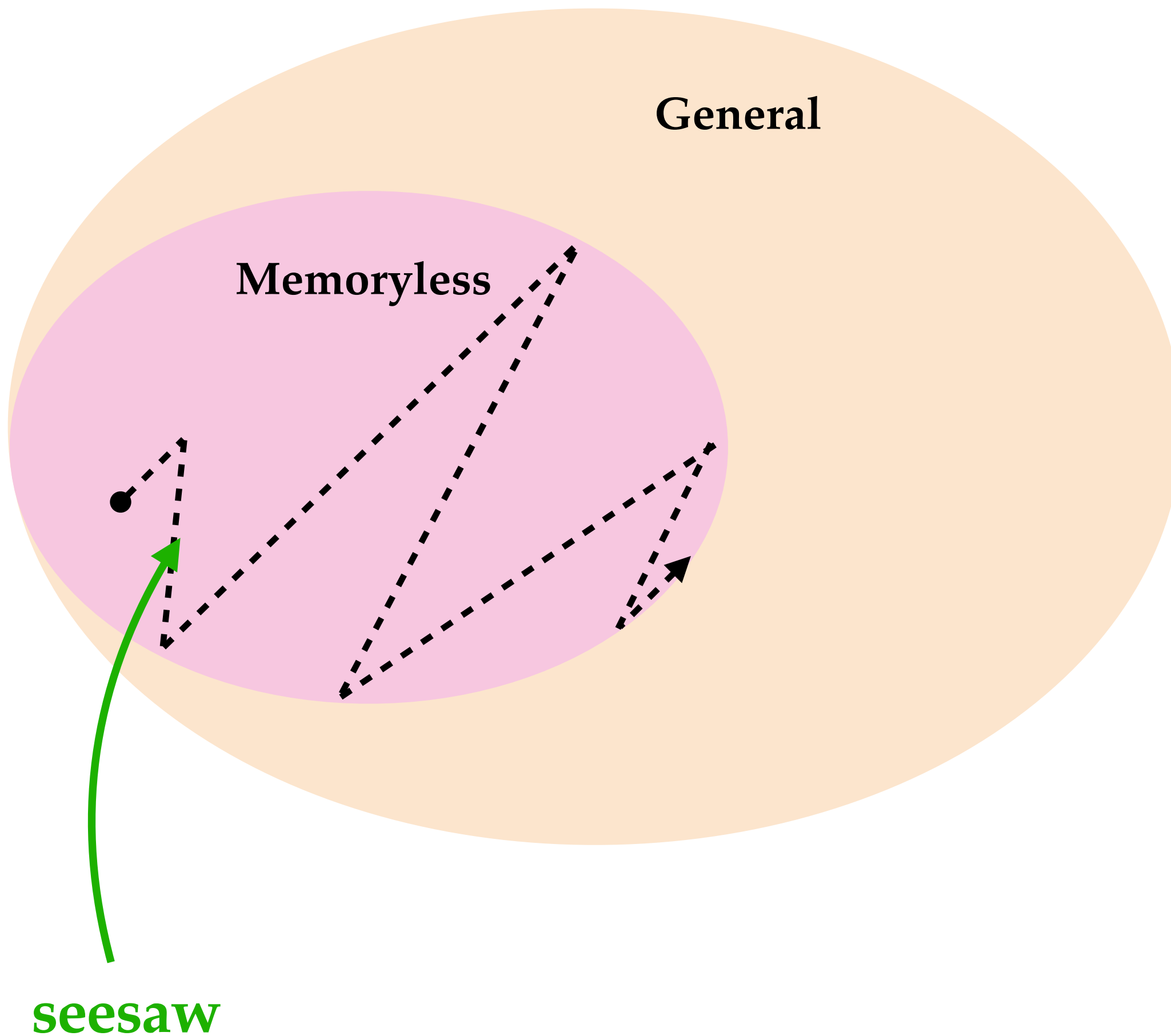
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$$P_{\text{seesaw}} \leq P_{\text{ML}}$$

lower
bound

II. Approximation: Attainability

Channel discrimination without memory



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lower
bound

$$P_{\text{seesaw}} \leq P_{\text{ML}}$$

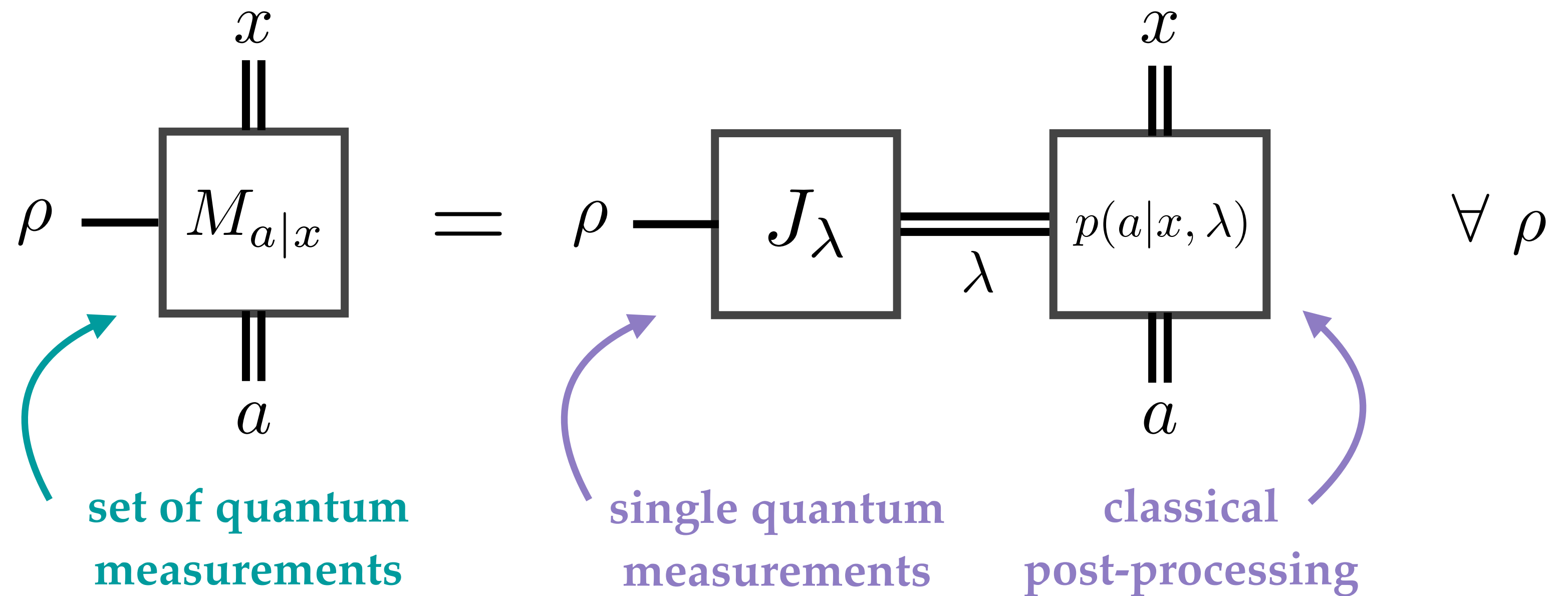
II. Approximation: Attainability

Most incompatible set of quantum measurements

II. Approximation: Attainability

Most incompatible set of quantum measurements

compatible set of
quantum measurements:



II. Approximation: Attainability

Most incompatible set of quantum measurements

Question. What is the most incompatible set of N d -dimensional measurements with k outcomes?

II. Approximation: Attainability

Most incompatible set of quantum measurements

Question. What is the most incompatible set of N d -dimensional measurements with k outcomes?

given $\{M_{a|x}\}$

$$\tau(N, k, d) = \max_{\{M_{a|x}\}} t^*(\{M_{a|x}\})$$

$$t^*(\{M_{a|x}\}) = \min t$$

$$\text{s.t. } \sum_{\mu} D(a|x, \mu) \overline{G}_{\mu} - M_{a|x} \geq 0$$

$$t \geq 0$$

$$\overline{G}_{\mu} \geq 0, \quad \sum_{\mu} \overline{G}_{\mu} = (t + 1)\mathbb{1}$$

N = number of measurements

k = number of outcomes

d = dimension

II. Approximation: Attainability

Most incompatible set of quantum measurements

given N, k, d

N = number of measurements

k = number of outcomes

d = dimension

$$\tau(N, k, d) = \max_{\{M_{a|x}\}} \min_{t, \{\bar{G}_\mu\}} t$$

$$\text{s.t.} \quad \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0$$

$$t \geq 0$$

$$\bar{G}_\mu \geq 0, \quad \sum_{\mu} \bar{G}_\mu = (t + 1) \mathbb{1}$$

$$M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

can I tackle this with a seesaw algorithm?

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

given $\{M_{a|x}\}$

$$t_1^* = \min_{t, \{\bar{G}_\mu\}} t$$

$$\text{s.t.} \quad \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0$$

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given $t, \{\bar{G}_\mu\}$

$$t_2^* = \max_{\{M_{a|x}\}} t$$

$$\text{s.t.} \quad \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0$$

$$M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

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$$t_1^* = \min_{t, \{\bar{G}_\mu\}} t$$

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$$t \geq 0$$

$$\bar{G}_\mu \geq 0, \quad \sum_{\mu} \bar{G}_\mu = (t + 1)\mathbb{1}$$

given $t, \{\bar{G}_\mu\}$

$$t_2^* = \max_{\{M_{a|x}\}} t \xrightarrow{\text{constant}}$$

$$\text{s.t.} \quad \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0$$

$$M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

seesaw doesn't go anywhere

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

N = number of measurements

k = number of outcomes

d = dimension

given N, k, d

$$\tau(N, k, d) = \max_{\{M_{a|x}\}} \min_{t, \{\bar{G}_\mu\}} t$$

$$\text{s.t.} \quad \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0$$

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$$\bar{G}_\mu \geq 0, \quad \sum_{\mu} \bar{G}_\mu = (t + 1) \mathbb{1}$$

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II. Approximation: Attainability

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duality theory

II. Approximation: Attainability

Most incompatible set of quantum measurements

given $\{M_{a|x}\}$

$$t^* = \min_{t, \{\bar{G}_\mu\}} t$$

$$\text{s.t.} \quad \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0$$

$$t \geq 0$$

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II. Approximation: Attainability

Most incompatible set of quantum measurements

$$\begin{array}{ccc} \text{given } \{M_{a|x}\} & & \text{given } \{M_{a|x}\} \\ t^* = \min_{t, \{\bar{G}_\mu\}} t & \xrightarrow{\text{Dual problem}} & t^* = \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) - \text{tr}(\Gamma) \\ \text{s.t. } \sum_{\mu} D(a|x, \mu) \bar{G}_\mu - M_{a|x} \geq 0 & & \text{s.t. } \sum_{ax} D(a|x, \mu) F_{a|x} \leq \Gamma \\ t \geq 0 & & \text{tr}(\Gamma) \leq 1 \\ \bar{G}_\mu \geq 0, \quad \sum_{\mu} \bar{G}_\mu = (t+1)\mathbb{1} & & \end{array}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Original

given N, k, d

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Rewritten

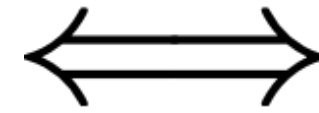
given N, k, d

$$\tau(N, k, d) = \max_{\{M_{a|x}\}} \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t.} \quad \sum_{ax} D(a|x, \mu) F_{a|x} \leq -\Gamma$$

$$\text{tr}(\Gamma) \geq -1$$

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II. Approximation: Attainability

Most incompatible set of quantum measurements

Original

given N, k, d

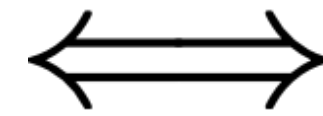
$$\tau(N, k, d) = \max_{\{M_{a|x}\}} \min_{t, \{\bar{G}_\mu\}} t$$

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Rewritten

given N, k, d

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II. Approximation: Attainability

Most incompatible set of quantum measurements

Rewritten

given N, k, d

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can I tackle this with a seesaw algorithm?

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

given $\{M_{a|x}\}$

$$\tau_1 = \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t.} \quad \sum_{ax} D(a|x, \mu) F_{a|x} \leq -\Gamma$$

$$\text{tr}(\Gamma) \geq -1$$

given $\Gamma, \{F_{a|x}\}$

$$\tau_2 = \max_{\{M_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t.} \quad M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

$$\begin{aligned} &\text{given } \{M_{a|x}\} \\ \tau_1 = &\max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma) \\ &\text{s.t. } \sum_{ax} D(a|x, \mu) F_{a|x} \leq -\Gamma \\ &\text{tr}(\Gamma) \geq -1 \end{aligned}$$

$$\begin{aligned} &\text{given } \Gamma, \{F_{a|x}\} \\ \tau_2 = &\max_{\{M_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma) \\ &\text{s.t. } M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1} \end{aligned}$$

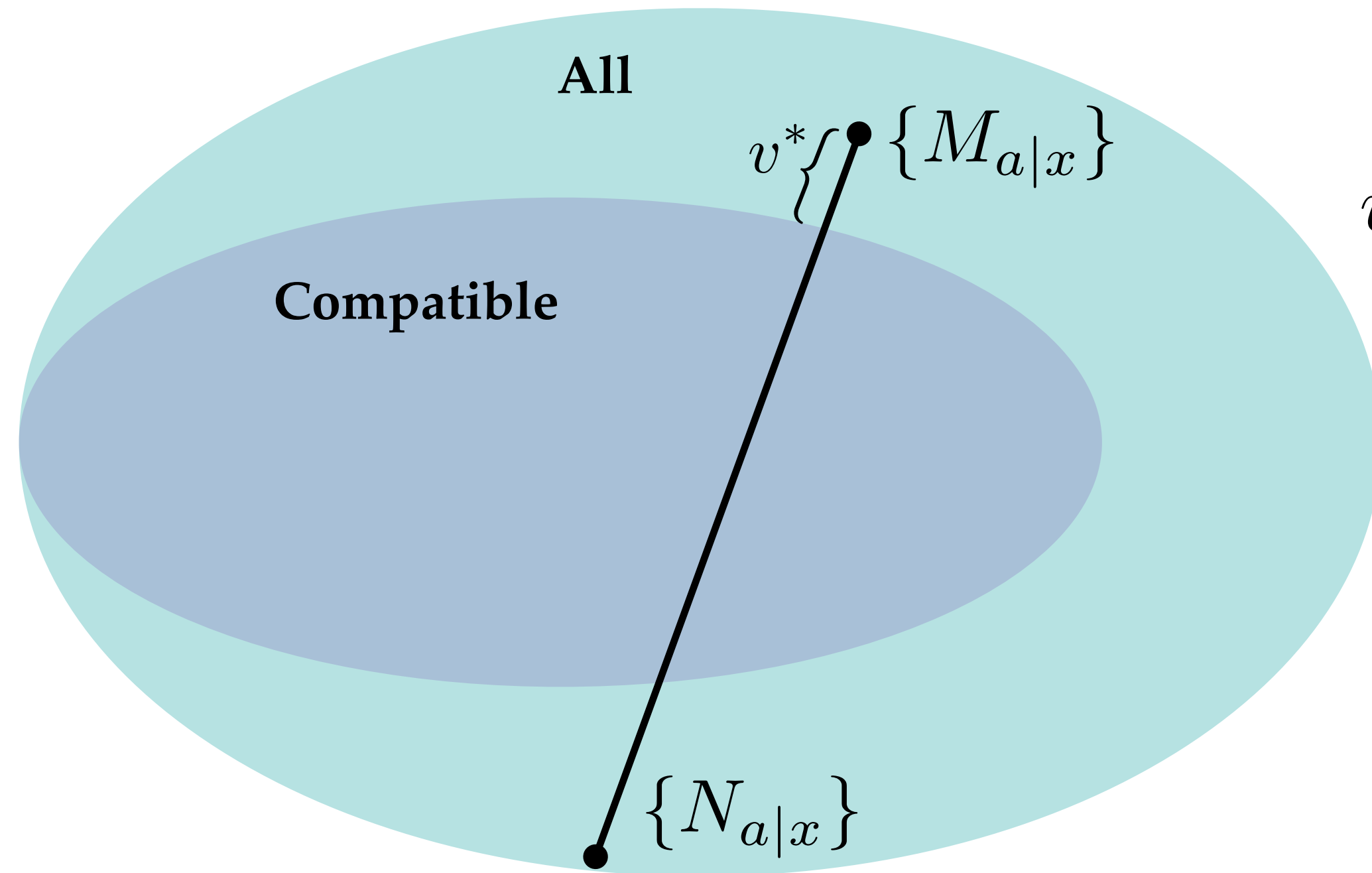
seesaw goes somewhere!

II. Approximation: Attainability

Most incompatible set of quantum measurements

Geometric interpretation

Primal problem



given $\{M_{a|x}\}$

$$v^* = \max v$$

$$\text{s.t. } v M_{a|x} + (1 - v) N_{a|x} = \sum_{\mu} D(a|x, \mu) G_{\mu}$$

$$0 \leq v \leq 1$$

$$N_{a|x} \geq 0, \quad \sum_a N_{a|x} = \mathbb{1}$$

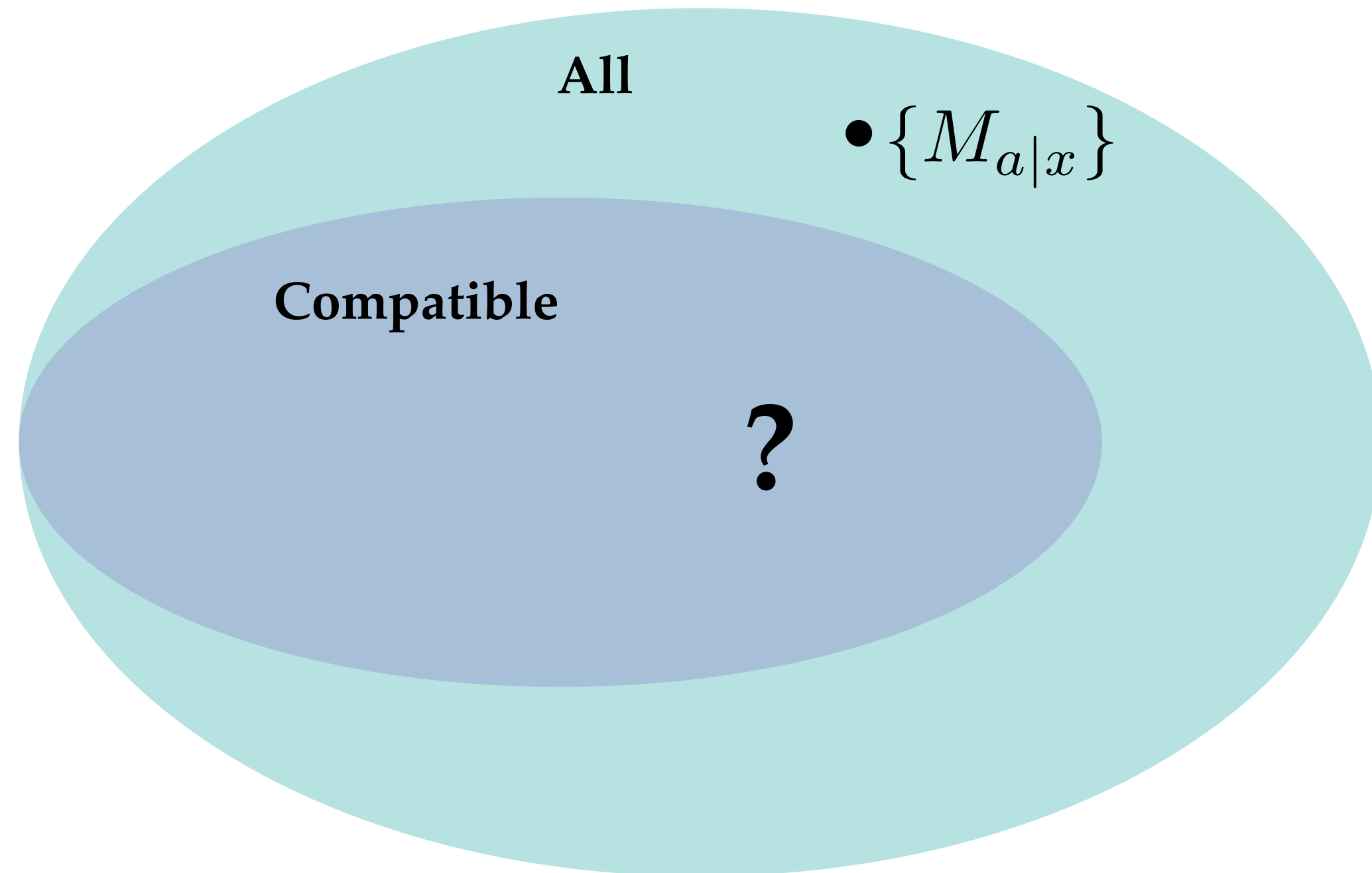
$$G_{\mu} \geq 0, \quad \sum_{\mu} G_{\mu} = \mathbb{1}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Geometric interpretation

Dual problem



$$\begin{aligned} & \text{given } \{M_{a|x}\} \\ t^* &= \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) - \text{tr}(\Gamma) \\ & \text{s.t. } \sum_{ax} D(a|x, \mu) F_{a|x} \leq \Gamma \\ & \text{tr}(\Gamma) \leq 1 \end{aligned}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

$$t^* = \max \left\{ t \mid \frac{M_{a|x} + tN_{a|x}}{t+1} = \sum_{\mu} D(a|x, \mu) G_{\mu} \right\}$$

$$t^* = \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) - \text{tr}(\Gamma^*) = 0 \iff \{M_{a|x}\} \text{ are compatible}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

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II. Approximation: Attainability

Most incompatible set of quantum measurements

$$t^* = \max \left\{ t \mid \frac{M_{a|x} + tN_{a|x}}{t+1} = \sum_{\mu} D(a|x, \mu) G_{\mu} \right\}$$

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$$t^* = \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) - \text{tr}(\Gamma^*) > 0 \iff \{M_{a|x}\} \text{ are **incompatible**}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

$$t^* = \max \left\{ t \mid \frac{M_{a|x} + tN_{a|x}}{t+1} = \sum_{\mu} D(a|x, \mu) G_{\mu} \right\}$$

$$t^* = \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) - \text{tr}(\Gamma^*) = 0 \iff \{M_{a|x}\} \text{ are compatible} \iff \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) = \text{tr}(\Gamma^*)$$

$$t^* = \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) - \text{tr}(\Gamma^*) > 0 \iff \{M_{a|x}\} \text{ are **incompatible**} \iff \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) > \text{tr}(\Gamma^*)$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

$$t^* = \max \left\{ t \mid \frac{M_{a|x} + tN_{a|x}}{t+1} = \sum_{\mu} D(a|x, \mu) G_{\mu} \right\}$$

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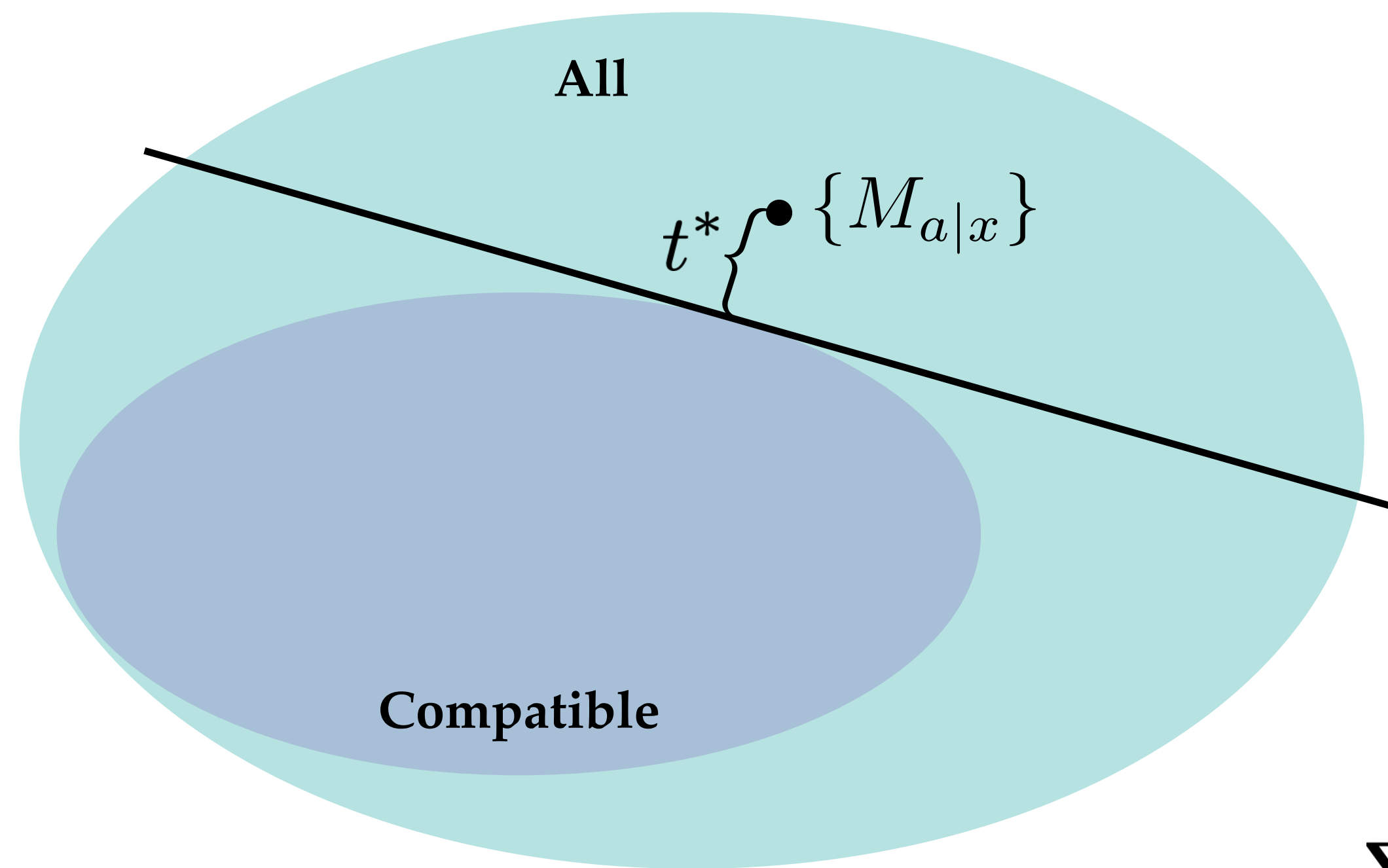
$$t^* = \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) - \text{tr}(\Gamma^*) > 0 \iff \{M_{a|x}\} \text{ are **incompatible**} \iff \sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) > \text{tr}(\Gamma^*)$$

$$\sum_{ax} \text{tr} \left(F_{a|x}^* M_{a|x} \right) \leq \text{tr}(\Gamma^*)$$

incompatibility witness

II. Approximation: Attainability

Most incompatible set of quantum measurements



Dual problem

given $\{M_{a|x}\}$

$$t^* = \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) - \text{tr}(\Gamma)$$

$$\text{s.t.} \quad \sum_{ax} D(a|x, \mu) F_{a|x} \leq \Gamma$$

$$\text{tr}(\Gamma) \leq 1$$

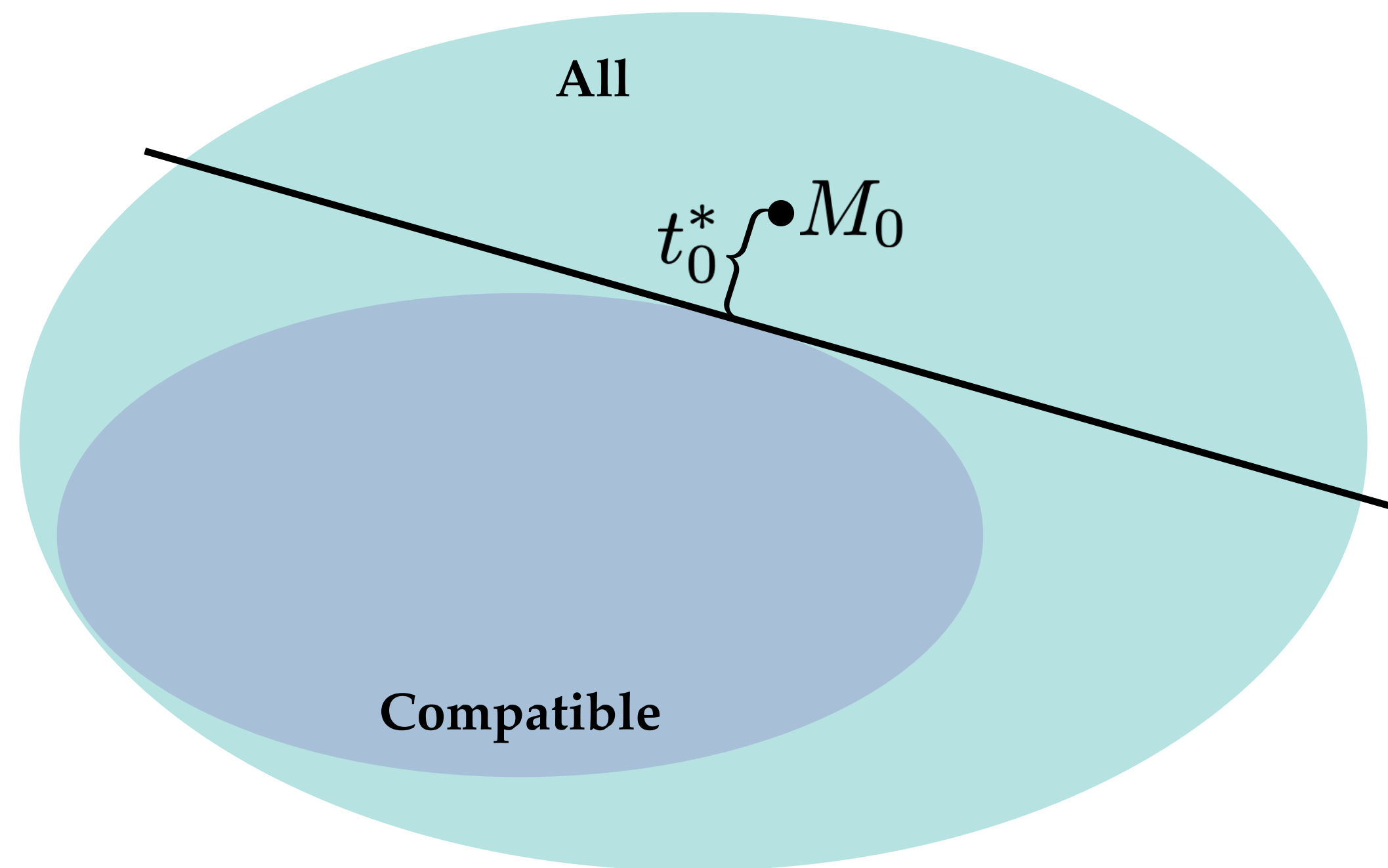
$$\sum_{ax} \text{tr}(F_{a|x}^* M_{a|x}) = \text{tr}(\Gamma^*)$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

SDP 1



given $\{M_{a|x}\}$

$$t^* = \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t.} \quad \sum_{ax} D(a|x, \mu) F_{a|x} \leq -\Gamma$$

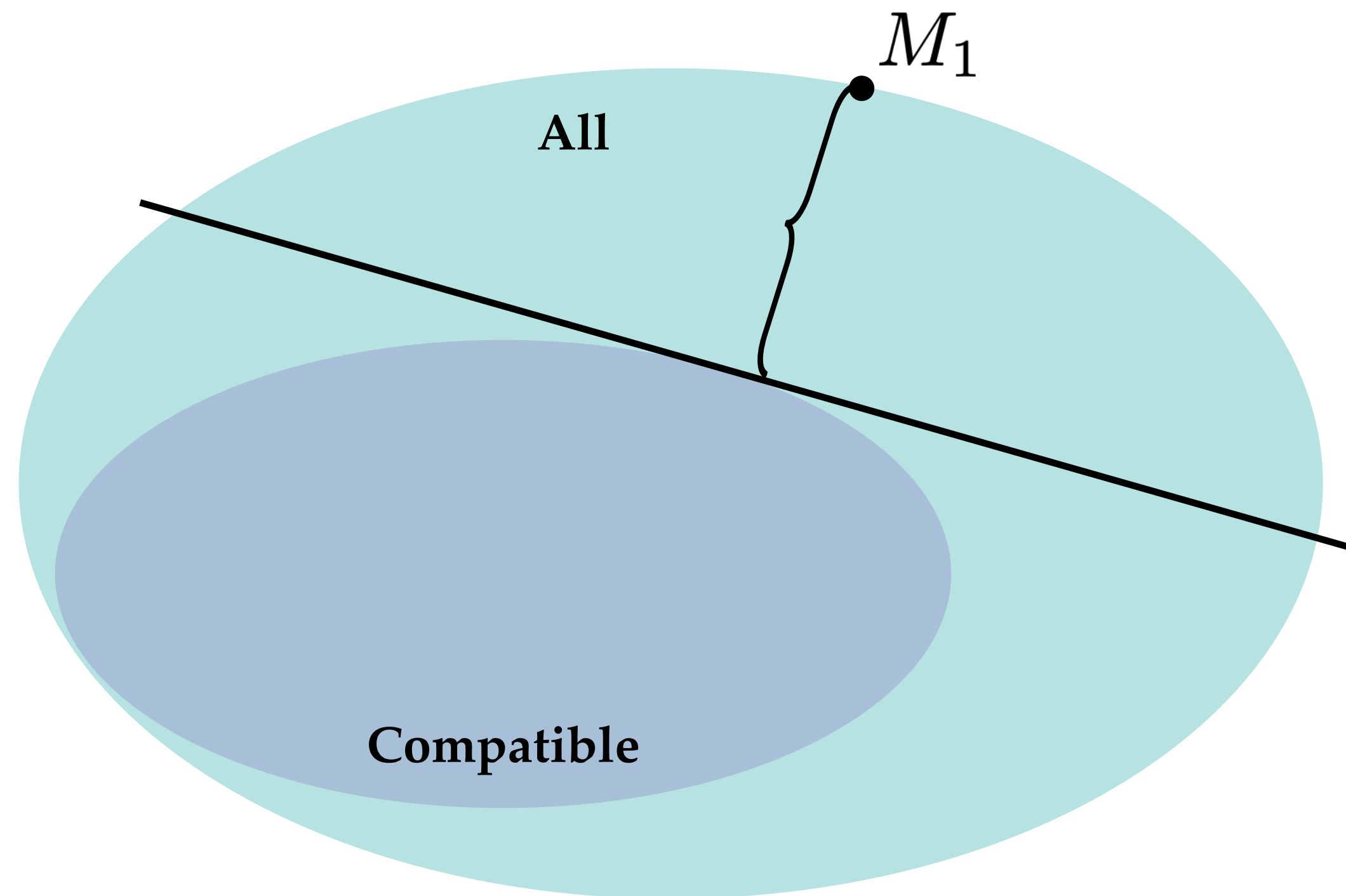
$$\text{tr}(\Gamma) \geq -1$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

SDP 2



given $\Gamma, \{F_{a|x}\}$

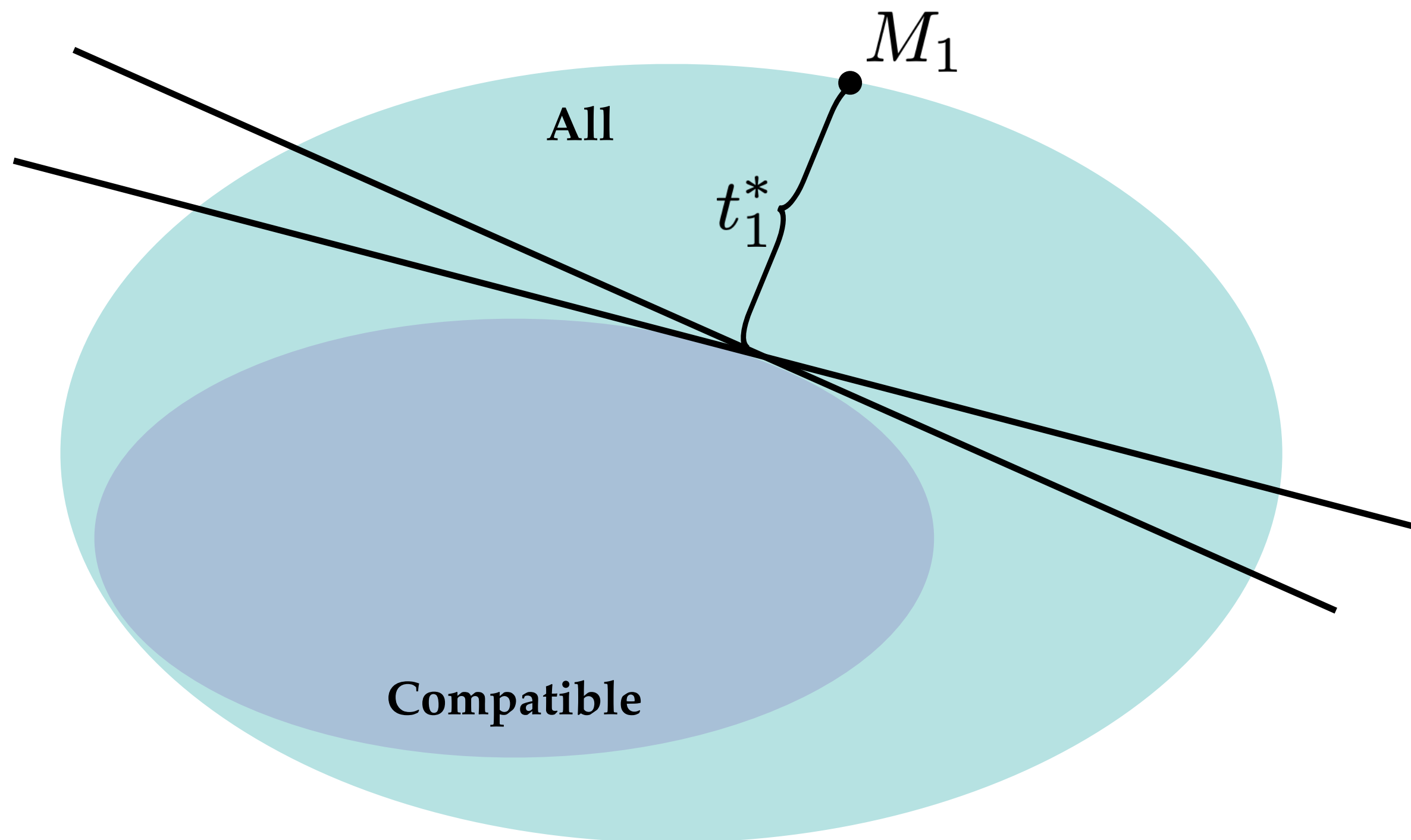
$$\max_{\{M_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t. } M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm



SDP 1

given $\{M_{a|x}\}$

$$t^* = \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t.} \quad \sum_{ax} D(a|x, \mu) F_{a|x} \leq -\Gamma$$

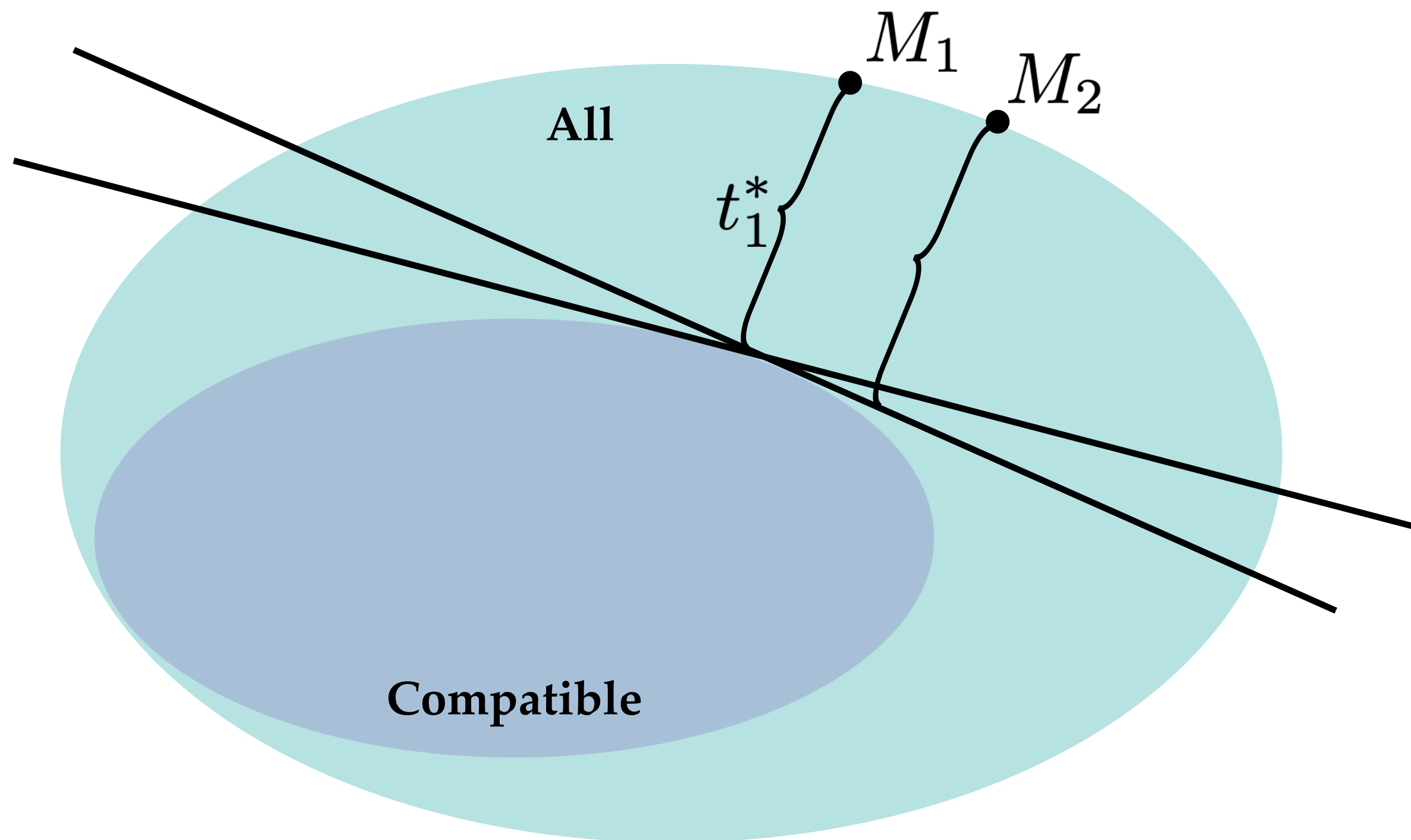
$$\text{tr}(\Gamma) \geq -1$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

SDP 2



given $\Gamma, \{F_{a|x}\}$

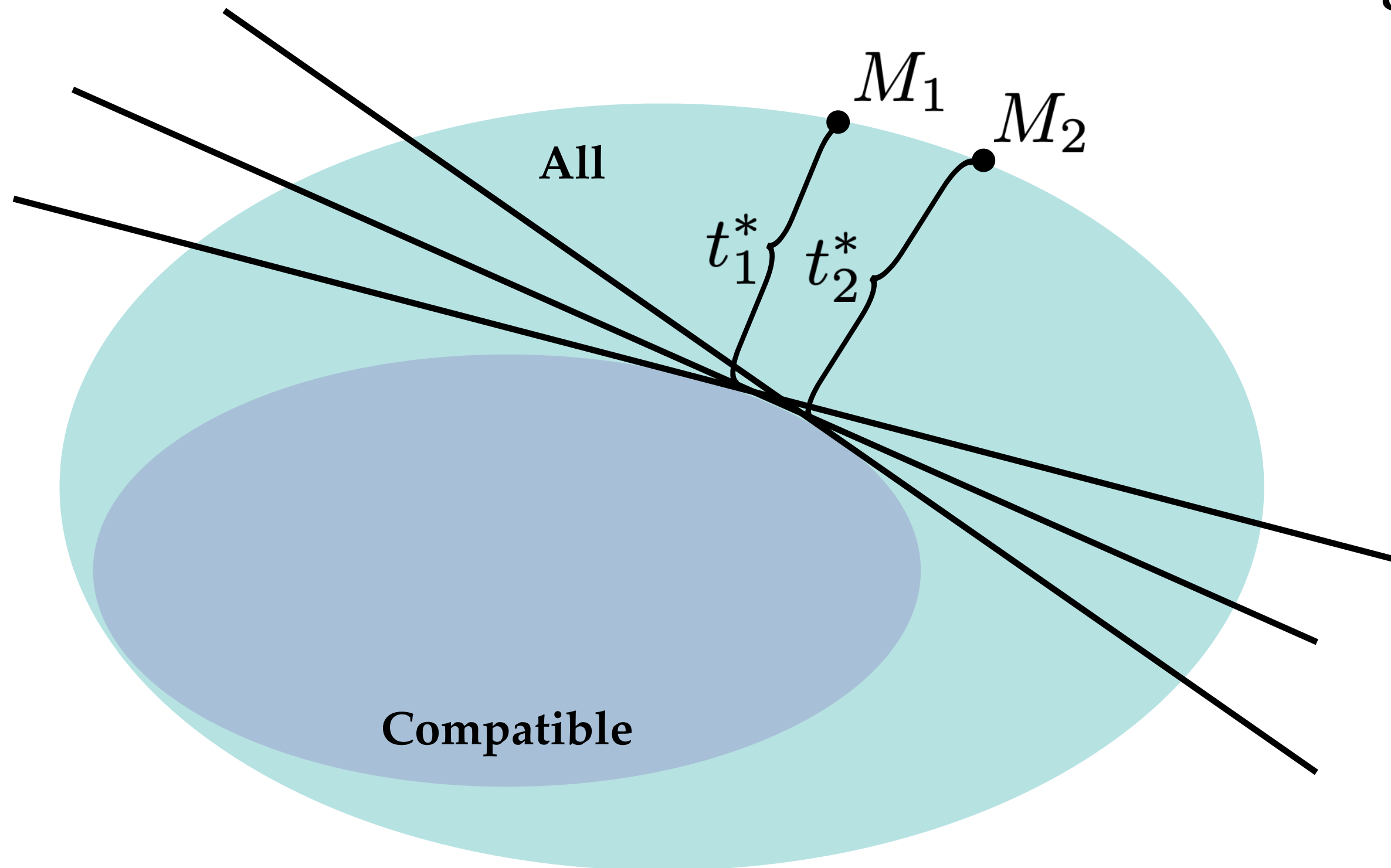
$$\max_{\{M_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t. } M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm



SDP 1

given $\{M_{a|x}\}$

$$t^* = \max_{\Gamma, \{F_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

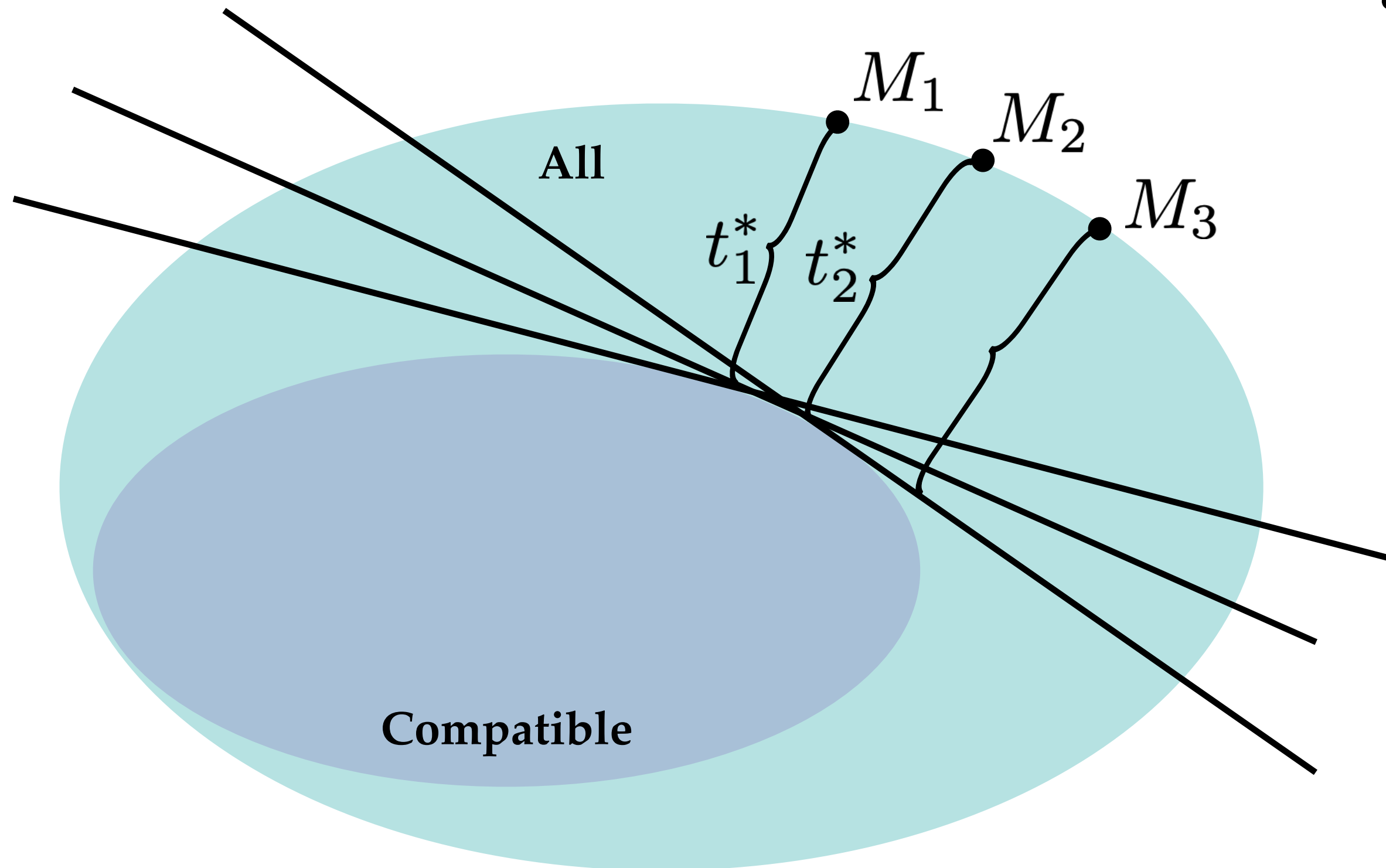
$$\text{s.t.} \quad \sum_{ax} D(a|x, \mu) F_{a|x} \leq -\Gamma$$

$$\text{tr}(\Gamma) \geq -1$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm



SDP 2

given $\Gamma, \{F_{a|x}\}$

$$\max_{\{M_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

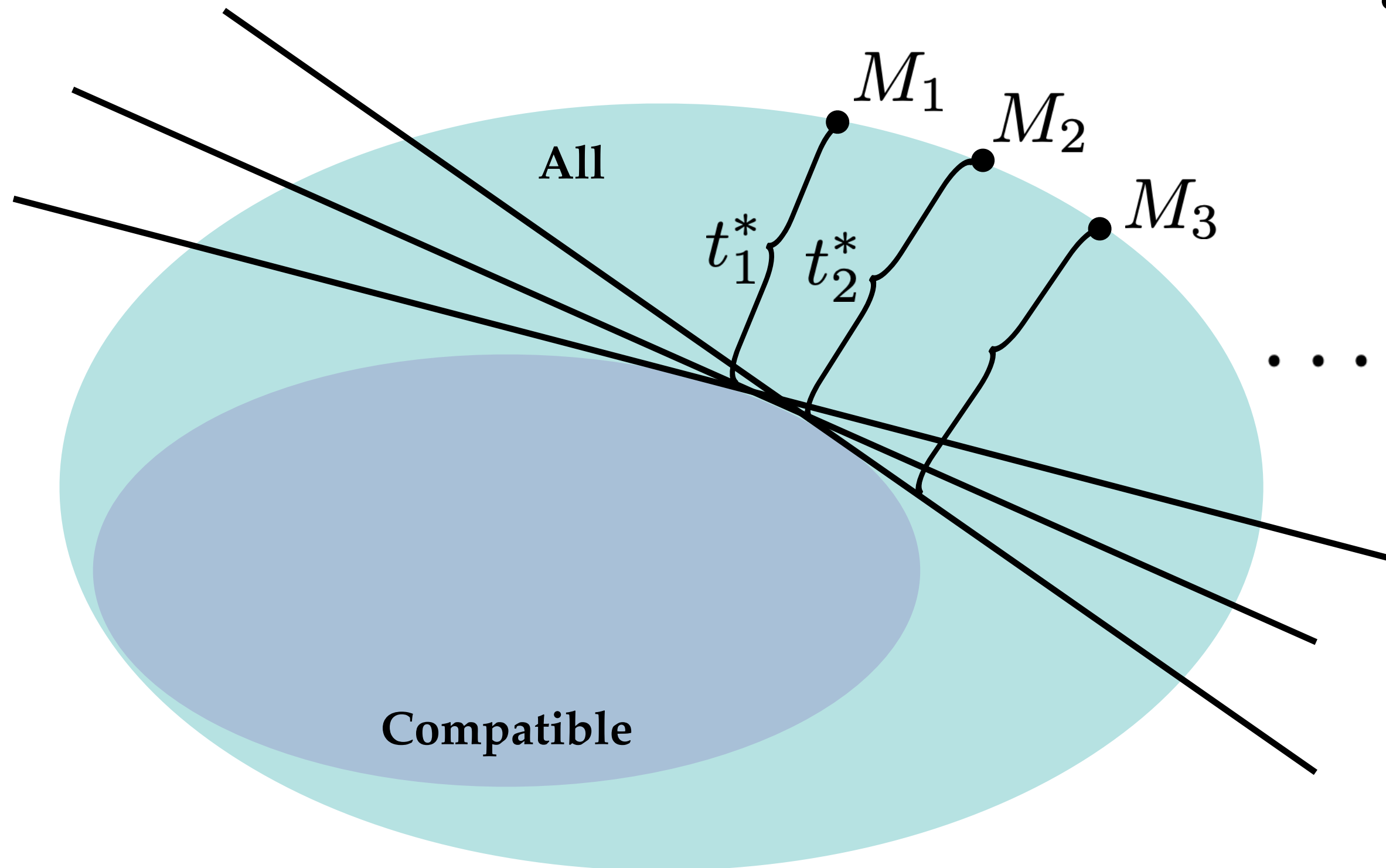
$$\text{s.t. } M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

II. Approximation: Attainability

Most incompatible set of quantum measurements

Seesaw algorithm

SDP 2



given $\Gamma, \{F_{a|x}\}$

$$\max_{\{M_{a|x}\}} \sum_{ax} \text{tr}(F_{a|x} M_{a|x}) + \text{tr}(\Gamma)$$

$$\text{s.t. } M_{a|x} \geq 0, \quad \sum_a M_{a|x} = \mathbb{1}$$

II. Approximating: **Attainability**

MAIN MESSAGE:

Seesaws are useful tools to tackle multilinear problems

Duality theory can help you convert max-min problems into max-max problems and tackle it with seesaws.

II. Approximating: **Impossibility**

II. Approximation: **Impossibility**

Channel discrimination without memory

II. Approximation: **Impossibility**

Channel discrimination without memory

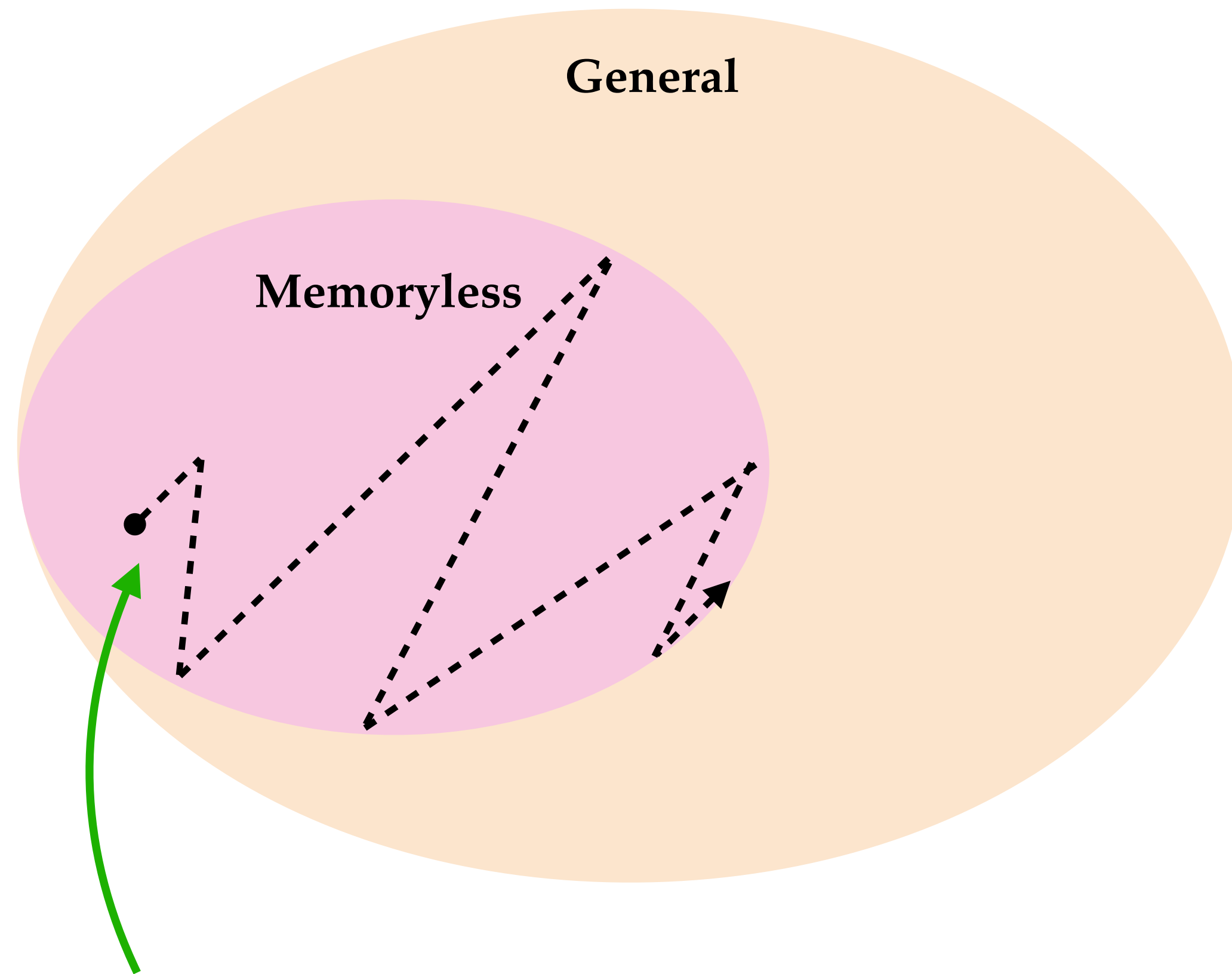
$$\mathcal{E} = \{p_i, \tilde{C}_i\}$$



$$P_{\text{ML}} = \max_{\rho, \{M_i\}} \sum_i p_i \operatorname{tr} [\tilde{C}_i(\rho) M_i]$$

II. Approximation: **Impossibility**

Channel discrimination without memory

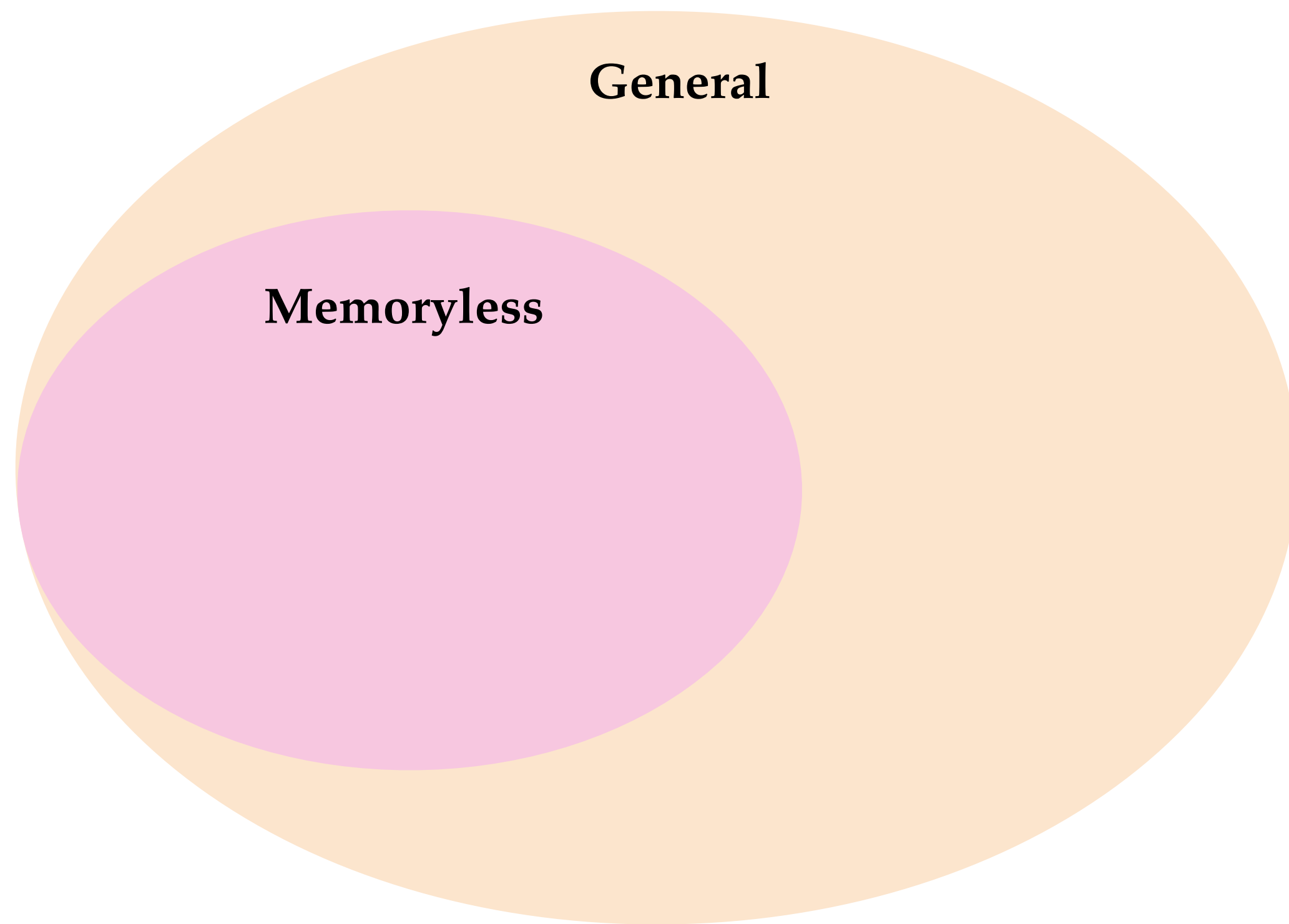


$$T_i = \rho \otimes M_i^T$$

inner search

II. Approximation: **Impossibility**

Channel discrimination without memory

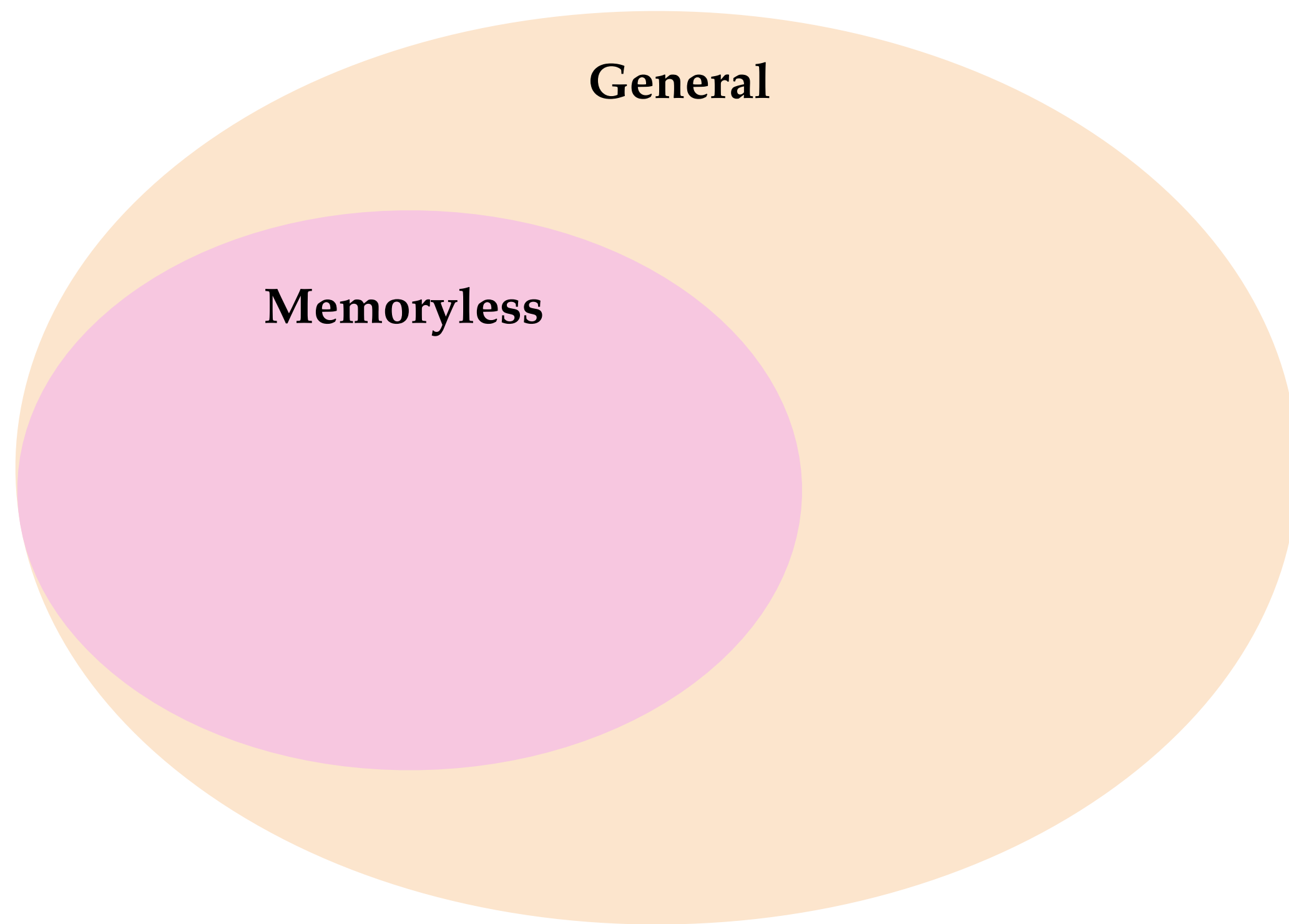


$$T_i = \rho \otimes M_i^T$$

How can I show that the optimal tester **requires** memory for implementation (does not accept such decomposition)?

II. Approximation: Impossibility

Channel discrimination without memory



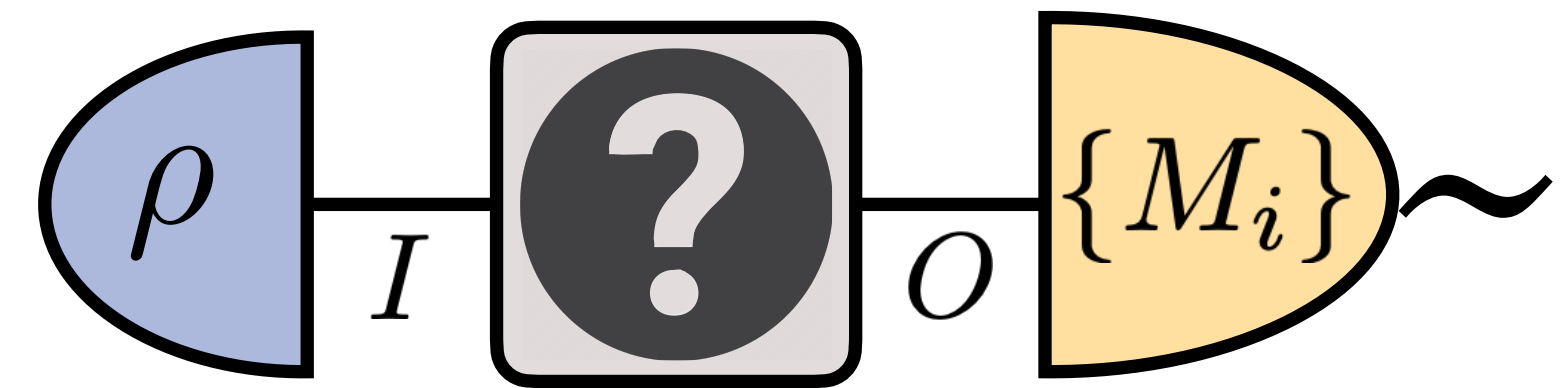
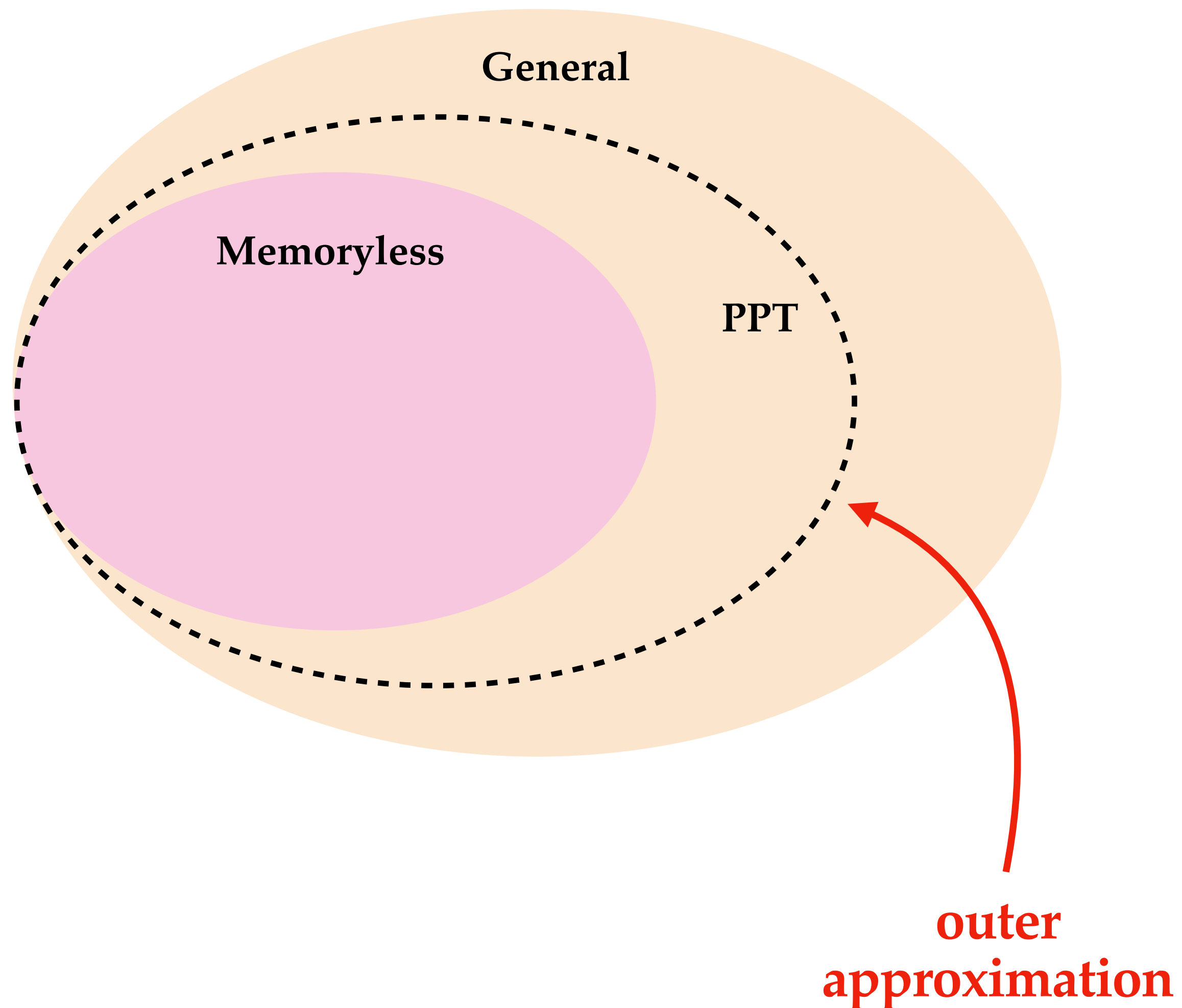
$$T_i = \rho \otimes M_i^T$$



$$(T_i)^{T_O} = \rho \otimes (M_i^T)^T = \rho \otimes M_i \geq 0$$

II. Approximation: **Impossibility**

Channel discrimination without memory



$$T_i = \rho \otimes M_i^T$$

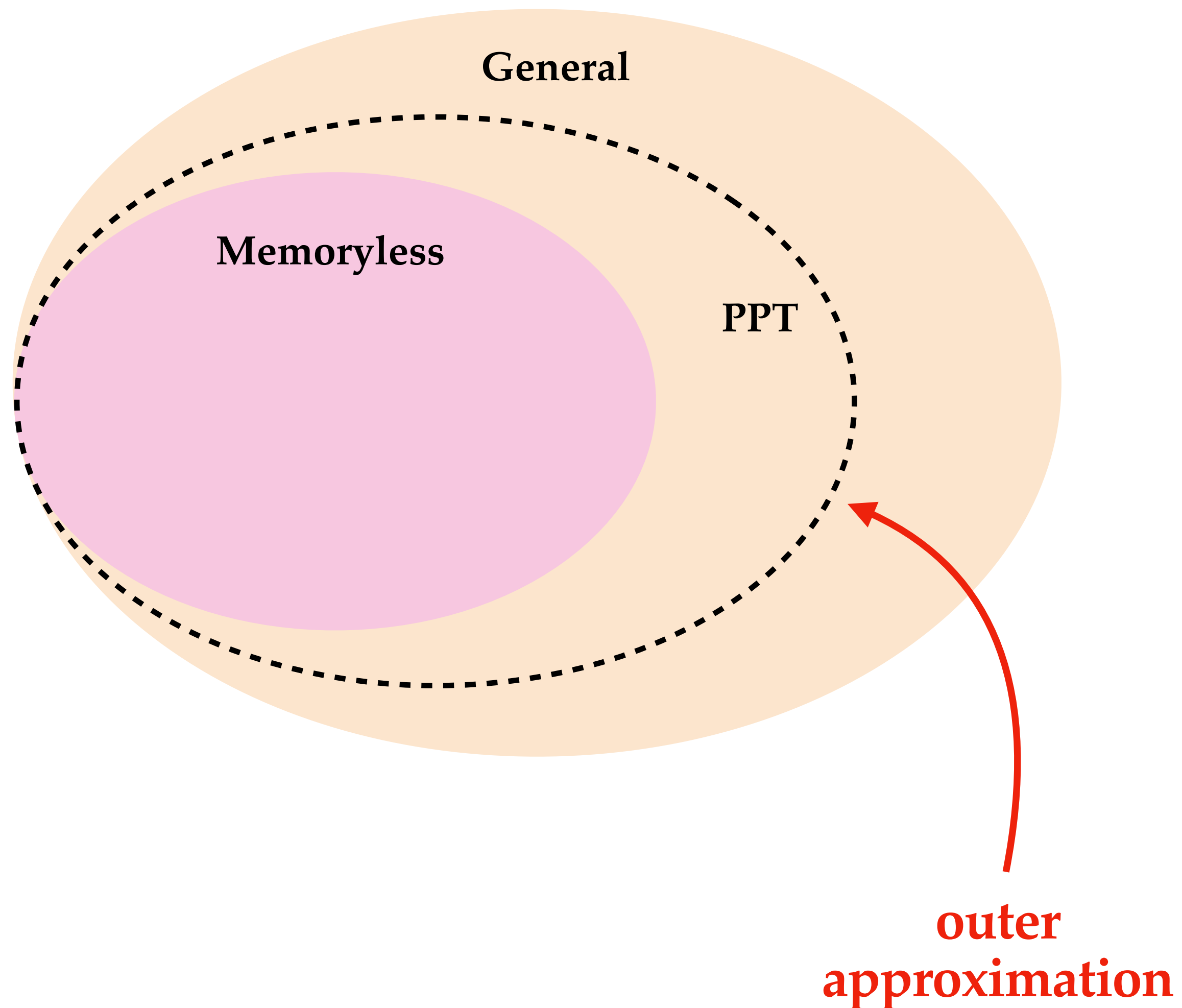
$$\Downarrow \quad \Uparrow$$

$$(T_i)^{T_O} = \rho \otimes (M_i^T)^T = \rho \otimes M_i \geq 0$$

positive partial transpose (PPT)

II. Approximation: **Impossibility**

Channel discrimination without memory



given $\{p_i, C_i\}$

$$P_{\text{PPT}} = \max_{\{T_i\}} \sum_i p_i \text{tr}(C_i^T T_i)$$

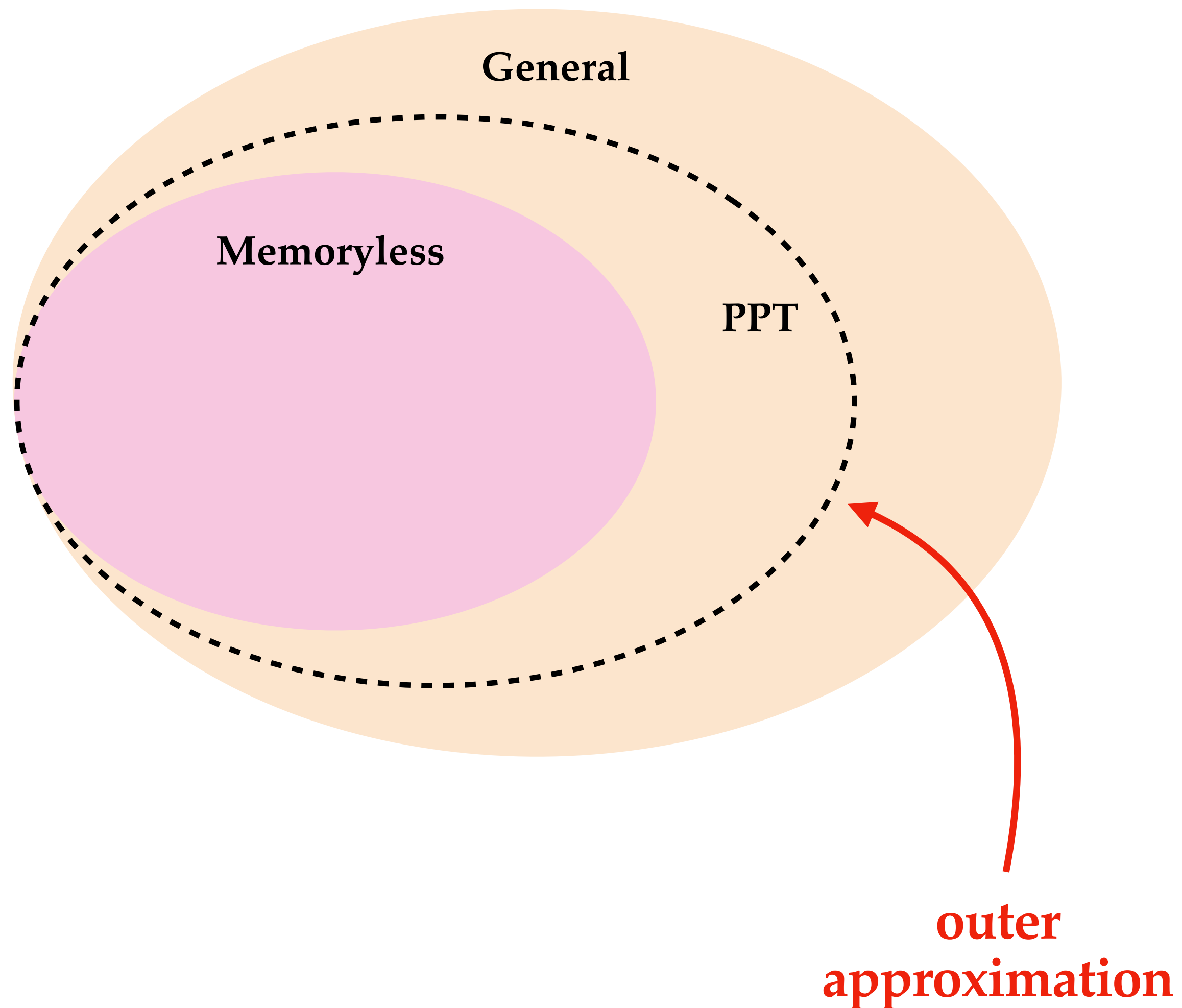
s.t. $T_i \geq 0, \quad (T_i)^{T_O} \geq 0$

$$\text{tr} \left(\sum_i T_i \right) = d_O$$

$$\sum_i T_i = \text{tr}_O \left(\sum_i T_i \right) \otimes \frac{\mathbb{1}_O}{d_O}$$

II. Approximation: **Impossibility**

Channel discrimination without memory



given $\{p_i, C_i\}$

$$P_{\text{PPT}} = \max_{\{T_i\}} \sum_i p_i \text{tr}(C_i^T T_i)$$

$$\text{s.t. } T_i \geq 0, \quad (T_i)^{T_O} \geq 0$$

$$\text{tr} \left(\sum_i T_i \right) = d_O$$

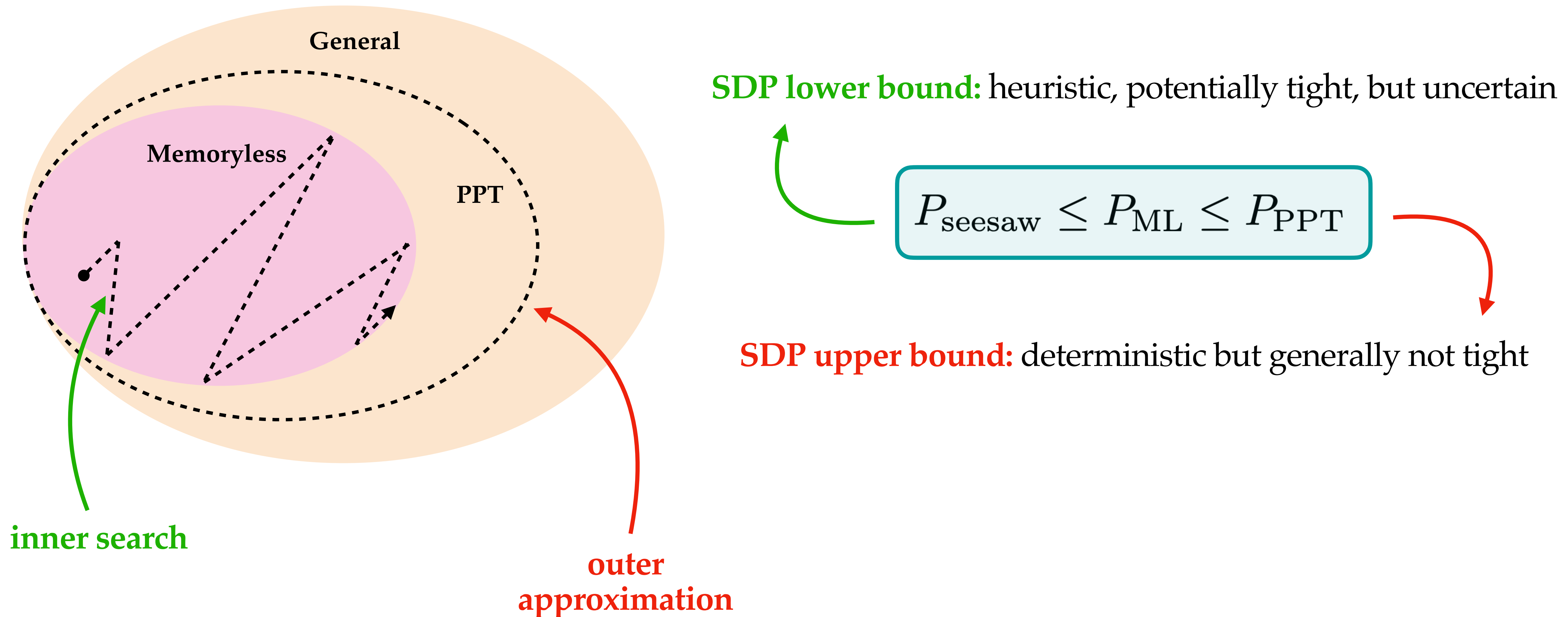
$$\sum_i T_i = \text{tr}_O \left(\sum_i T_i \right) \otimes \frac{\mathbb{1}_O}{d_O}$$

$$P_{\text{ML}} \leq P_{\text{PPT}}$$

upper bound

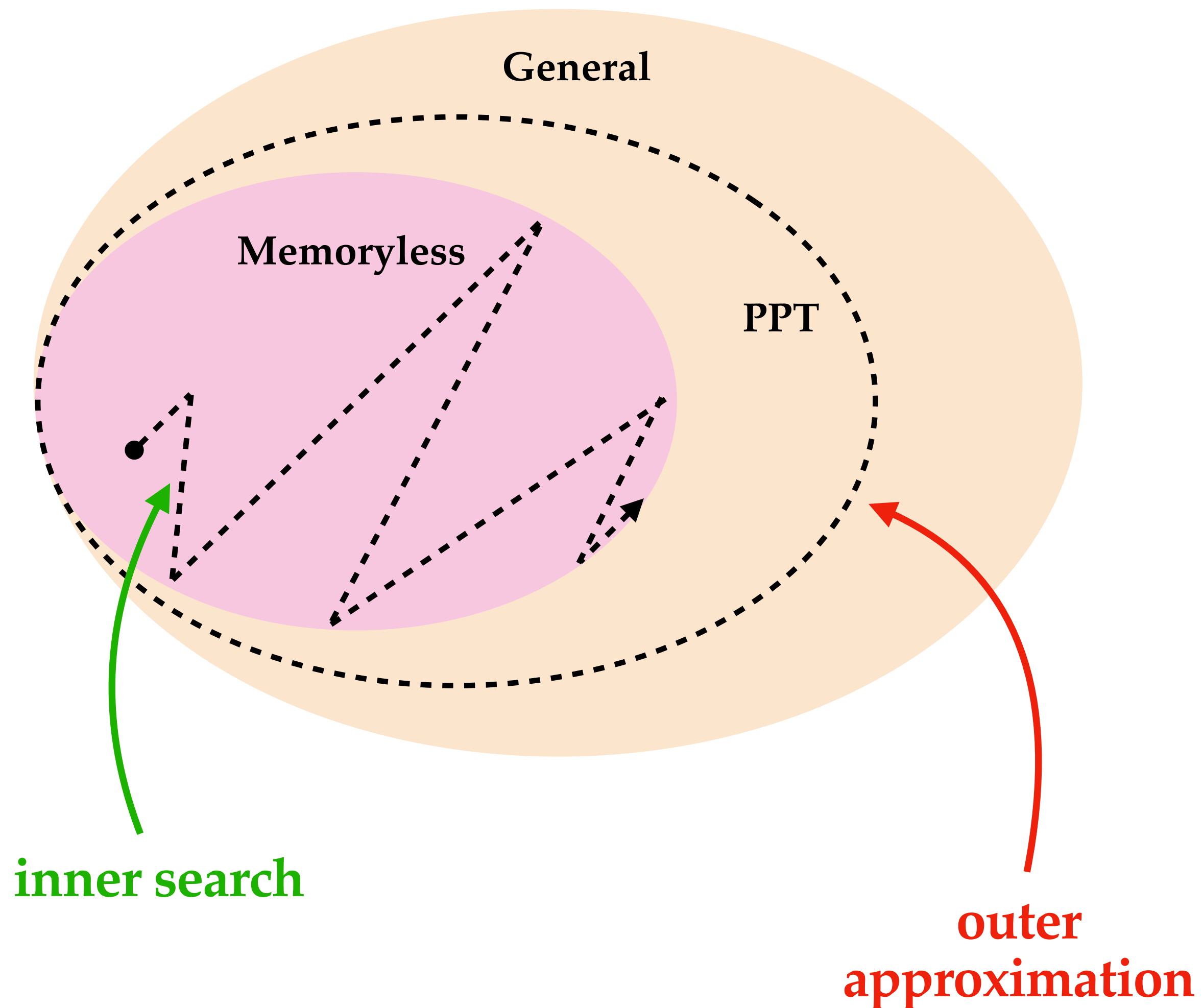
II. Approximation: **Impossibility**

Channel discrimination without memory



II. Approximation: **Impossibility**

Channel discrimination without memory



if $P_{\text{seesaw}} = P \implies$ optimal strategy
is **memoryless**

$$P_{\text{seesaw}} \leq P_{\text{ML}} \leq P_{\text{PPT}} \leq P$$

if $P_{\text{PPT}} < P \implies$ optimal strategy
requires memory

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

separable quantum state:

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

separable quantum state:

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

given ρ_{AB}

find $\{p_i\}, \{\rho_i^A\}, \{\rho_i^B\}$

s.t. $\rho_{AB} = \sum p_i \rho_i^A \otimes \rho_i^B$

$$p_i \geq 0, \sum_i p_i = 1$$

$$\rho_i^A \geq 0, \text{tr}(\rho_i^A) = 1$$

$$\rho_i^B \geq 0, \text{tr}(\rho_i^B) = 1$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

separable quantum state:

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

A **seesaw** approach might prove it is separable if it **finds** a separable decomposition;
But it **can never prove** such decomposition does **not exist**.

given ρ_{AB}

find $\{p_i\}, \{\rho_i^A\}, \{\rho_i^B\}$

s.t. $\rho_{AB} = \sum p_i \rho_i^A \otimes \rho_i^B$

$$p_i \geq 0, \sum_i p_i = 1$$

$$\rho_i^A \geq 0, \text{tr}(\rho_i^A) = 1$$

$$\rho_i^B \geq 0, \text{tr}(\rho_i^B) = 1$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

PTT revisited:

separable quantum state:

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B \implies (\rho_{\text{sep}})^{T_B} = \sum_i p_i \rho_i^A \otimes (\rho_i^B)^{T_B} \geq 0$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

$$\rho_{AB} \text{ is separable} \implies (\rho_{AB})^{T_B} \geq 0$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

$$\rho_{AB} \text{ is separable} \implies (\rho_{AB})^{T_B} \geq 0$$

$$(\rho_{AB})^{T_B} \not\geq 0 \implies \rho_{AB} \text{ is } \mathbf{entangled}$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

$$\rho_{AB} \text{ is separable} \implies (\rho_{AB})^{T_B} \geq 0$$

$$(\rho_{AB})^{T_B} \not\geq 0 \implies \rho_{AB} \text{ is entangled}$$

$$(\rho_{AB})^{T_B} \geq 0 \stackrel{?}{\implies} \rho_{AB} \text{ is separable}$$

What do you do when your state is PPT but you cannot find a separable decomposition via seesaw?

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

Question. How can I determine whether a given quantum state is entangled?

$$\rho_{AB} \text{ is separable} \implies (\rho_{AB})^{T_B} \geq 0$$

$$(\rho_{AB})^{T_B} \not\geq 0 \implies \rho_{AB} \text{ is entangled}$$

$$(\rho_{AB})^{T_B} \geq 0 \not\Rightarrow \rho_{AB} \text{ is separable}$$

What do you do when your state is PPT but you cannot find a separable decomposition via seesaw?

II. Approximation: **Impossibility**

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

$$k = 1 : \implies \rho^{(1)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1}$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

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$$k = 1 : \implies \rho^{(1)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \underbrace{\rho_i^{B_1}}_{\text{1 copy}}$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

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II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

$$k = 1 : \implies \rho^{(1)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1} \left(\rho^{(1)} \right) = \text{tr}_B \left(\rho^{(1)} \right)$$

$$k = 2 : \implies \rho^{(2)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \underbrace{\rho_i^{B_1} \otimes \rho_i^{B_2}}_{\text{2 copies}}$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

$$k = 1 : \implies \rho^{(1)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1} \left(\rho^{(1)} \right) = \text{tr}_B \left(\rho^{(1)} \right)$$

$$k = 2 : \implies \rho^{(2)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \underbrace{\rho_i^{B_1} \otimes \rho_i^{B_2}}_{\text{2 copies}} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1 B_2} \left(\rho^{(2)} \right) = \text{tr}_{B B_1} \left(\rho^{(2)} \right) \\ = \text{tr}_{B B_2} \left(\rho^{(2)} \right)$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

$$k = 1 : \quad \Longrightarrow \quad \rho^{(1)} = \sum_i p_i \, \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1} \quad ; \quad \rho_{\text{sep}} = \text{tr}_{B_1} \left(\rho^{(1)} \right) = \text{tr}_B \left(\rho^{(1)} \right)$$

$$k = 2 : \quad \Rightarrow \quad \rho^{(2)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1} \otimes \rho_i^{B_2} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1 B_2} \left(\rho^{(2)} \right) = \text{tr}_{B B_1} \left(\rho^{(2)} \right)$$

$= \text{tr}_{B B_2} \left(\rho^{(2)} \right)$

... k copies

$$k \in \mathbb{N} : \quad \Rightarrow \quad \rho^{(k)} = \sum_i p_i \, \rho_i^A \otimes \rho_i^B \otimes \overbrace{\rho_i^{B_1} \otimes \dots \otimes \rho_i^{B_k}}$$

II. Approximation: Impossibility

Entanglement and the k-symmetric extension hierarchy

k-symmetric extension hierarchy—a.k.a the Doherty-Parrilo-Spedalieri (DPS) hierarchy

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

$$k = 1 : \implies \rho^{(1)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1} \left(\rho^{(1)} \right) = \text{tr}_B \left(\rho^{(1)} \right)$$

$$k = 2 : \implies \rho^{(2)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \rho_i^{B_1} \otimes \rho_i^{B_2} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1 B_2} \left(\rho^{(2)} \right) = \text{tr}_{B B_1} \left(\rho^{(2)} \right) \\ = \text{tr}_{B B_2} \left(\rho^{(2)} \right)$$

...

$$k \in \mathbb{N} : \implies \rho^{(k)} = \sum_i p_i \rho_i^A \otimes \rho_i^B \otimes \overbrace{\rho_i^{B_1} \otimes \dots \otimes \rho_i^{B_k}}^{k \text{ copies}} ; \quad \rho_{\text{sep}} = \text{tr}_{B_1 \dots B_k} \left(\rho^{(k)} \right),$$

$$\text{symmetric} \longrightarrow \rho^{(k)} = (\mathbb{1}_A \otimes U_\pi) \rho^{(k)} (\mathbb{1}_A \otimes U_\pi), \quad U_\pi \in S_k$$

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II. Approximation: **Impossibility**

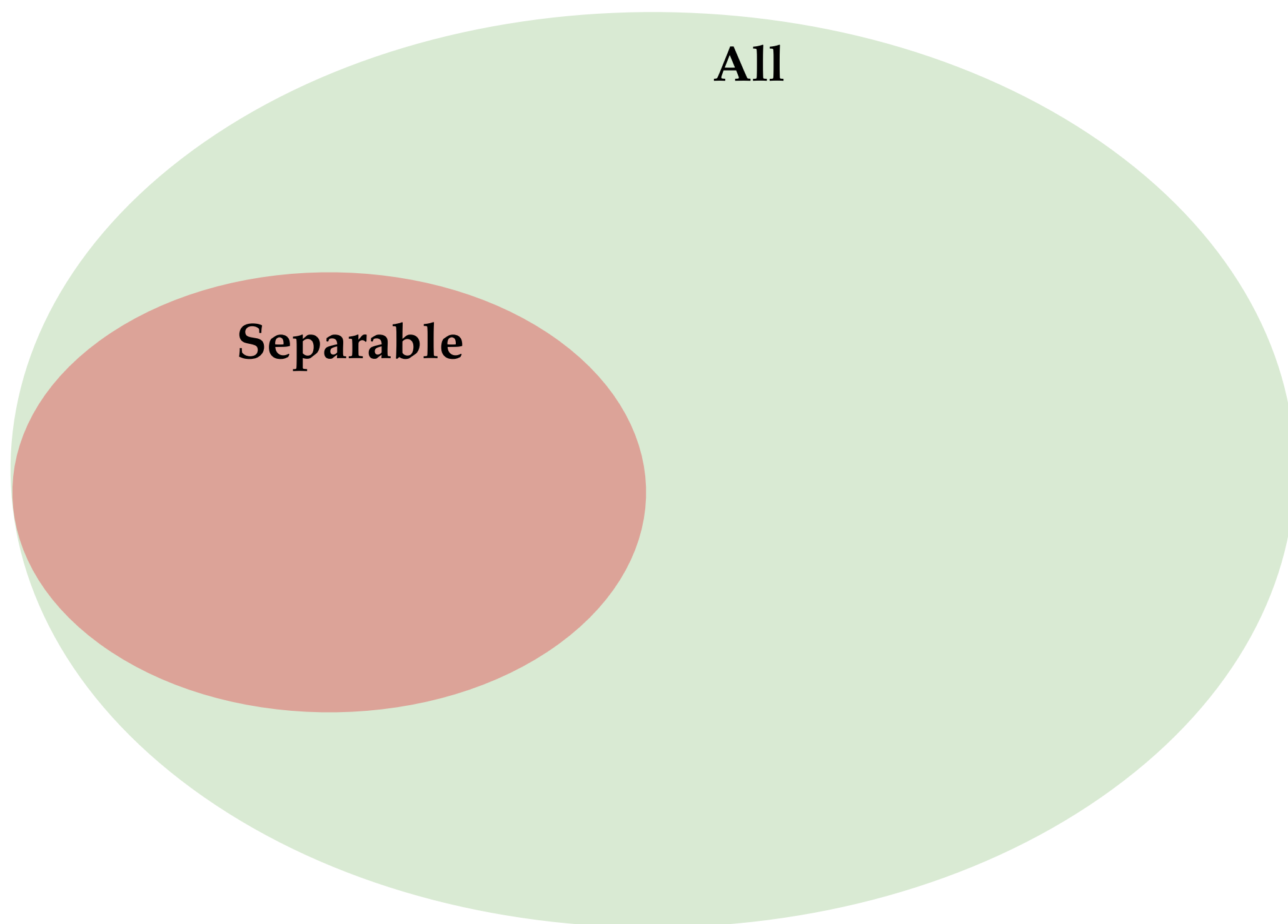
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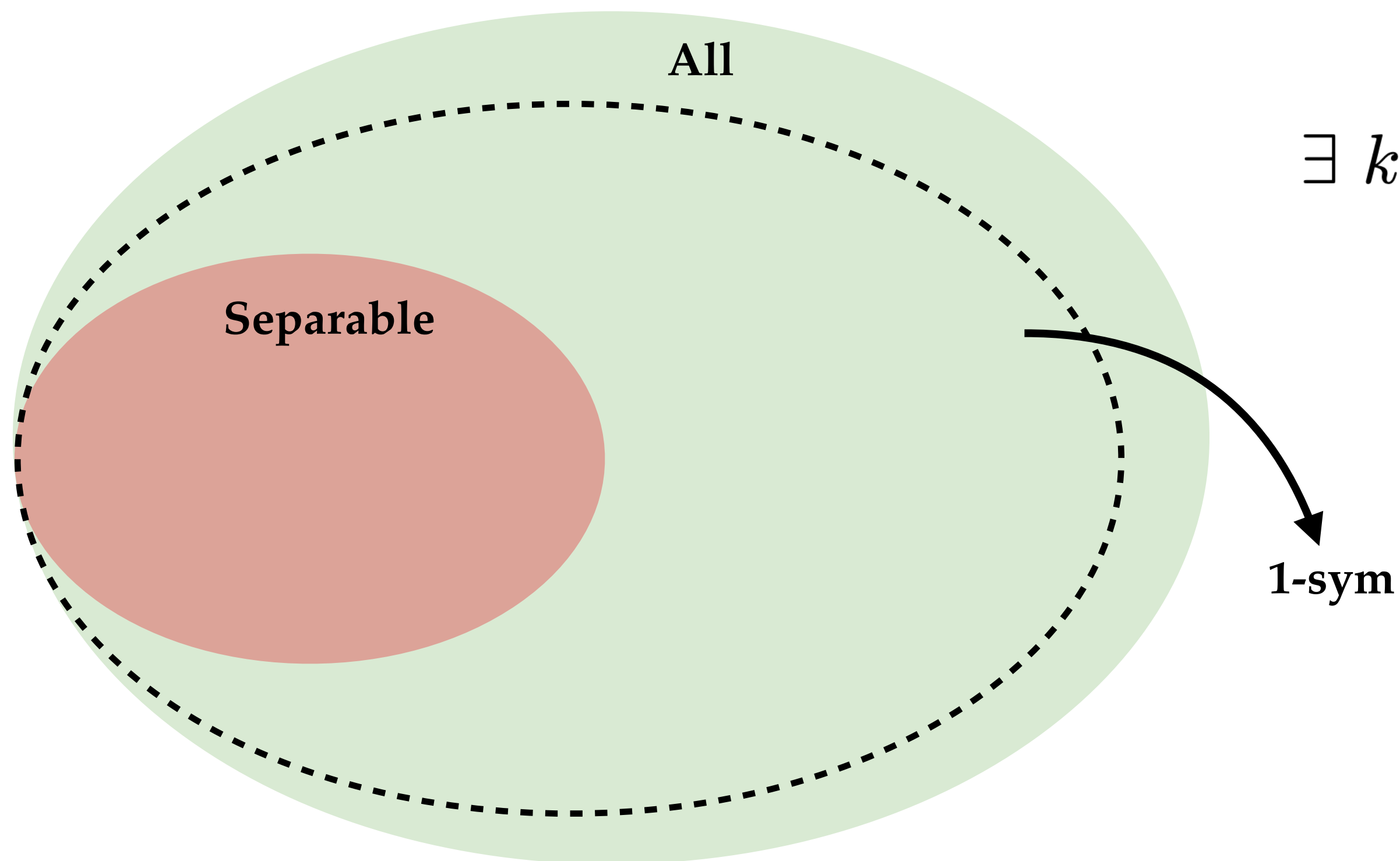
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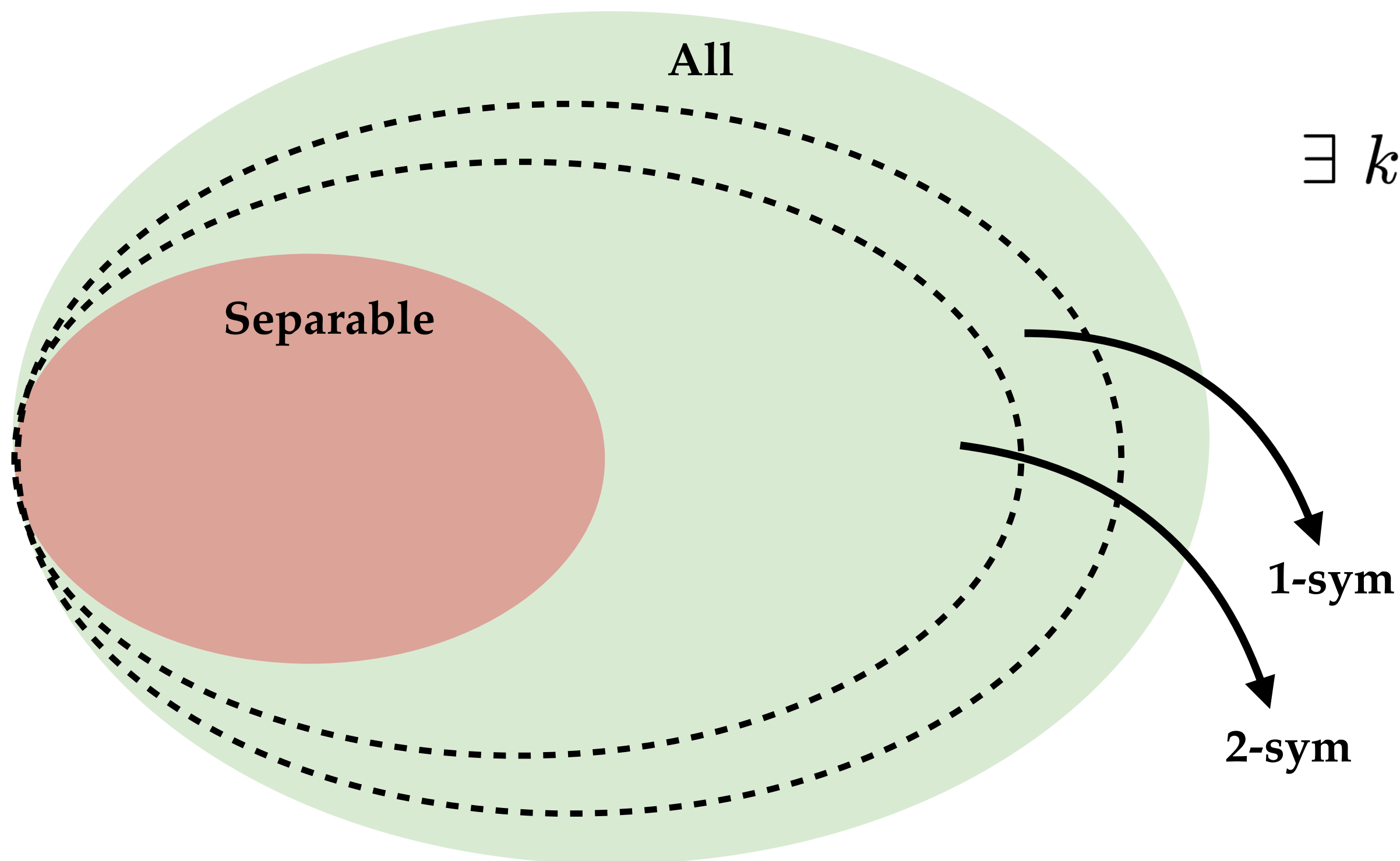
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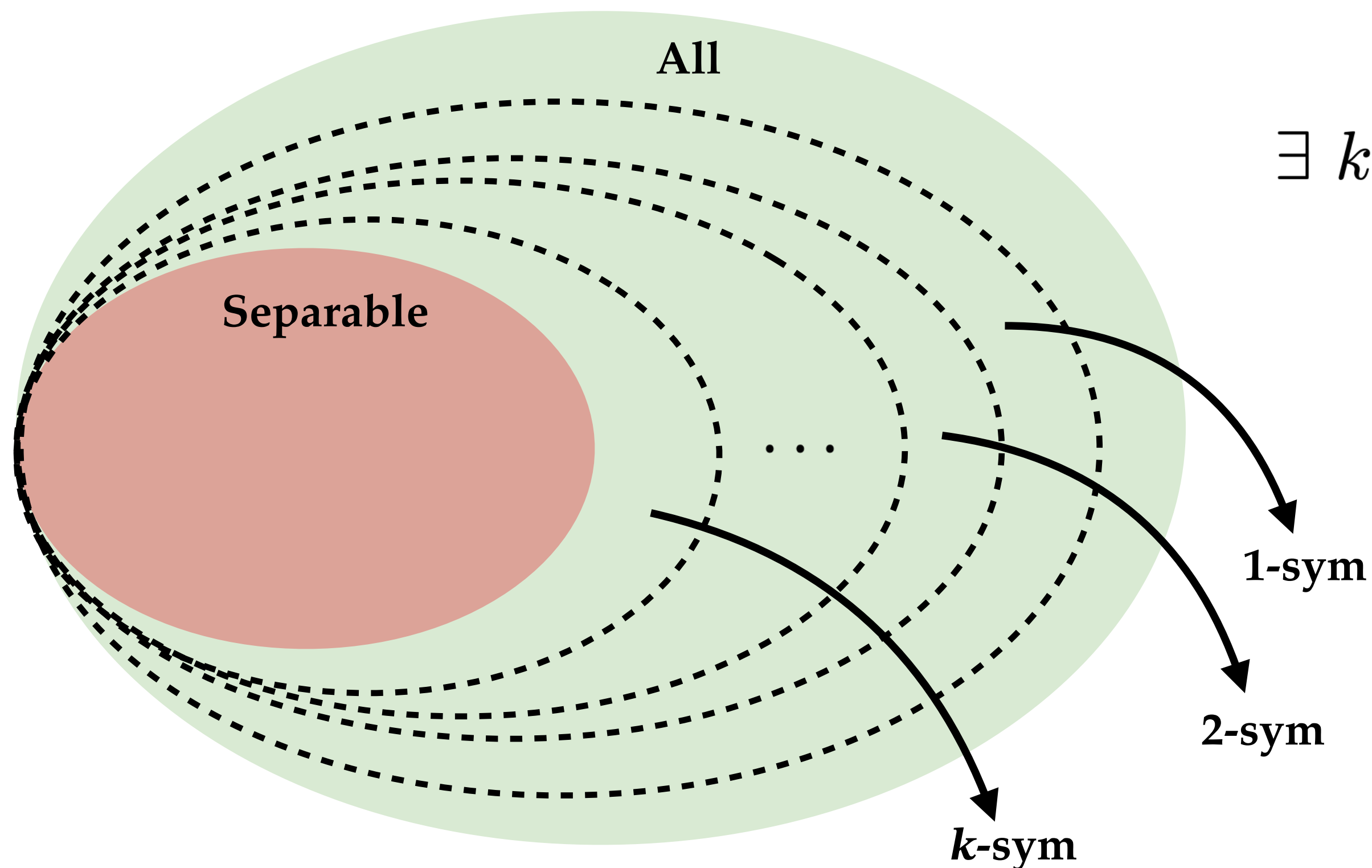
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II. Approximation: **Impossibility**

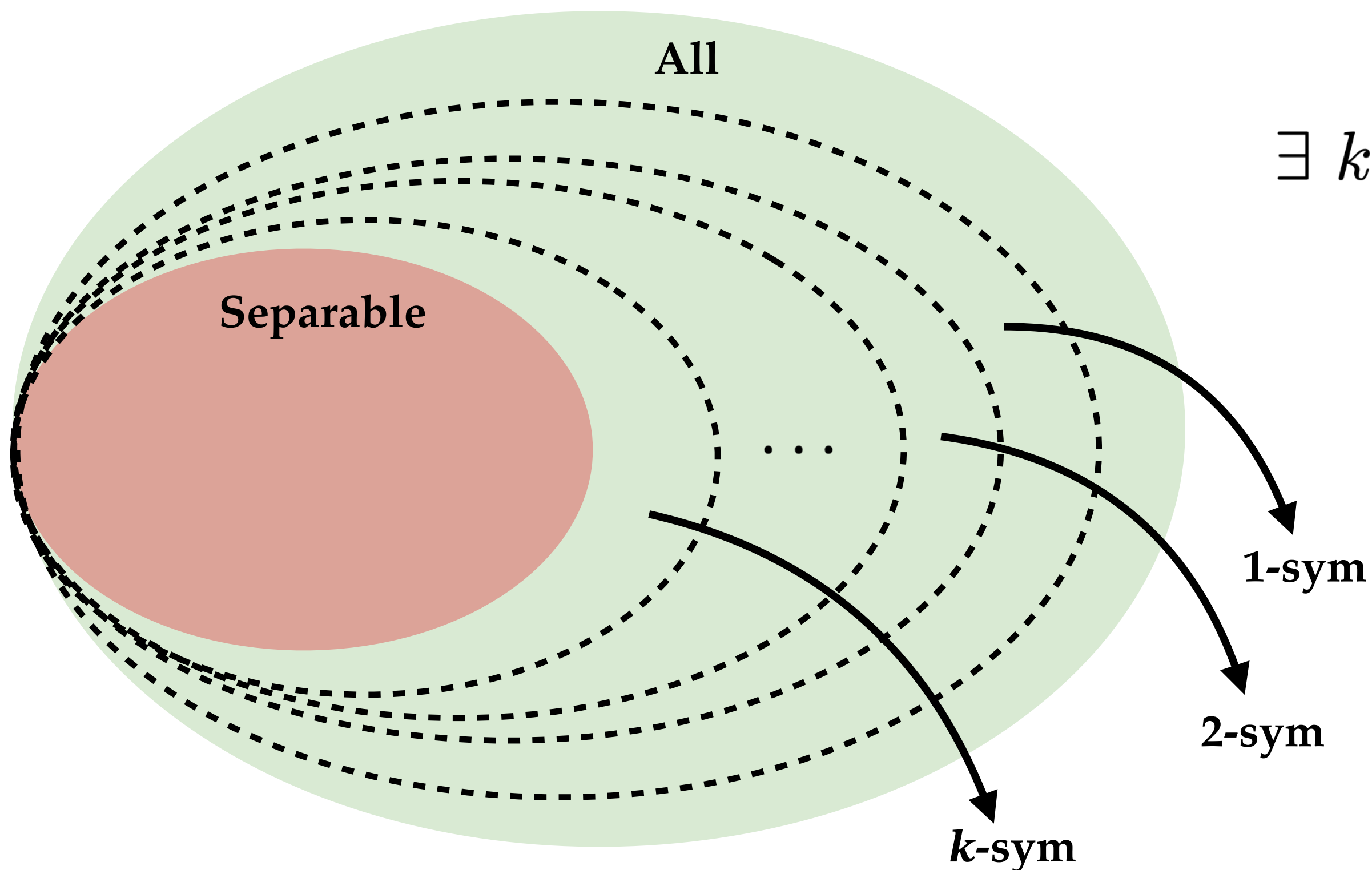
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$$\rho_{AB} \in k\text{-sym} \forall k \in \mathbb{N} \stackrel{?}{\implies} \rho_{AB} \text{ is separable}$$

II. Approximation: **Impossibility**

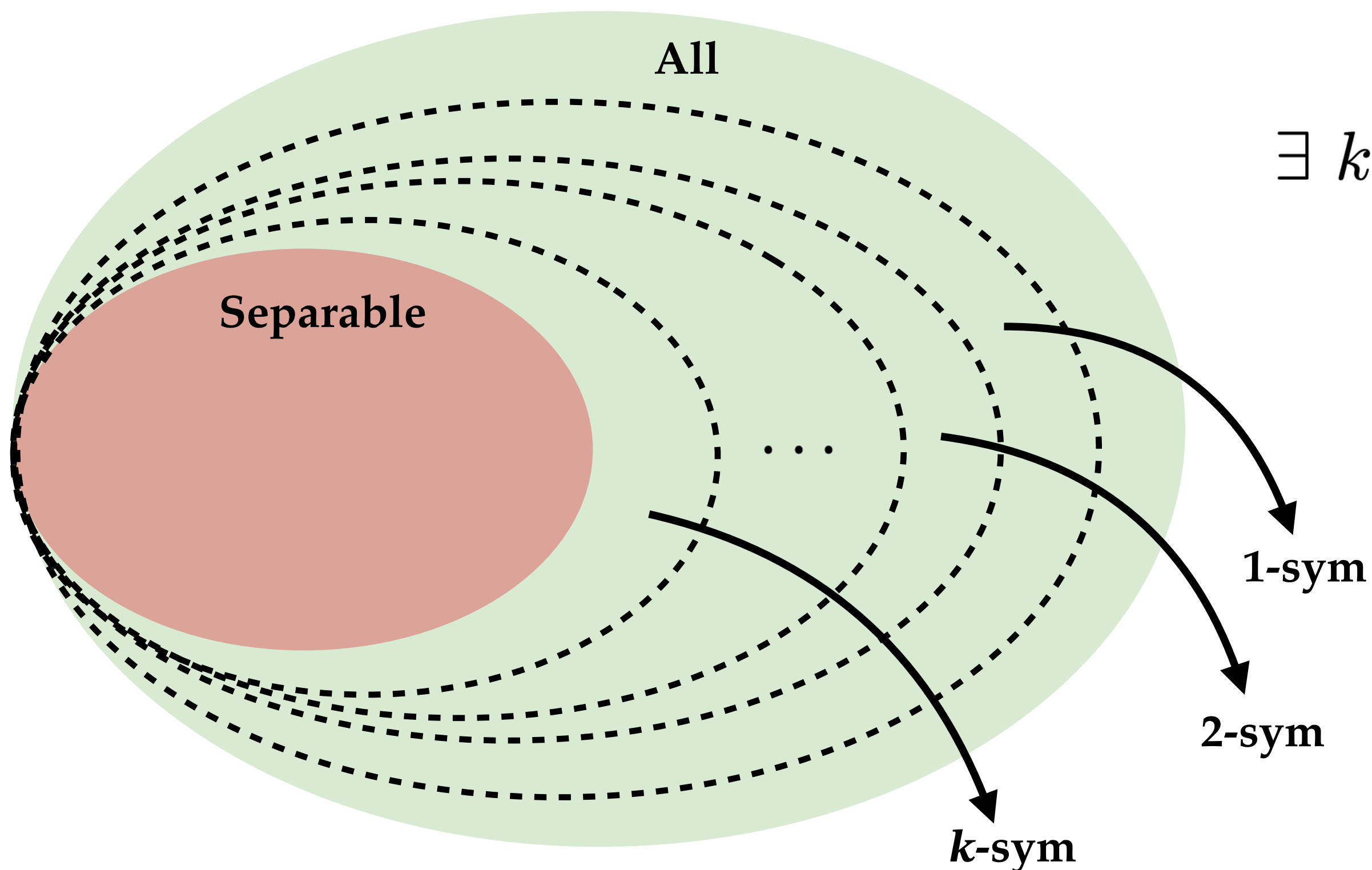
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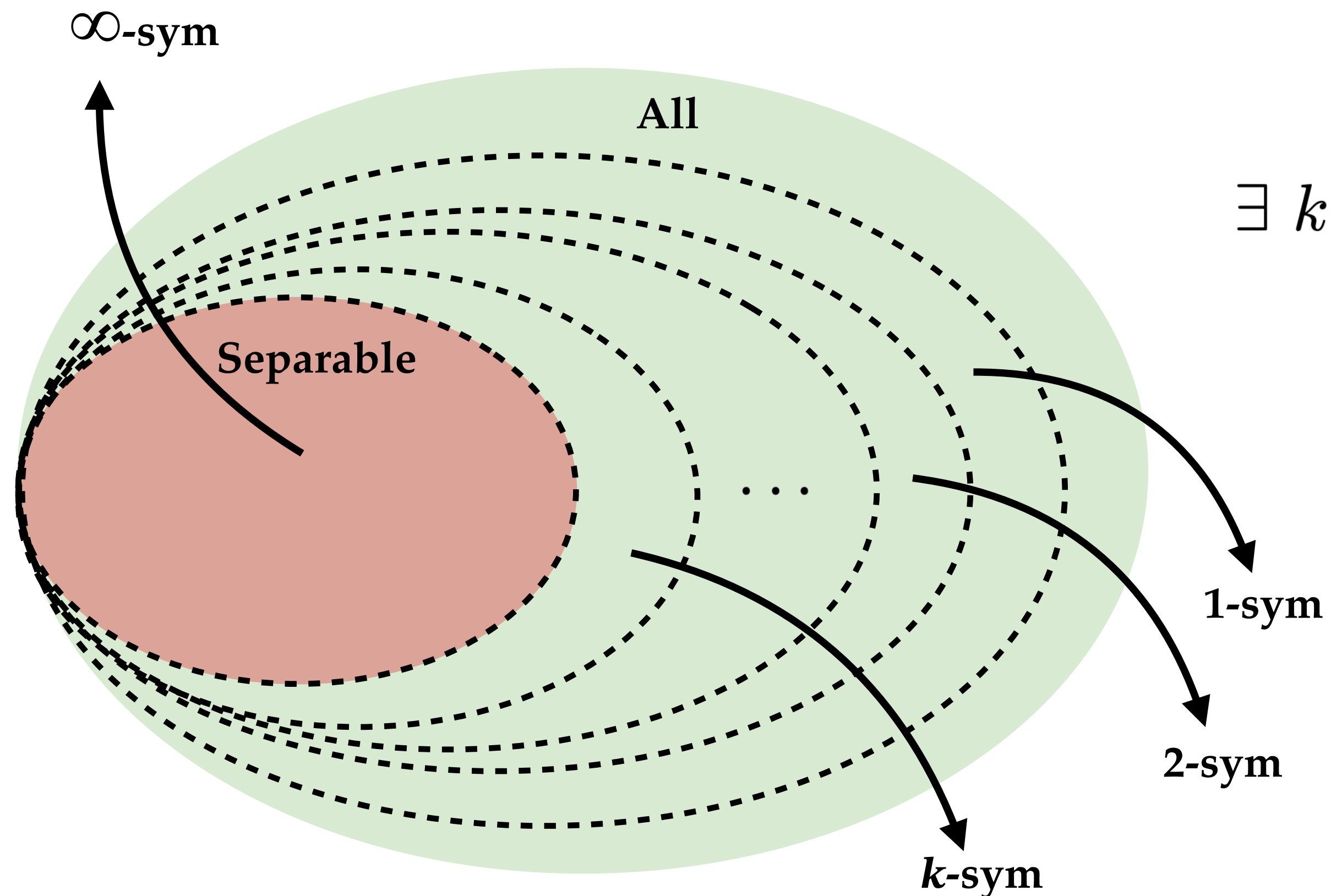
$$\rho_{AB} \in k\text{-sym} \forall k \in \mathbb{N} \implies \rho_{AB} \text{ is separable} \quad \checkmark$$

II. Approximation: **Impossibility**

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$$\rho_{AB} \in k\text{-sym} \forall k \in \mathbb{N} \xRightarrow{\checkmark} \rho_{AB} \text{ is separable}$$

$$\text{separable} = \infty\text{-sym}$$

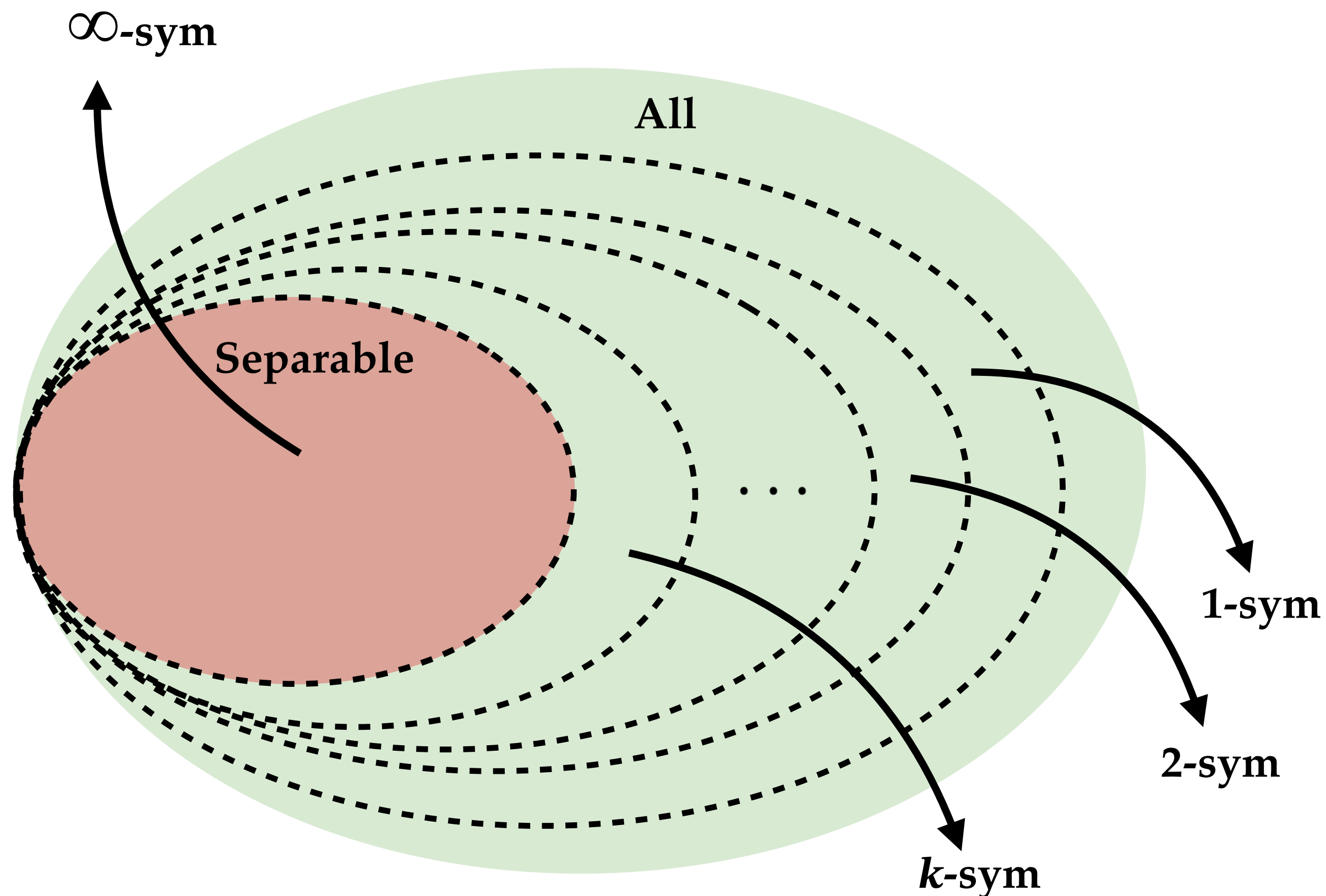
convergence

II. Approximation: **Impossibility**

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For all entangled states ρ_{AB}
 $\exists k \in \mathbb{N} ; \rho_{AB} \notin k\text{-sym}$

separable = ∞ -sym

convergence

II. Approximating: **Impossibility**

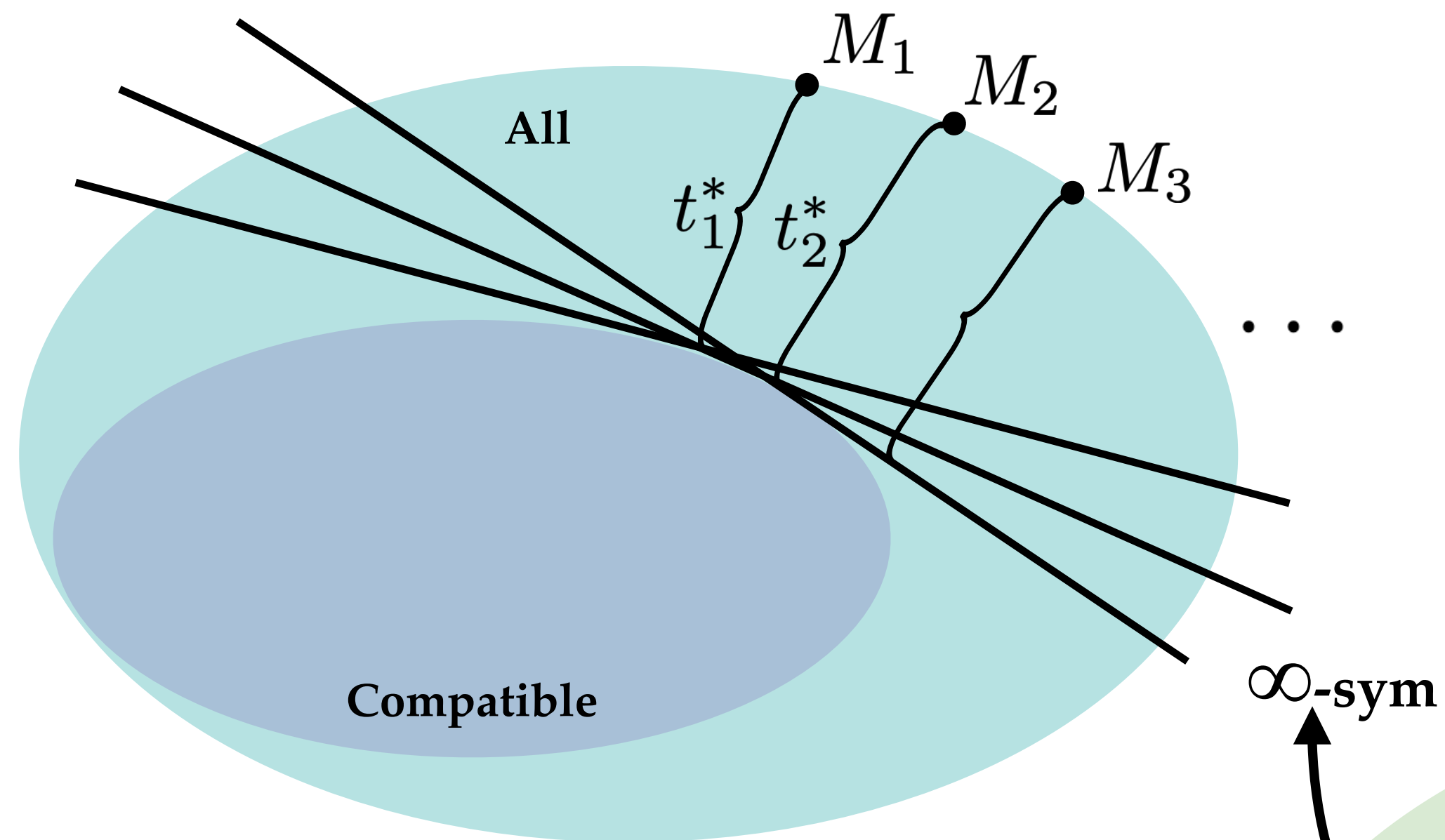
MAIN MESSAGE:

SDPs are useful tools to provide outer approximations of sets that do not accept a linear characterization.

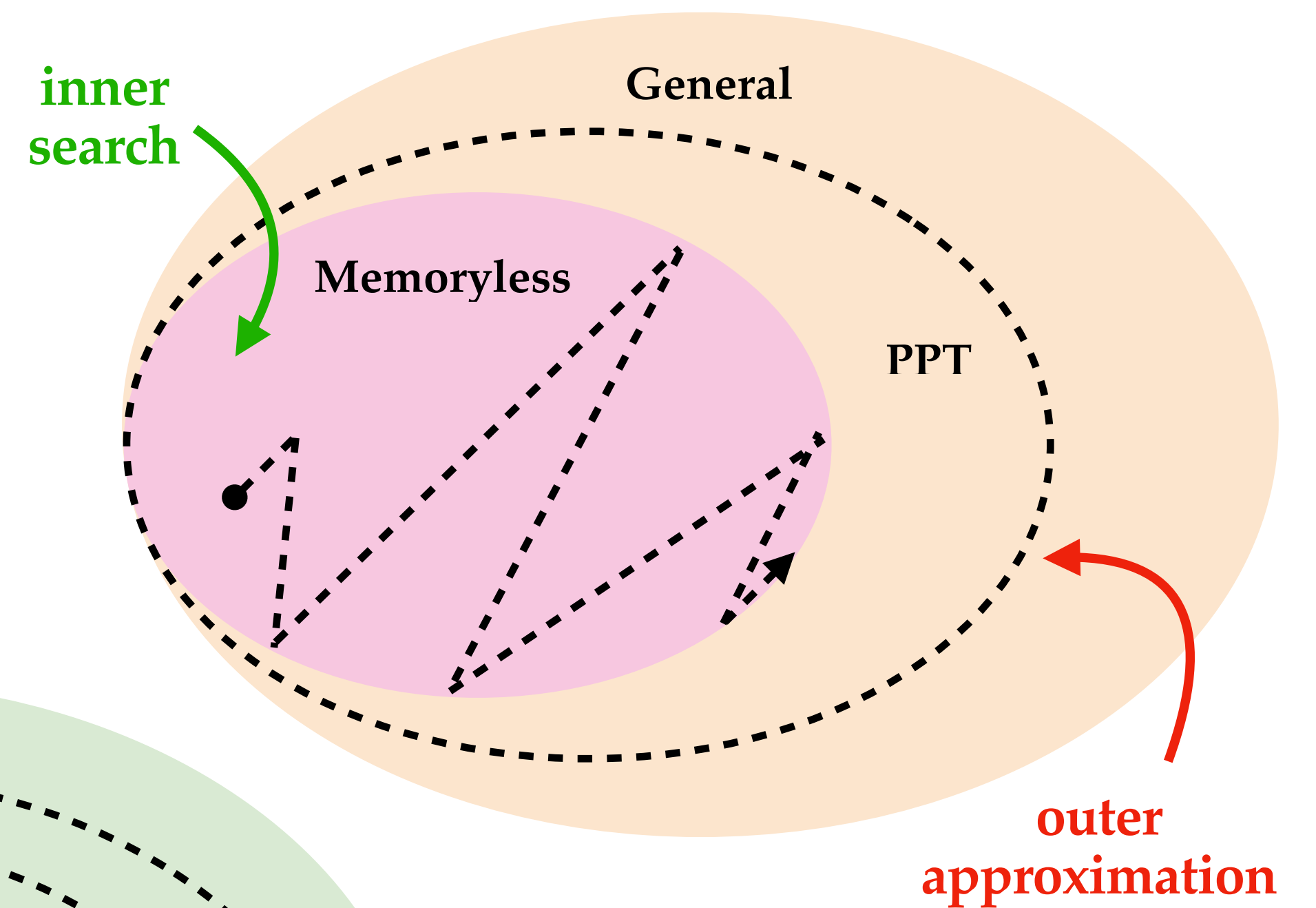
Converging SDP hierarchies can fully characterize some very complicated problems in quantum information.

Summary

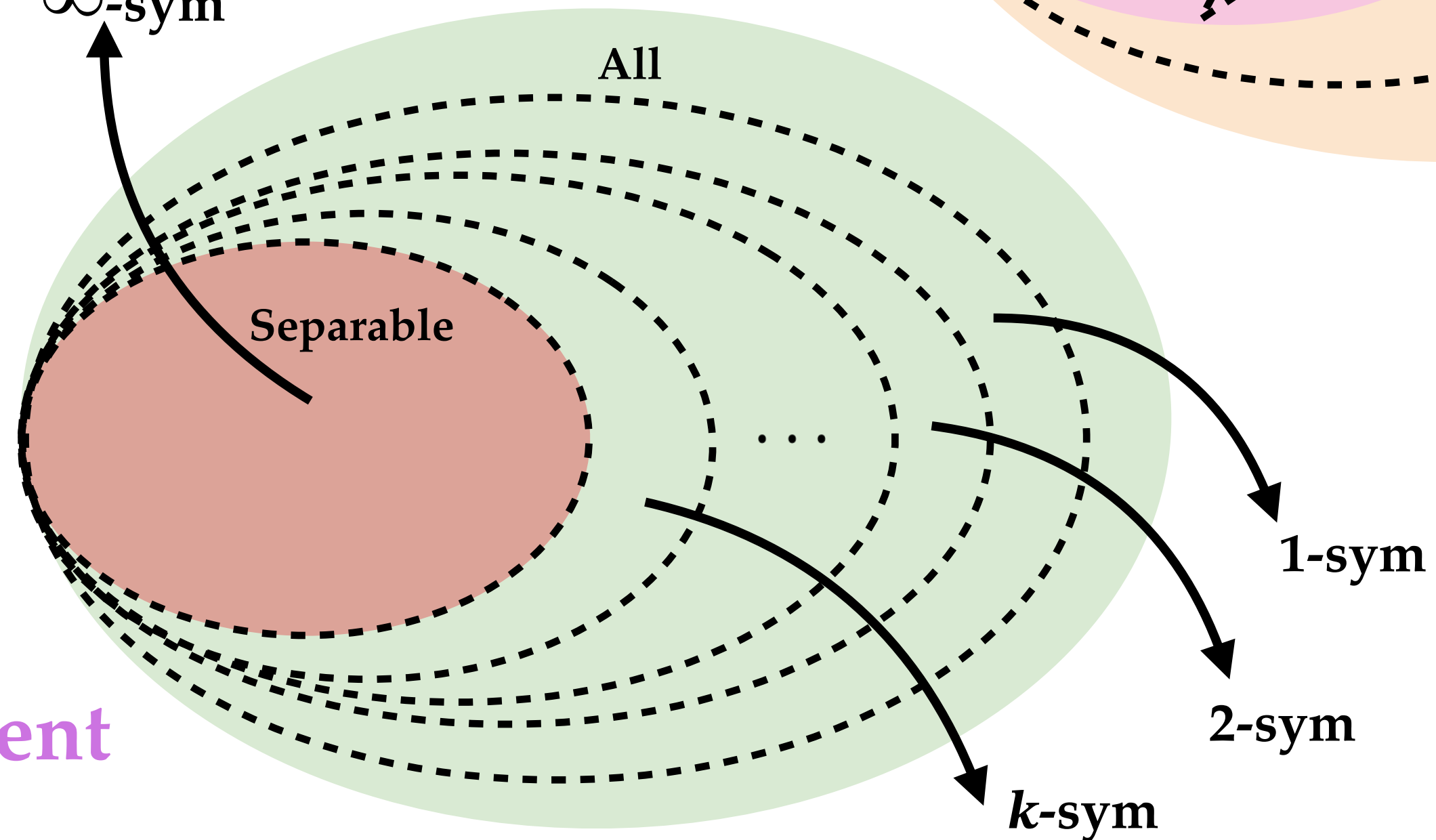
Measurement incompatibility



Channel discrimination



Entanglement

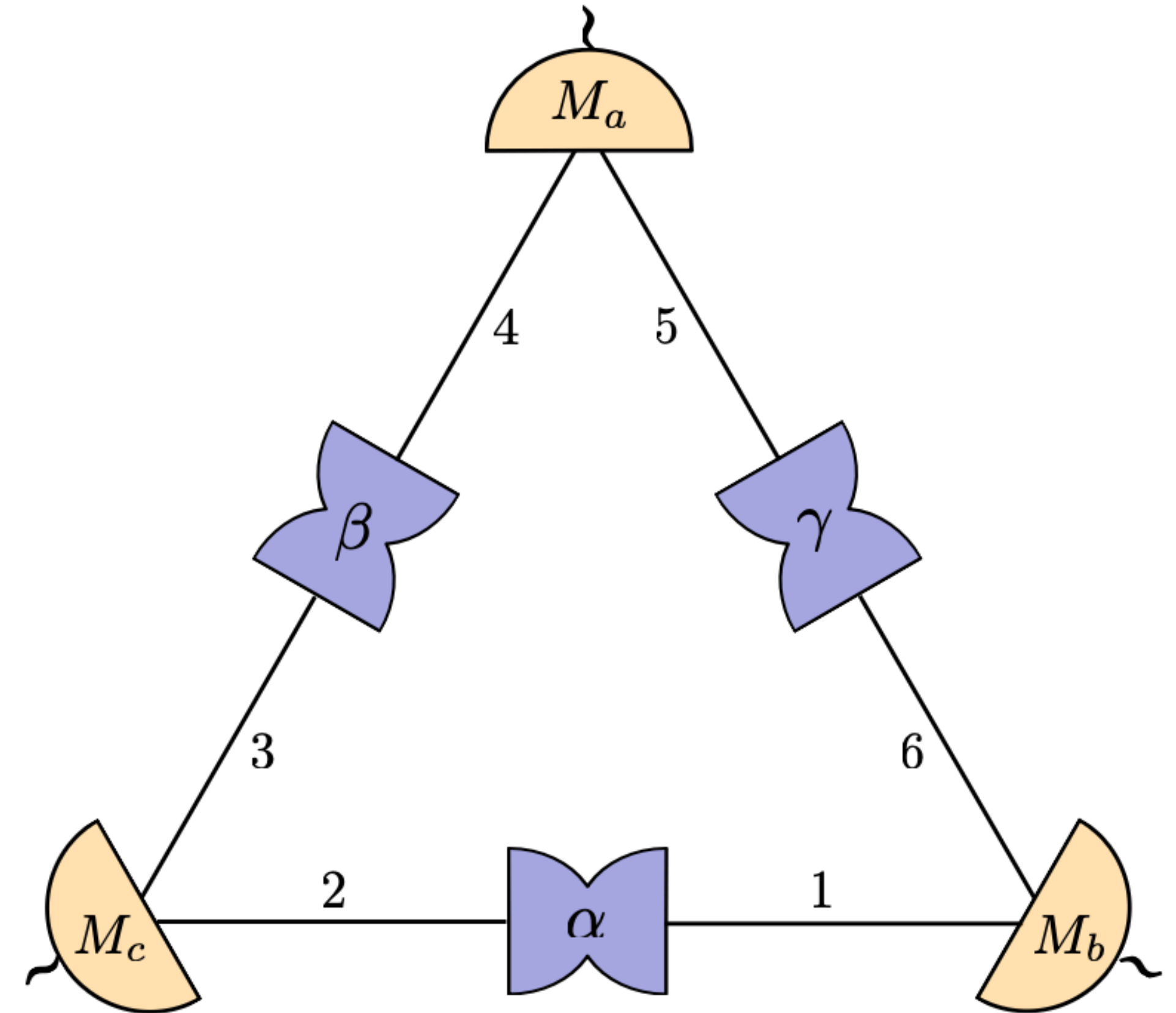


SDPs for optimization over non-convex sets

Quantum correlations in networks

$$p(abc) = \text{tr}[(\rho_\alpha \otimes \rho_\beta \otimes \rho_\gamma)(M_a^A \otimes M_b^B \otimes M_c^C)]$$

non-convex set



Branciard et al., Phys. Rev. A 85, 032119 (2012)

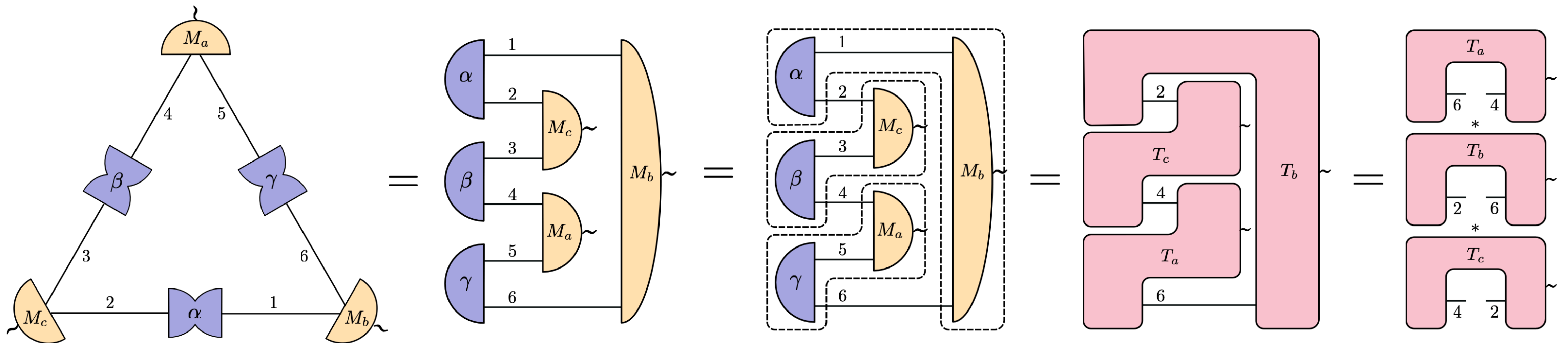
Fritz, New J. Phys. 14 103001 (2012)

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Inner search via SDP seesaw:

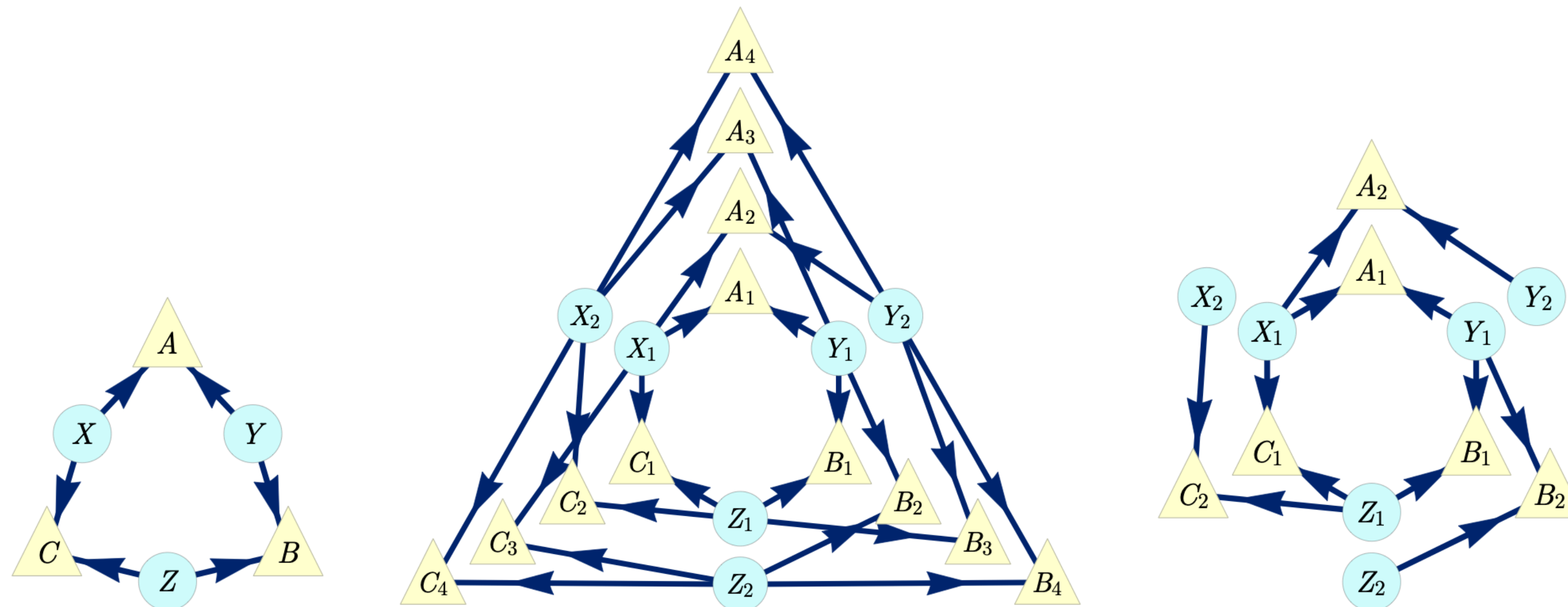


SDPs for optimization over non-convex sets

Quantum correlations in networks

$$p(abc) = \text{tr}[(\rho_\alpha \otimes \rho_\beta \otimes \rho_\gamma)(M_a^A \otimes M_b^B \otimes M_c^C)]$$

Outer approximation via SDP hierarchy:



Inflation technique

Rewriting infinitely many constraints

Unitary inversion

Does there exist a linear function that obeys some affine constraints such that

$$f(\text{Choi}(U)^{\otimes k}) = \text{Choi}(U^\dagger) \quad \forall U \in \text{SU}(d)$$

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$$f(\text{Choi}(U_i^{\otimes k})) = \text{Choi}(U_i^\dagger) \quad \forall i \in \{1, \dots, N\}$$

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Unitary inversion

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$$f(\text{Choi}(U_i^{\otimes k})) = \text{Choi}(U_i^\dagger) \quad \forall i \in \{1, \dots, N\}$$

where $\{U_i\}_{i=1}^N$, with $U_i \in \text{SU}(d)$, is such that

$$\text{span}(\{\text{Choi}(U_i)^{\otimes k}\}_{i=1}^N) = \text{span}(\{\text{Choi}(U)^{\otimes k} \mid U \in \text{SU}(d)\})$$

Further applications

- Quantum combs (quantum circuits with open slots)
Parallel, sequential, and indefinite-causal-order strategies for:
Hypothesis testing, discrimination, estimation...
& Universal channel transformations, storage-and-retrieval, purification...
& Quantum interactive proofs
- Metrology and quantum Fisher information
- Generalized DPS for memory-constrained problems
- Quantum correlations, NPA hierarchy, quantum bounds for Bell inequality violations
- Quantum tomography, fidelity estimation, distances & norms
- Device-independent protocols, entropy lower bounds for key rates and randomness generation
- Resources theories and robustness measures
- ...

Conclusions

- 1) Not every SDP is created in an SDP form.
- 2) Seesaws are useful tools to tackle multilinear problems, duality theory can help solve min-max problems.
- 3) SDPs are useful tools to provide non-trivial outer approximations of sets that do not accept a linear characterization (even non-convex sets).

Thank you!